

يونيورسيتي سلطان زين العابدين

UNISZA

UNIVERSITI SULTAN ZAINAL ABIDIN

TKS4163 : WORKSHOP

HOMESTAY BOOKING SYSTEM DETAIL DESIGN

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CHAPTER 1 : SCOPE

1.1 System Overview

Nowadays, homestay is a trend of tourism or study abroad program that allows the visitor to rent a room from a local family to learn more about the local lifestyle. While homestays can located in any destination worldwide, some countries prefer homestay than others as a means of developing their tourism industry.

Travelling using homestay service has been an increasingly popular trend worldwide that is encouraged by many social organizations and governments. Homestay service also supports the cultural exchange between customer and owner which makes them understand each other's better and creates friendships. In some respect, homestay service plays an active role in the globalization process. This type of service brings about an additional flow of income to the host families and advertises for the culture and people of a nation. Homestay is very effective option to students who want to improve their foreign language and knowledge about local culture. Tourist prefers homestay more than hotel because it is more comfortable as they can choose what type of home they need for their family. An online system is believed to facilitate and upgrade the current manual system and brought a new level in enhancing the management system, beside creating a user-friendly interface for user ease.

In this system, there are three stakeholders that involved which are user, owner and admin. Users use the system to book homestay while owner will apply partnership to rent the homestay. Admin is in charge to manage homestay booking system.

There are few existed systems that have been referred;
Booking.com, Agoda.com, CariHomestay.net, HolidayHour.com.my

1.2 Problem Statement

In manual system, user finds it is difficult to compare homestay information because they have to open more than one site to make comparison information. Next, manual system also does not promising accurate and latest information because they need to call to get information that may cause misunderstanding in term communication and deliverance of info. Apart from that, manuals system does not provide the maps of the location so it becomes so hard to user to find out the exact location of homestay.

1.3 Objectives

The main objectives for the homestay booking system are to make ease for user to compare about the homestay information. Without homestay booking system, user cannot compare all the information about homestay thus with homestay booking system, user can make comparison and gain more beneficial detail. Besides that, it also provide more accurate and latest information to the user because in manually tourists need to call to know the current information. From the homestay booking system also user's information can be saved directly and accurately into database. It is no need to jot down all the detail on a paper because risk of lost or false information either to customer or admin can be reduce when the system develop.

1.4 Scope

1.4.1 User

The target user for this system is tourists that want to find the best accommodations.

1.4.2 System

The target for this system only covers area in Malaysia.

CHAPTER 2 : PRELIMINARY DESIGN

2.1 CONTEXTUAL DIAGRAM

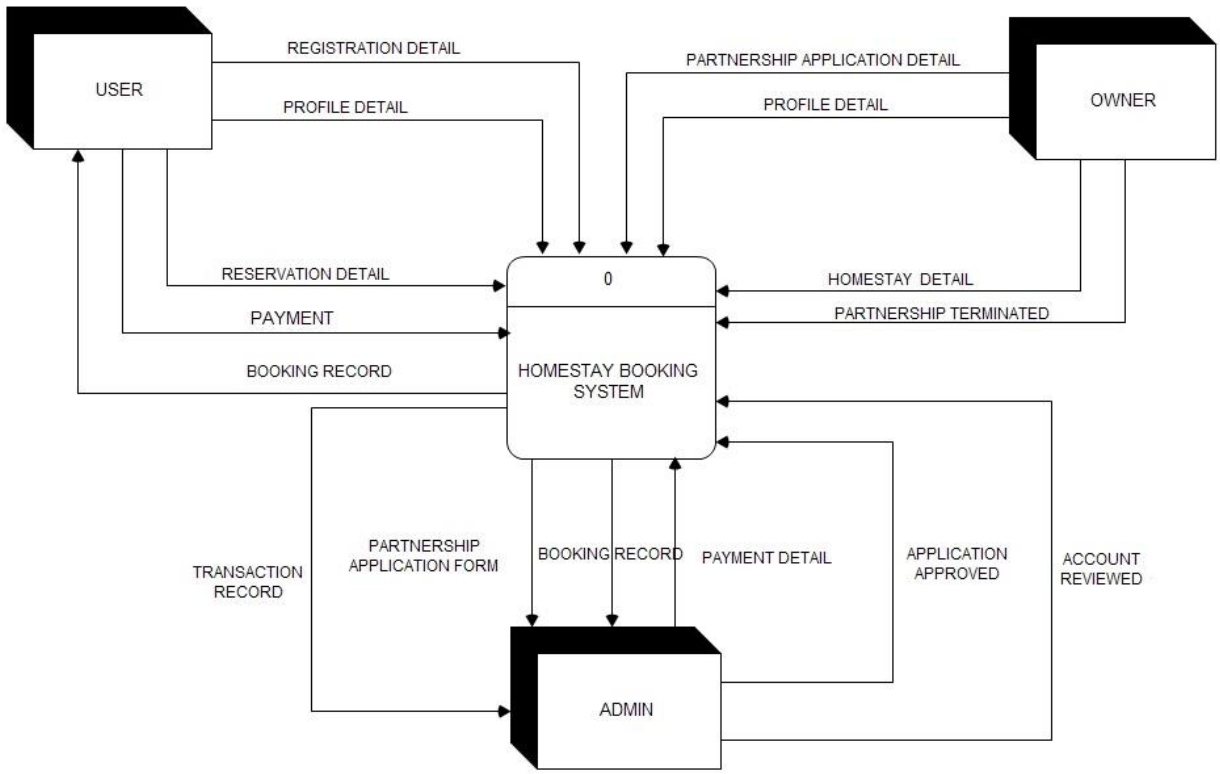


Figure 2.1

DESCRIPTION OF CONTEXT DIAGRAM

The context diagram for the homestay booking system is shown in the figure above. The HOMESTAY BOOKING SYSTEM (HBS) process is at the center of the diagram. The three entities (USER, OWNER and ADMINISTRATOR) are placed around the central process. Fifteenth data flows are involved in the interaction between the central process and the entities. The USER entity has one incoming data flow, BOOKING RECORD, but has four outgoing data flows, REGISTRATION DETAIL, PROFILE DETAIL, RESERVATION DETAIL and PAYMENT. The OWNER entity has no incoming data flow and four outgoing data flow, PARTNERSHIP APPLICATION DETAIL, PROFILE DETAIL, HOMESTAY DETAIL and PARTNERSHIP TERMINATED. Finally, the ADMIN entity has three incoming data flow, PAYMENT DETAIL, APPLICATION APPROVED and ACCOUNT REVIEWED and three outgoing data flows, TRANSACTION RECORD and PARTNERSHIP APPLICATION FORM and BOOKING RECORD.

2.2 DATA FLOW DIAGRAM LEVEL 0

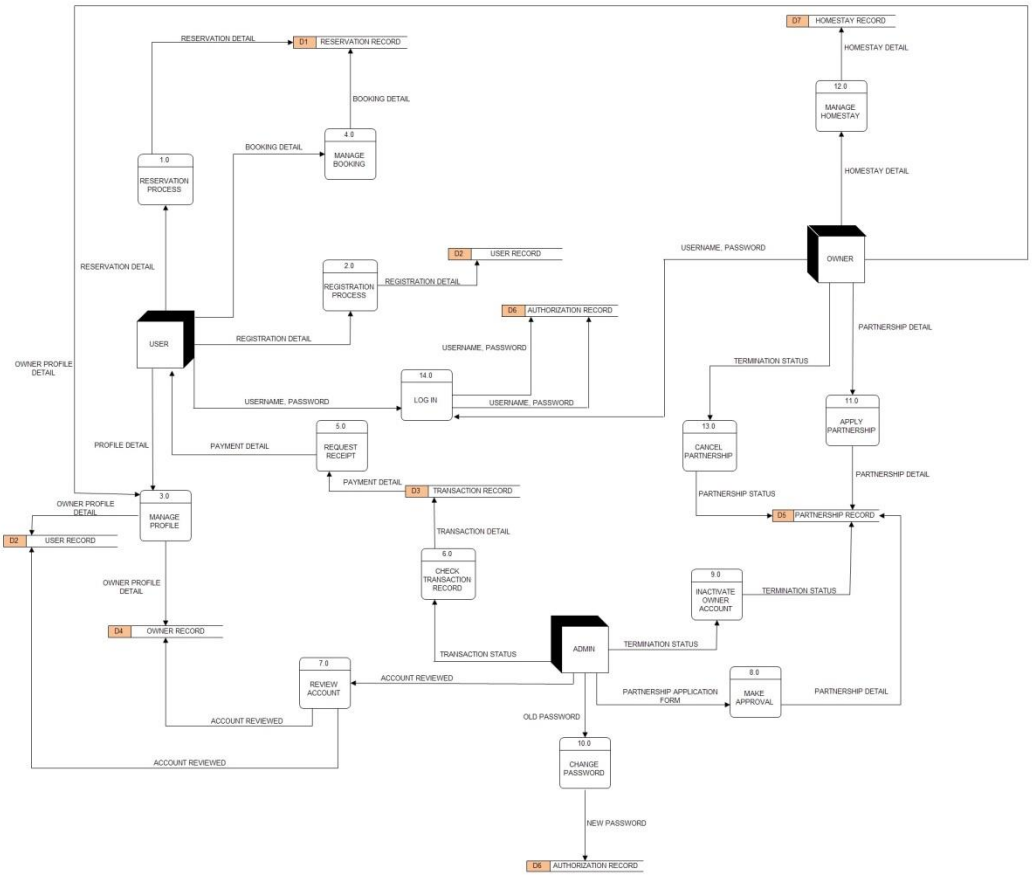


Figure 2.2

Description

The DFD has three entities which are USER, OWNER and ADMIN. REGISTRATION PROCESS, MANAGE PROFILE, MANAGE BOOKING, REQUEST RECEIPT, MAKE PAYMENT PROCESS, CHECK TRANSACTION RECORD, REVIEW ACCOUNT, MAKE APPROVAL, INACTIVATE OWNER ACCOUNT, CHANGE PASSWORD, APPLY PARTNERSHIP, CHECK PARTNERSHP APPROVAL, MANAGE HOMESTAY and CANCEL PARTNERSHIP are fifteen processes involve in the system. There are seven data stores created in the system which is BOOKING RECORD, USER RECORD, TRANSACTION RECORD, OWNER RECORD, PARTNERSHIP RECORD, AUTHORIZATION RECORD and HOMESTAY RECORD.

1. USER enters USERNAME and PASSWORD to LOG IN process which outputs USERNAME and PASSWORD into AUTHORIZATION RECORD data store.
2. A USER inputs RESERVATION DETAIL into RESERVATION PROCESS which outputs RESERVATION DETAIL into BOOKING RECORD data store.
3. USER enters REGISTRATION DETAIL and processes by REGISTRATION PROCESS. The REGISTRATION DETAIL then is store in USER RECORD data store.
4. When USER inputs the PROFILE DETAIL, a process named PROFILE DETAIL will sends PROFILE DETAIL to the USER RECORD data store.
5. USER enters the BOOKING DETAIL. The MANAGE BOOKING process sends the BOOKING DETAIL to the RESERVATION RECORD data store.
6. ADMIN checks TRANSACTION STATUS. CHECK TRANSACTION RECORD process then inputs the TRANSACTION DETAIL to TRANSACTION RECORD data store.
7. ADMIN selects ACCOUNT REVIEWED. ACCOUNT REVIEW process then sends ACCOUNT REVIEWED to USER RECORD and OWNER RECORD.
8. ADMIN changes OLD PASSWORD. The CHANGE PASSWORDS process the sends the NEW PASSWORD to the AUTHORIZATION RECORD data store.
9. USER enters USERNAME and PASSWORD to LOG IN process which outputs USERNAME and PASSWORD into AUTHORIZATION RECORD data store.
10. When OWNER inputs PARTNERSHIP DETAIL, the APPLY PARTNERSHIP process then sends the PARTNERSHIP DETAIL to PARTNERSHIP RECORD data store.
11. OWNER inputs HOMESTAY DETAIL. MANAGE HOMESTAY process then sends HOMESTAY DETAIL to HOMESTAY RECORD data store.
12. OWNER keys in TERMINATION STATUS. The CANCEL PARTNERSHIP process then sends PARTNERSHIP STATUS to PARTNERSHIP RECORD data store.
13. ADMIN inputs TERMINATION STATUS which processed by INACTIVATE OWNER ACCOUNT. The TERMINATION STATUS then is sent to PARTNERSHIP RECORD data store.

2.3 PROCESS DECOMPOSITION

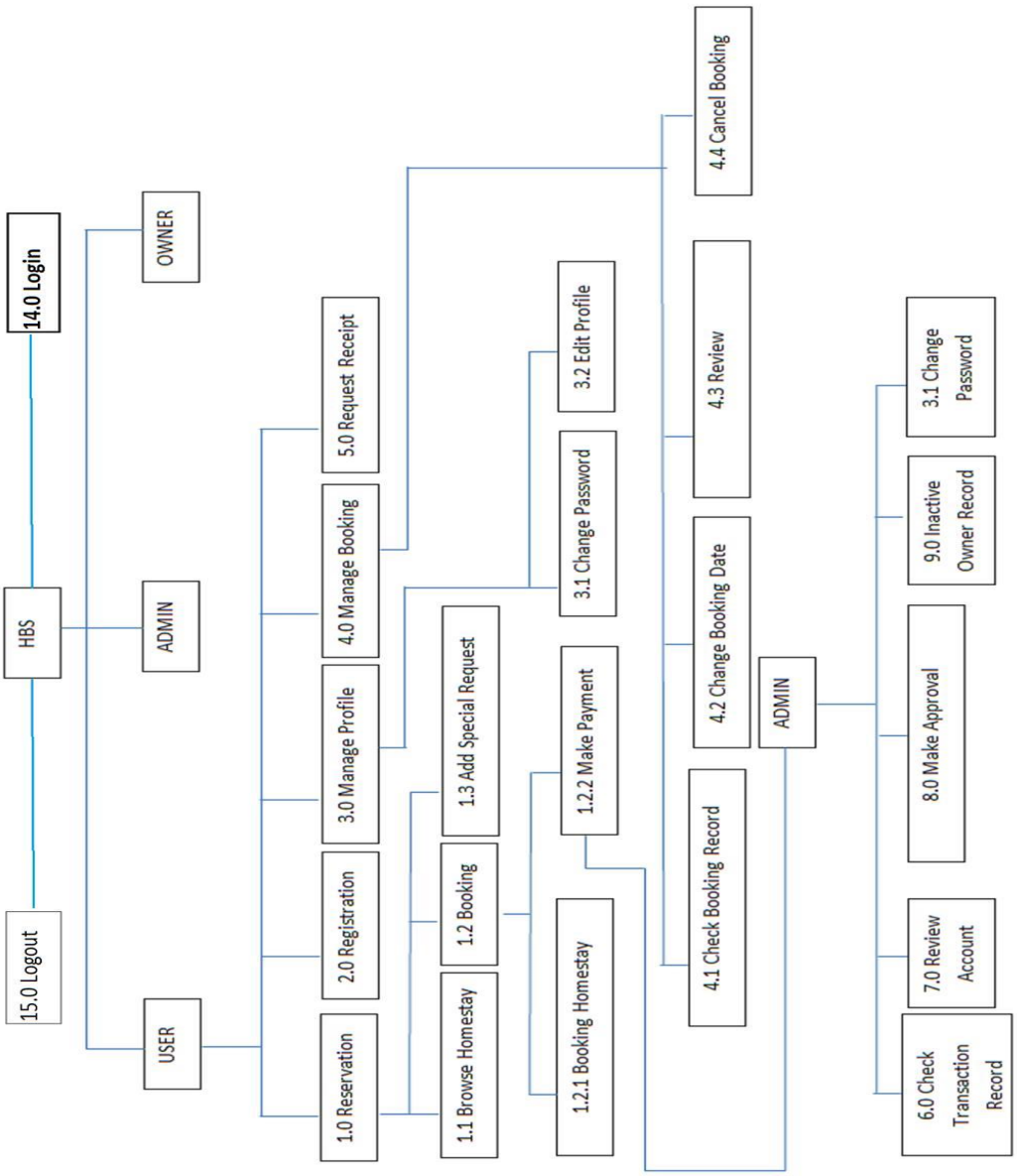


Figure 2.3

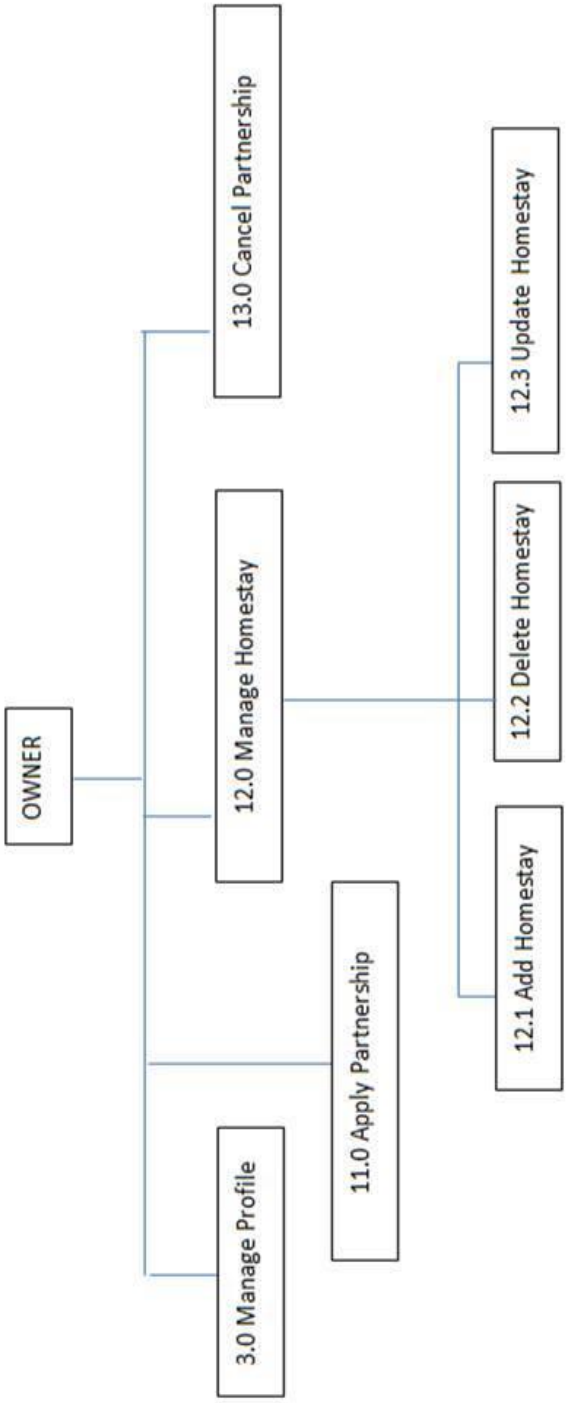


Figure 2.3(cont.)

CHAPTER 3 : DETAIL DESIGN

PSEUDOCODE, FLOW CHART AND INTERFACE FOR USER

3.1 Process 1.0 : Reservation Process

3.1.1 Process 1.1: Browse Homestay

a. Pseudo code

1. Start
2. Input destination, startDate, endDate
3. Match the request destination detail with database Homestay Record and database Booking Record
4. If (request match)
   4.1 Display the list of homestay
5. Else
   5.1 Display “No result found” message
6. End

b. Flow chart

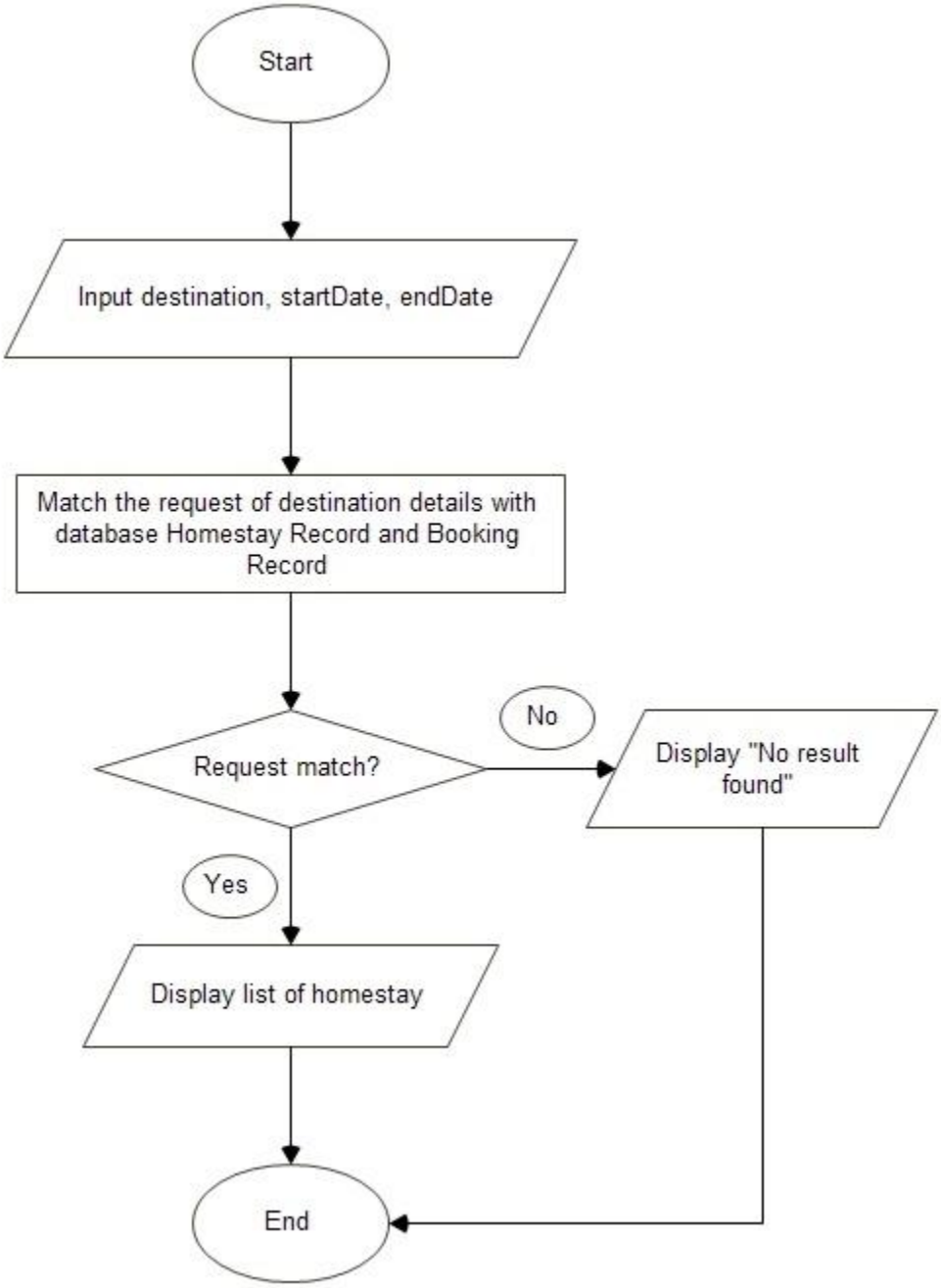
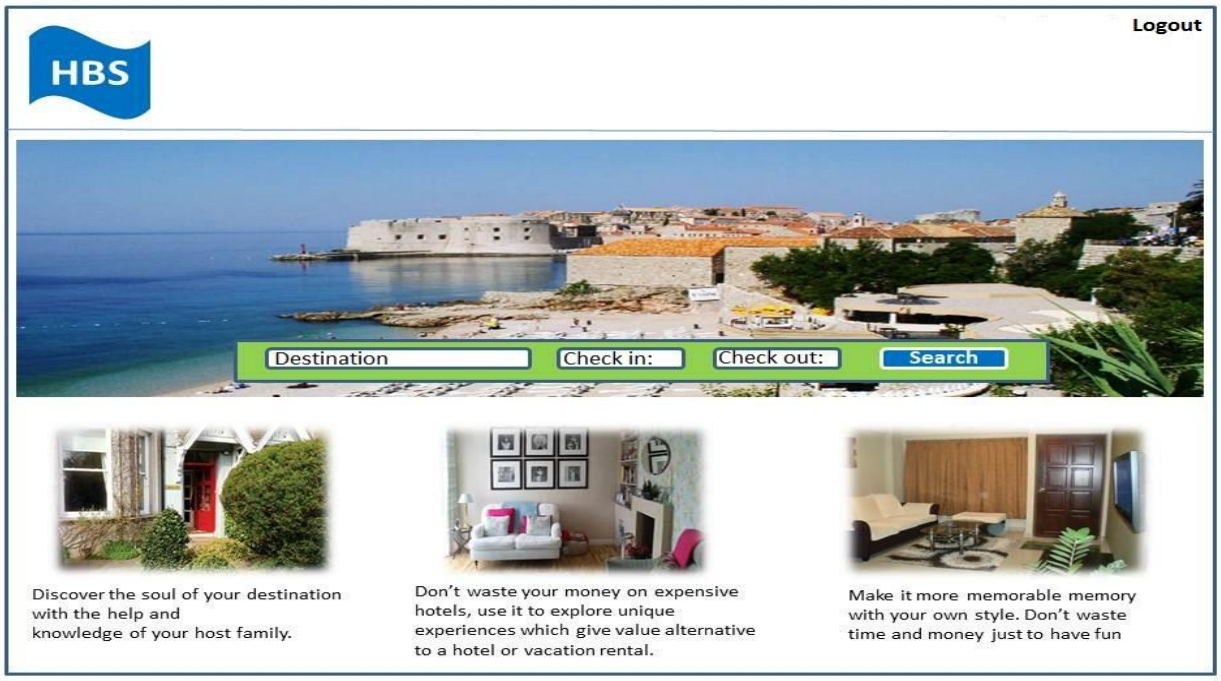
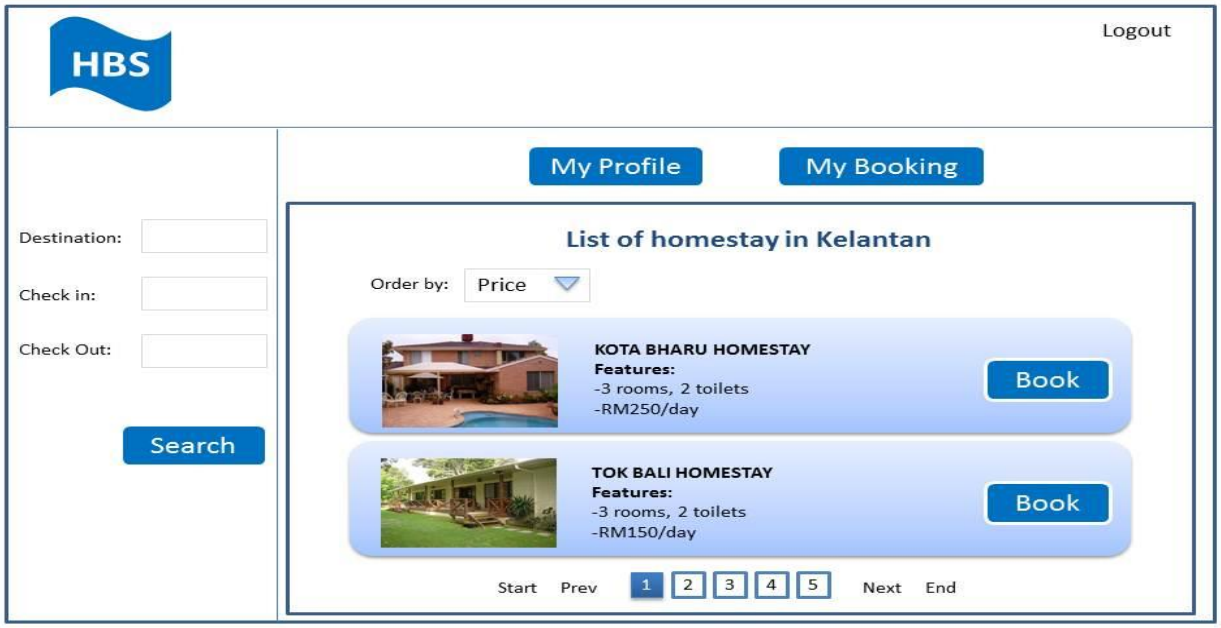


Figure 3.1.1

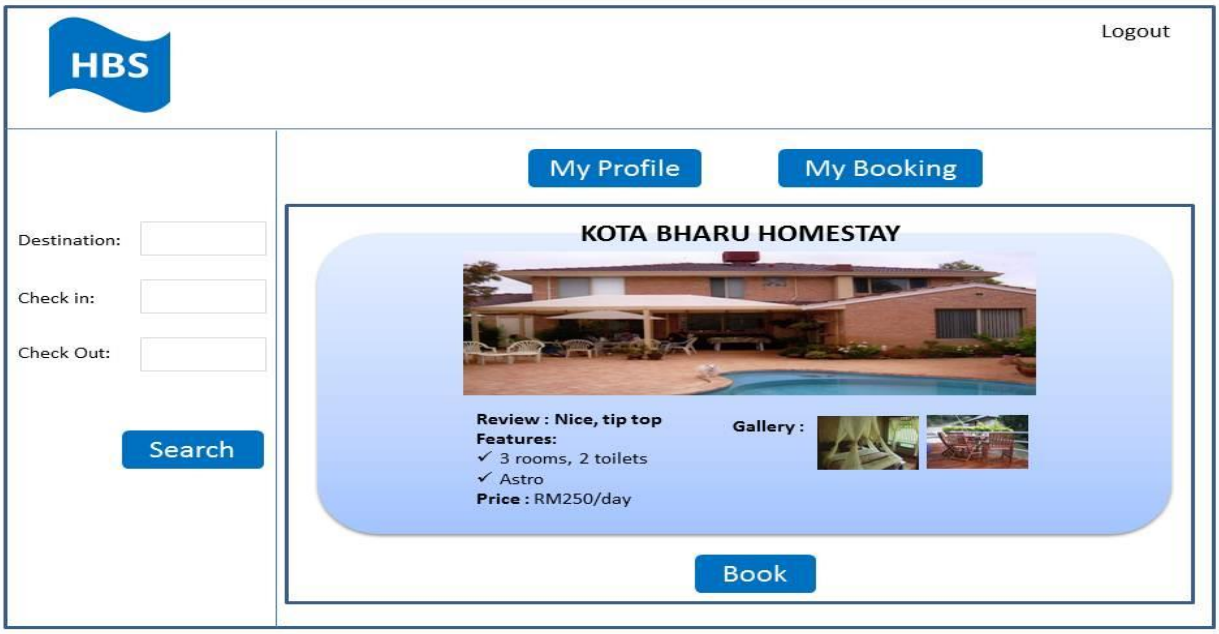
c. Graphic User Interface



Screen 1



Screen 2


HBS
Logout
My Profile
My Booking
Destination:
Check in:
Check Out:
Search
KOTA BHARU HOMESTAY
Review : Nice, tip top
Features:
✓ 3 rooms, 2 toilets
✓ Astro
Price : RM250/day
Gallery :
Book

Screen 3

3.1.2 Process 1.2.1: Make Payment

a. Pseudo code

1. Start
2. Select homestay
3. Input reservation details
4. If (select Submit)
   4.1 Save the reservation details into database Booking Record
   4.2 Load payment form
   4.3 Input payment details
   4.4 If (select Submit)
      4.4.1 Save the payment details into database Payment
      4.4.2 Display User Booking ID
5. Else (select Reset)
   5.1 Reset form
6. End

b. Flow chart

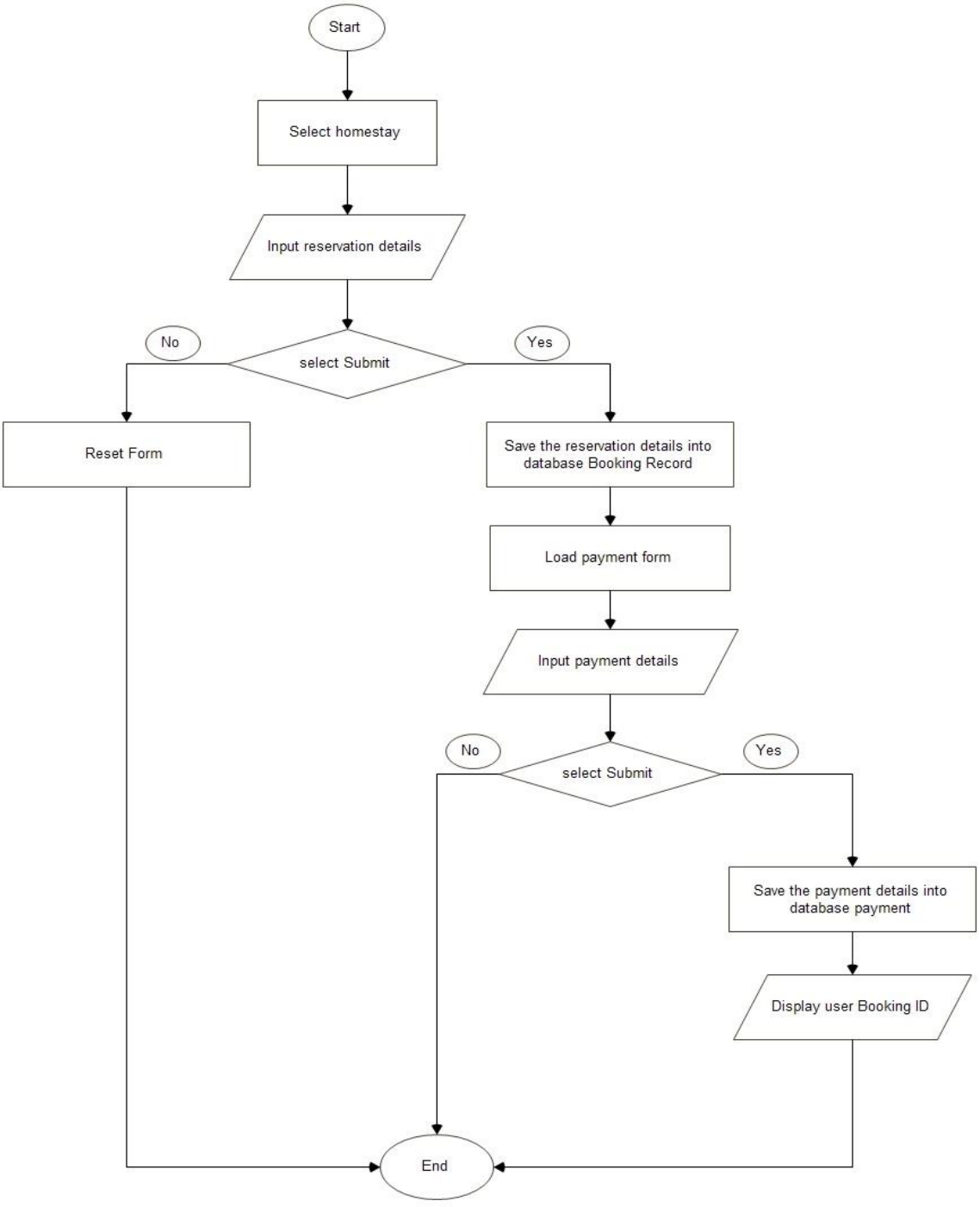
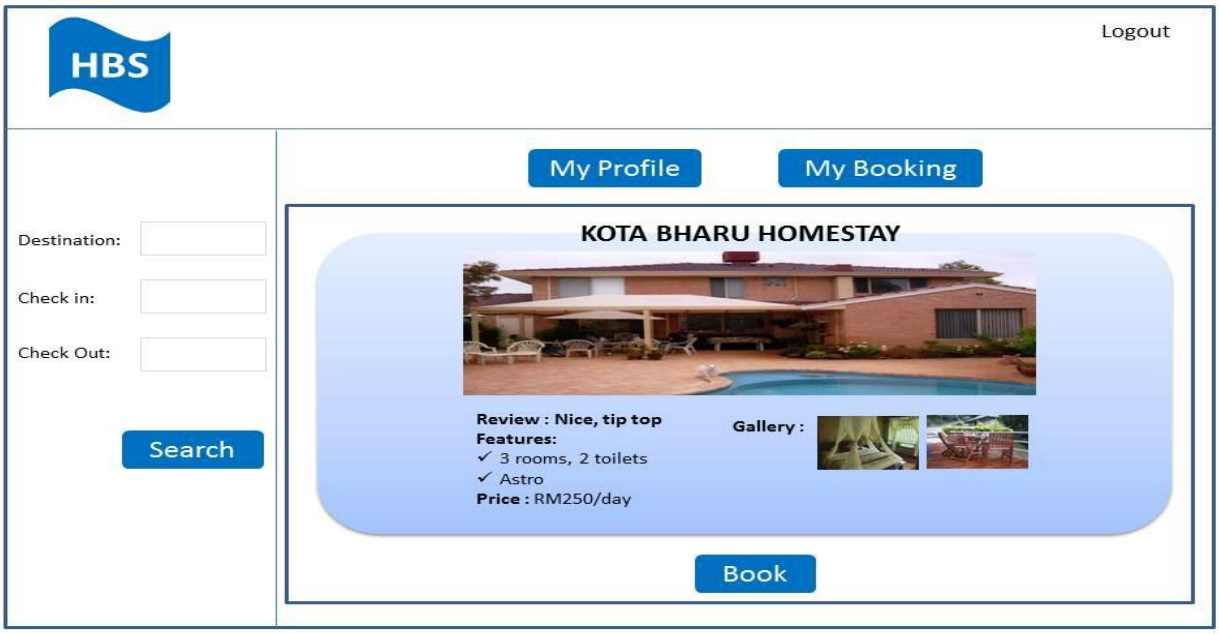
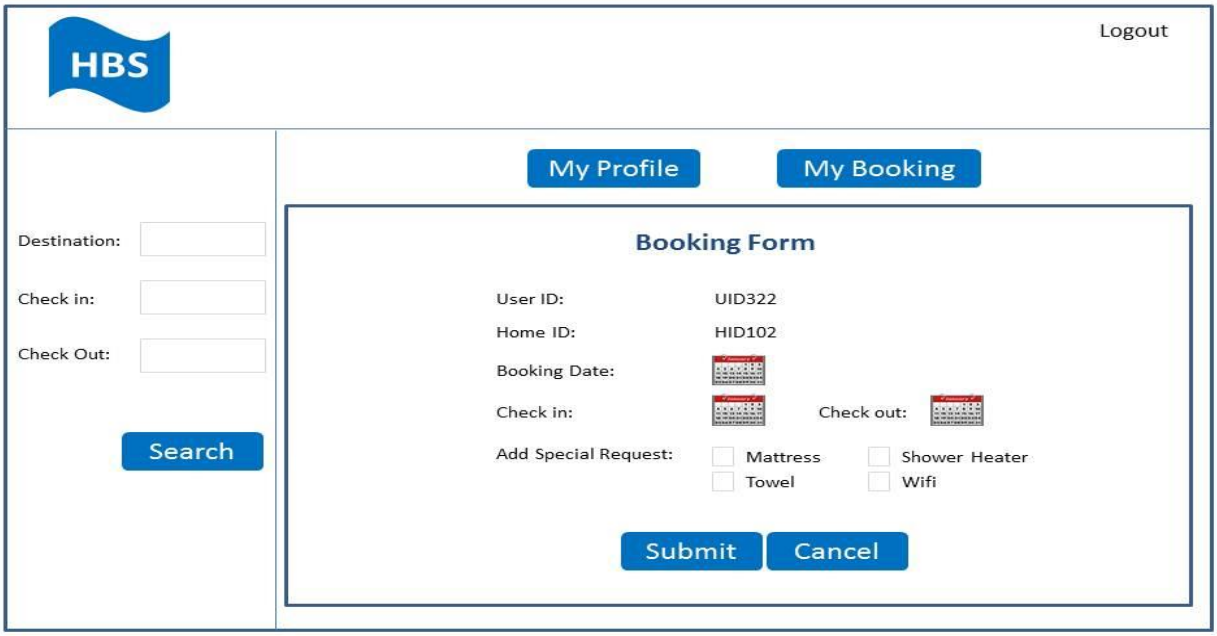


Figure 3.1.2

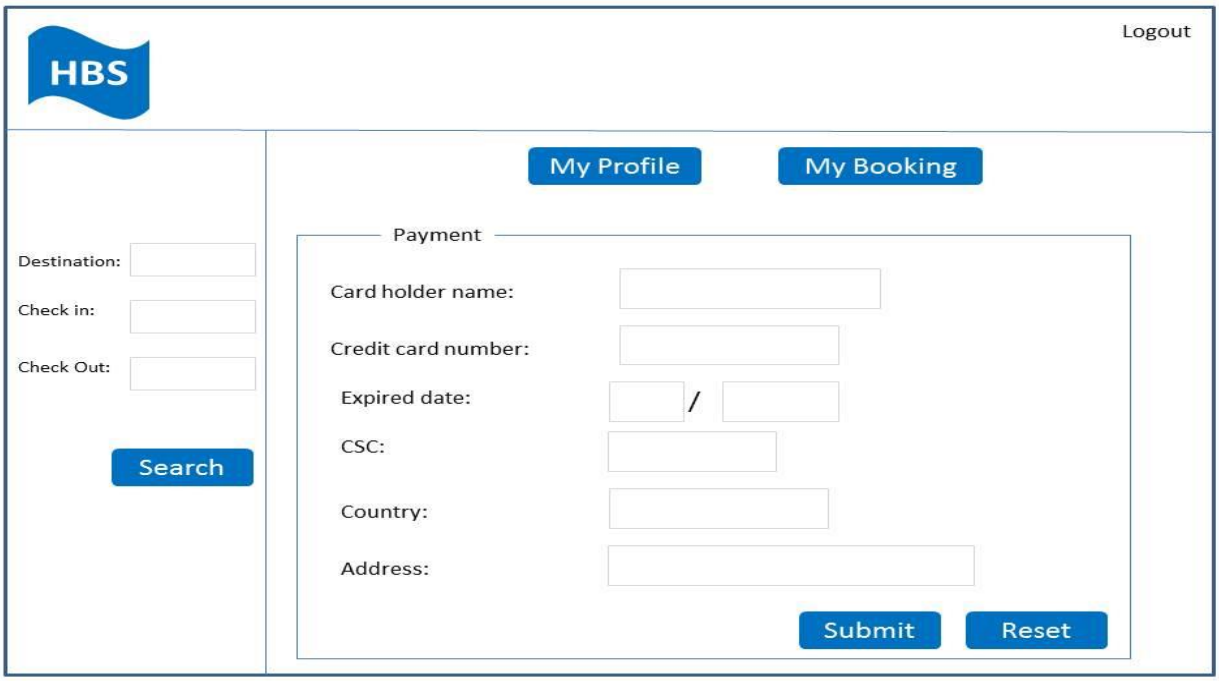
c. Graphic User Interface



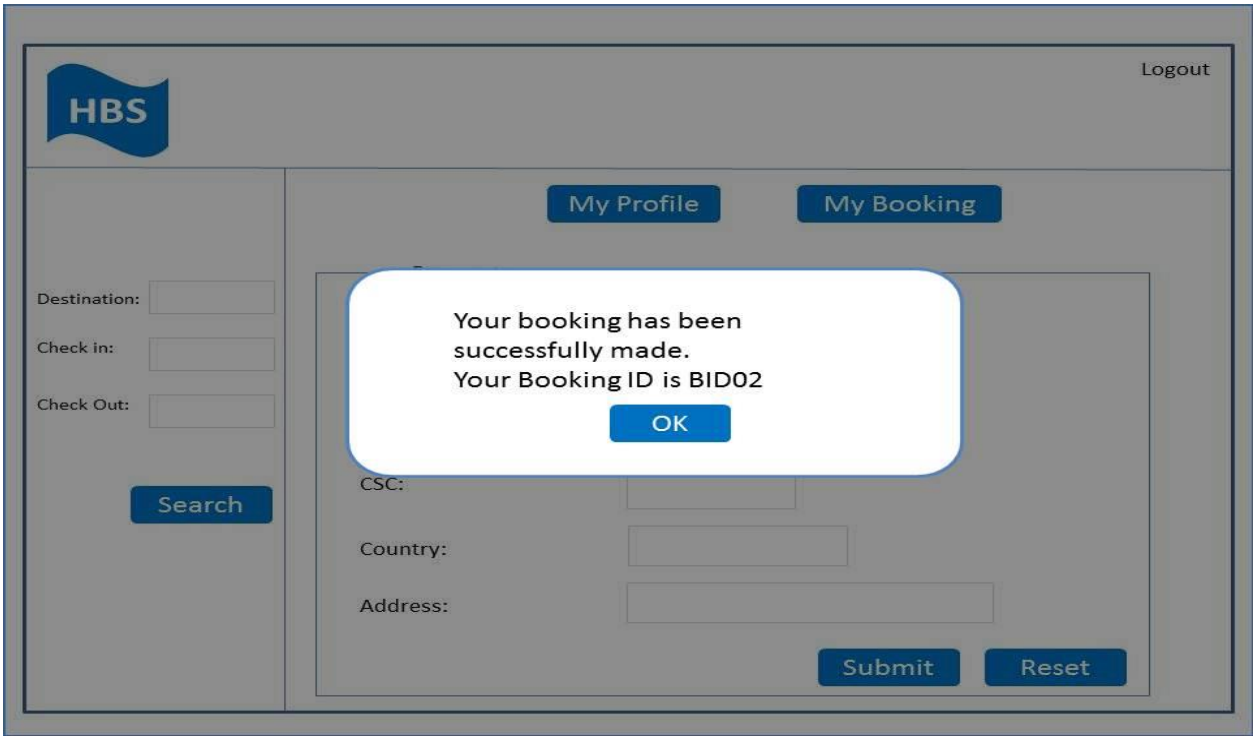
Screen 1



Screen 2



Screen 3



Screen 4

3.2 Process 2.0 : Registration Process

a. Pseudo code

1. Start
2. Load registration form
3. Input registration details
4. If (select Submit)
   4.1 Save user's registration details into database Authorization Record
   4.2 Display message registration successful
5. Else
   5.1 Display cancel message
6. End

b. Flow chart

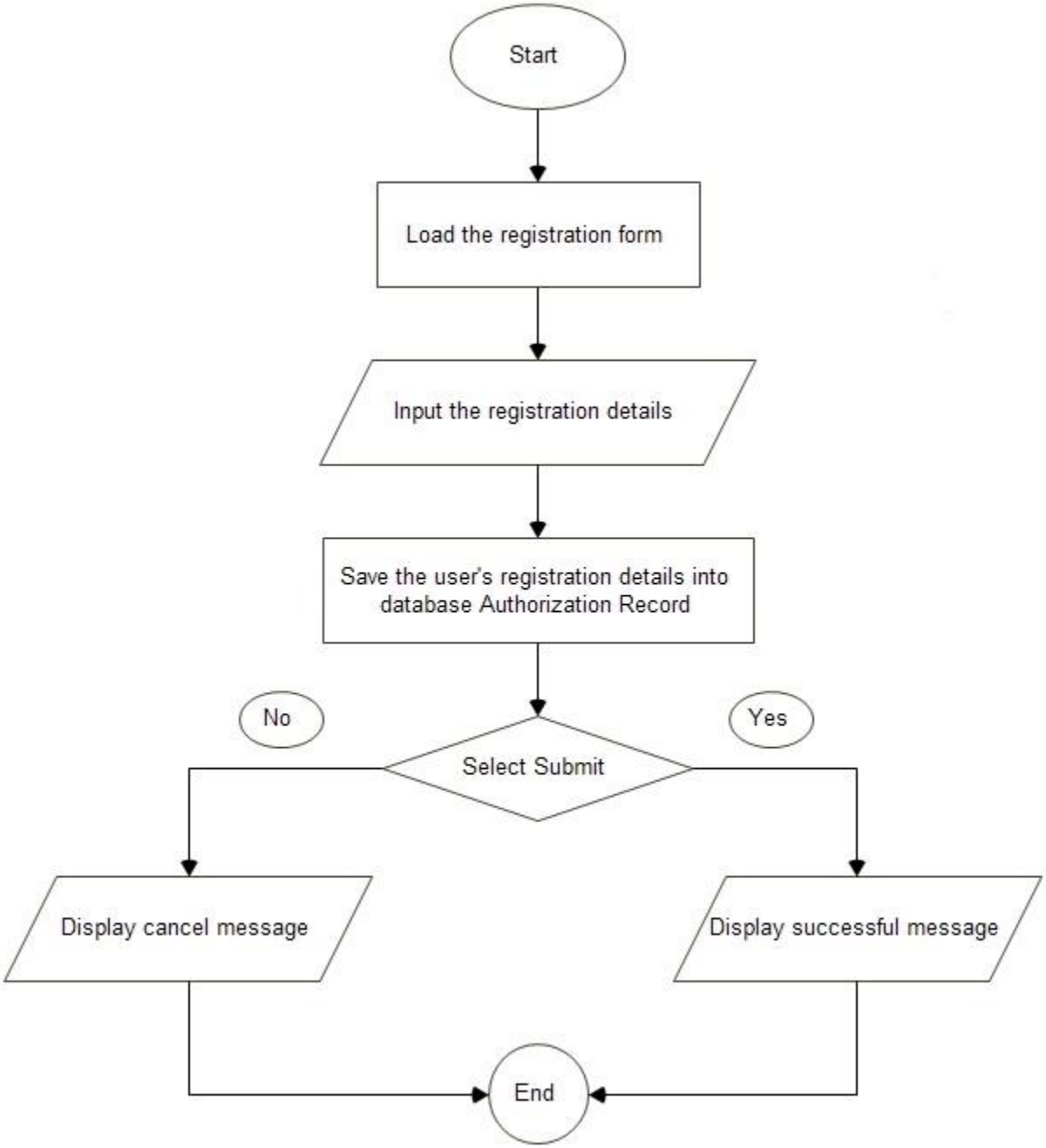
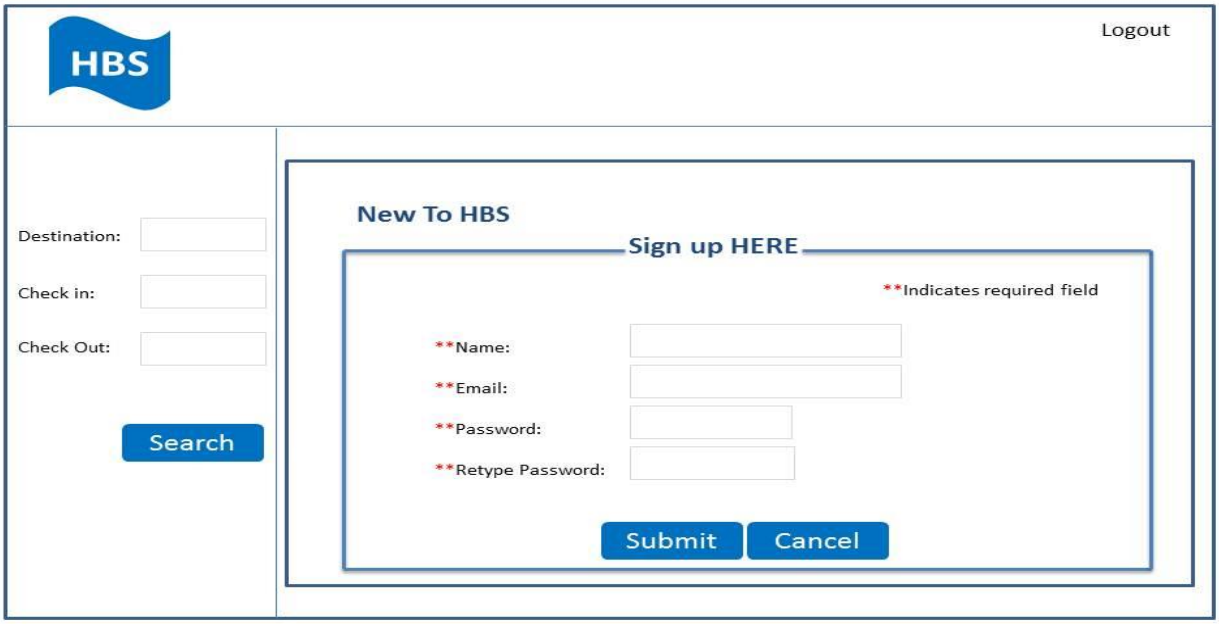
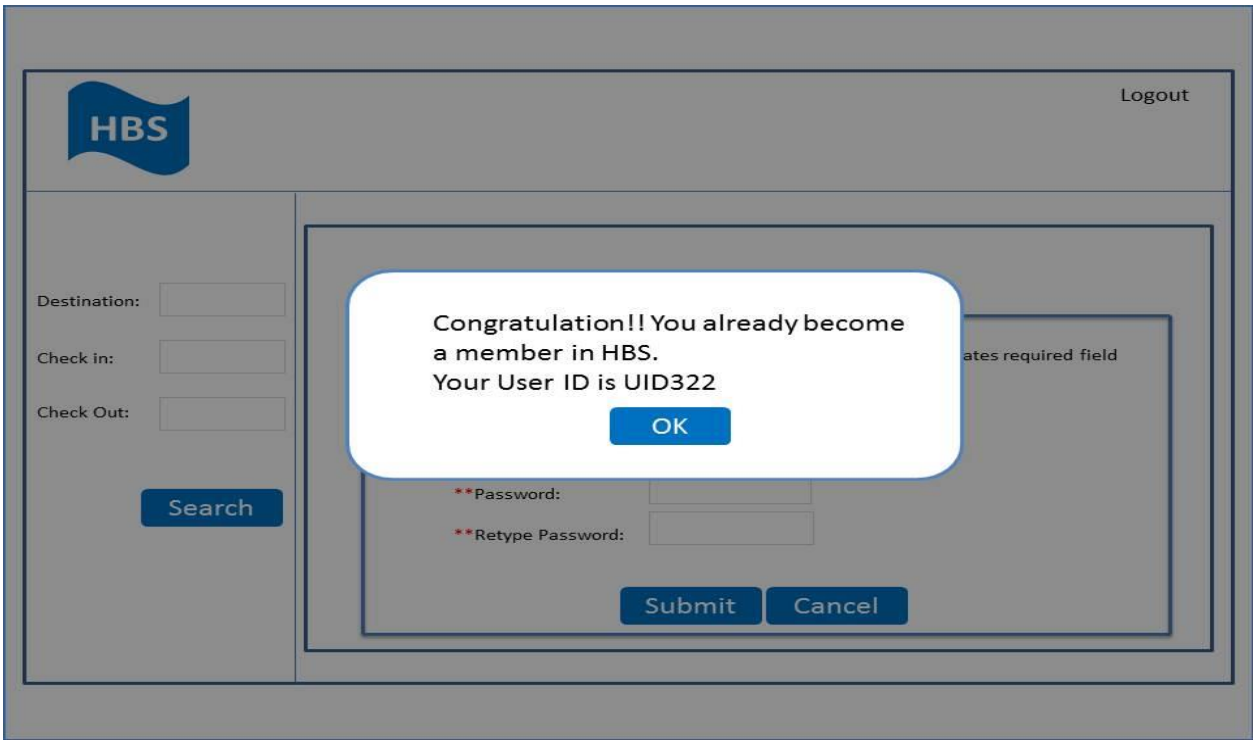


Figure 3.2

c. Graphic User Interface



Screen 1



Screen 2

3.3 Process 3.0 : Manage Profile

3.3.1 Process 3.1 : Change Password

a. Pseudocode

1. Start
2. Make access “Manage Profile”
3. Select “Change Password”
4. Input change password details
    4.1 Input old_password
    4.2 Input new_password
    4.3 Re-input new _password
5. Verify old_password with database Authorization Record
6. Verify new_password with re-input new_password
7. If (true)
    7.1 Update new password into database Authorization Record
    7.2 Display message change password successful
8. Else
    8.1 Display message change password fail
9. End

b. Flow chart

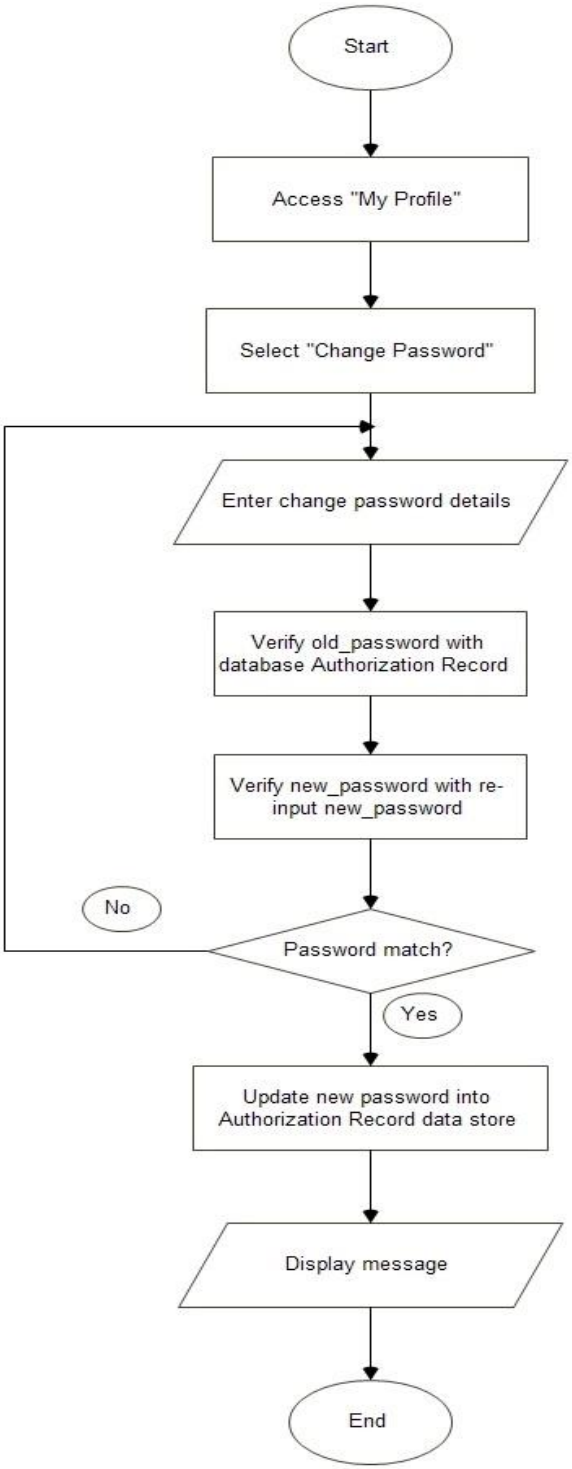
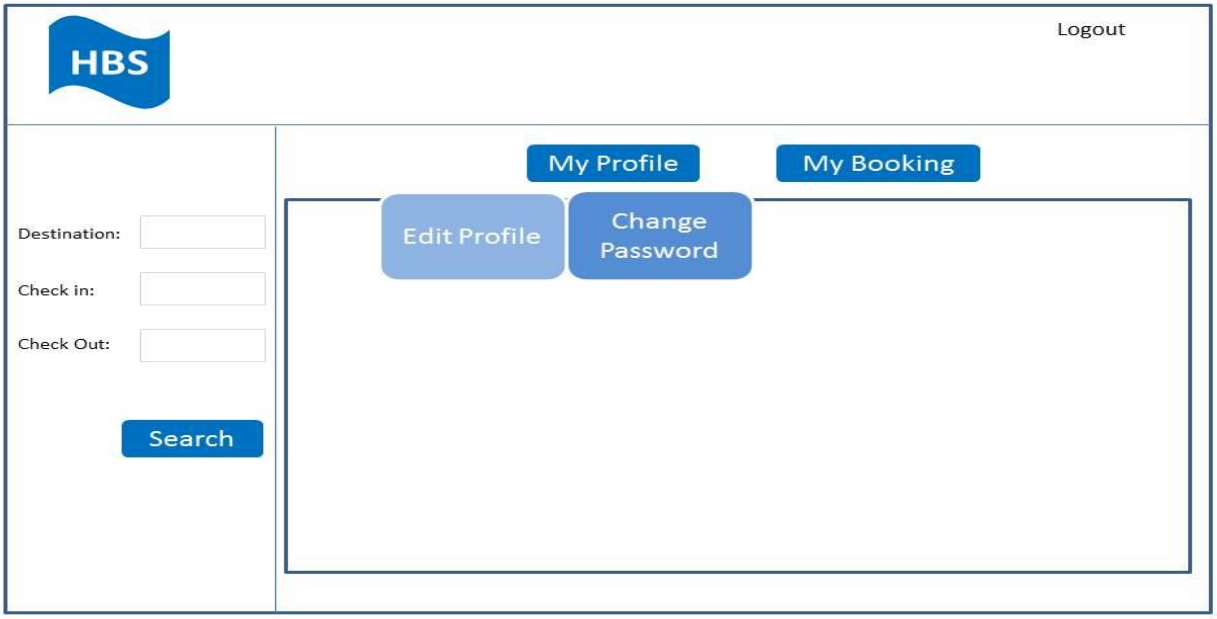
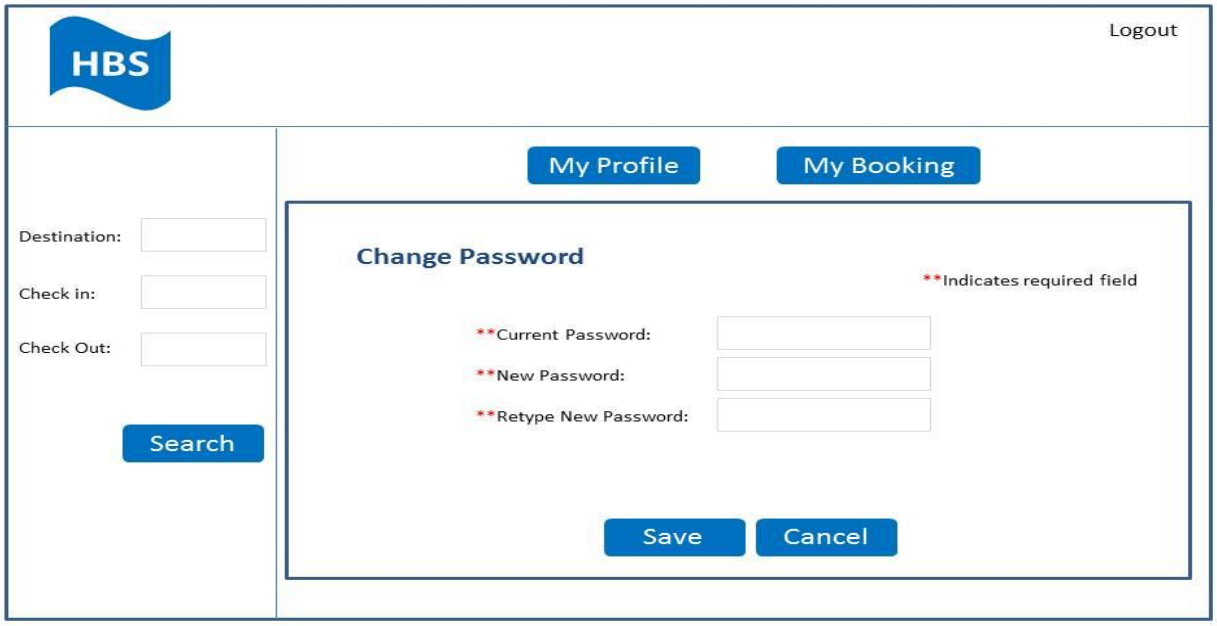


Figure 3.3.1

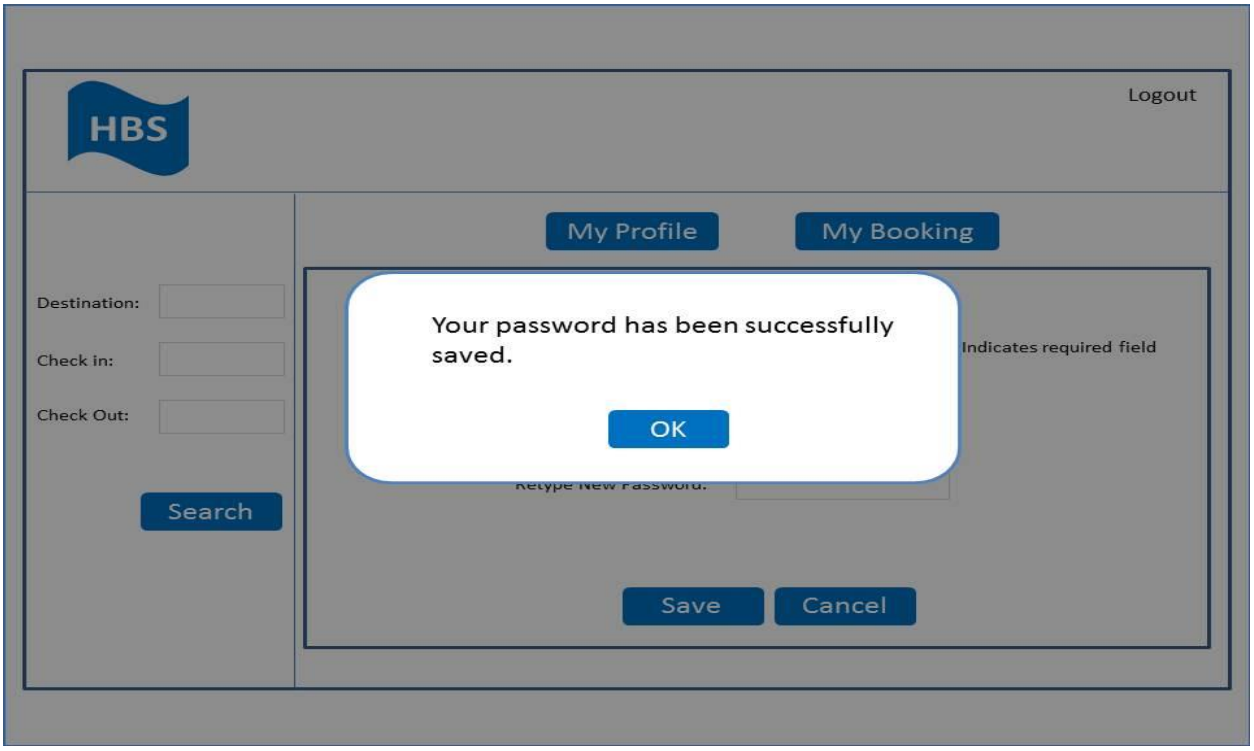
c. Graphic User Interface



Screen 1



Screen 2


HBS
Logout
My Profile
My Booking
Destination:
Check in:
Check Out:
Search
Your password has been successfully saved.
OK
Indicates required field
Save
Cancel

Screen 3

3.3.2 Process 3.2 : Edit Profile

a. Pseudo code

1. Start
2. Make access “My Profile”
3. Select “Edit Profile”
4. Load editable profile information form
5. User edit profile information
6. If (select Save
   6.1 Save the edited profile information in database User Record
   6.2 Display message edit profile is successful
7. Else
   7.1 Cancel edit profile
8. End

b. Flow chart

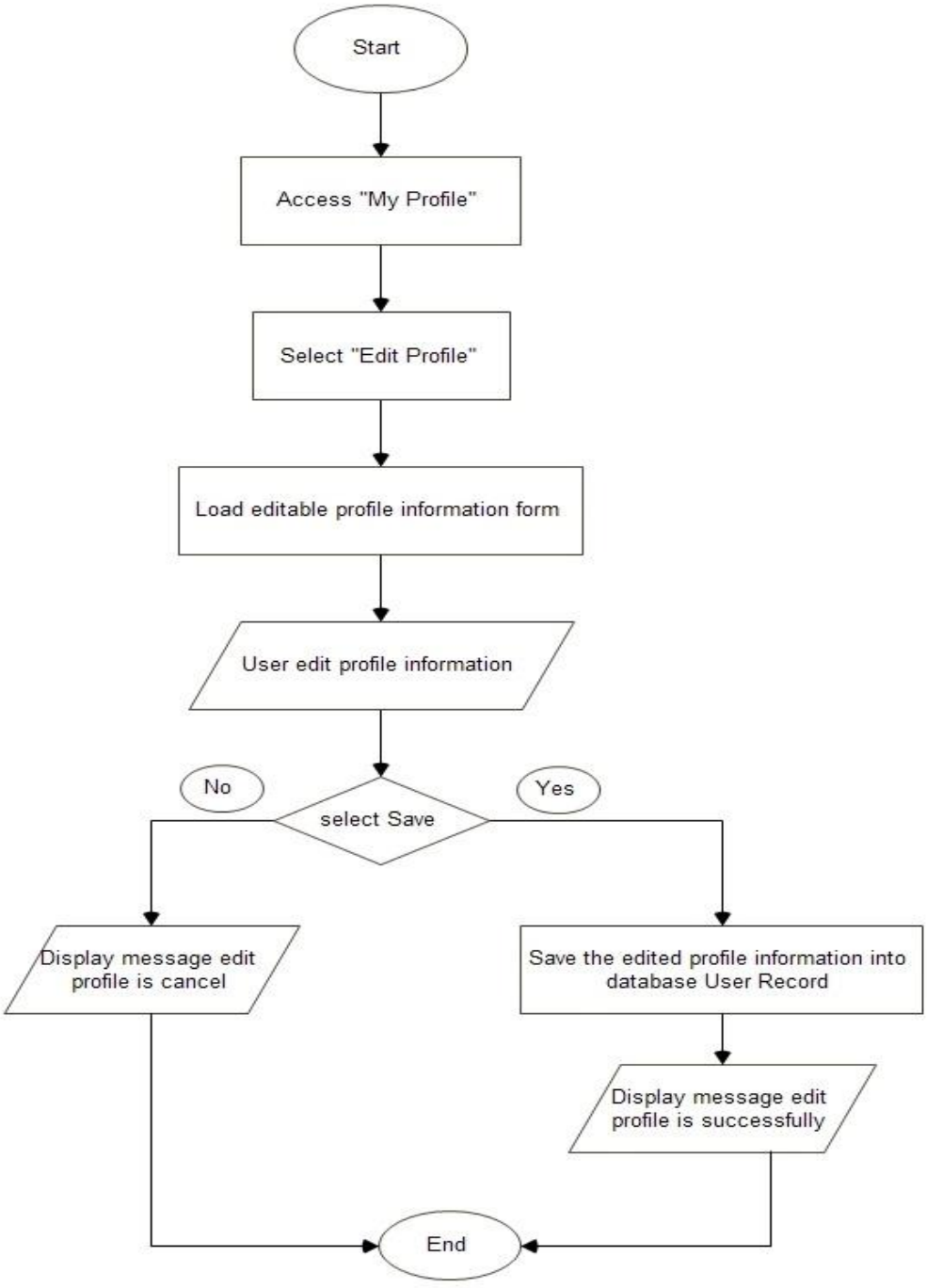
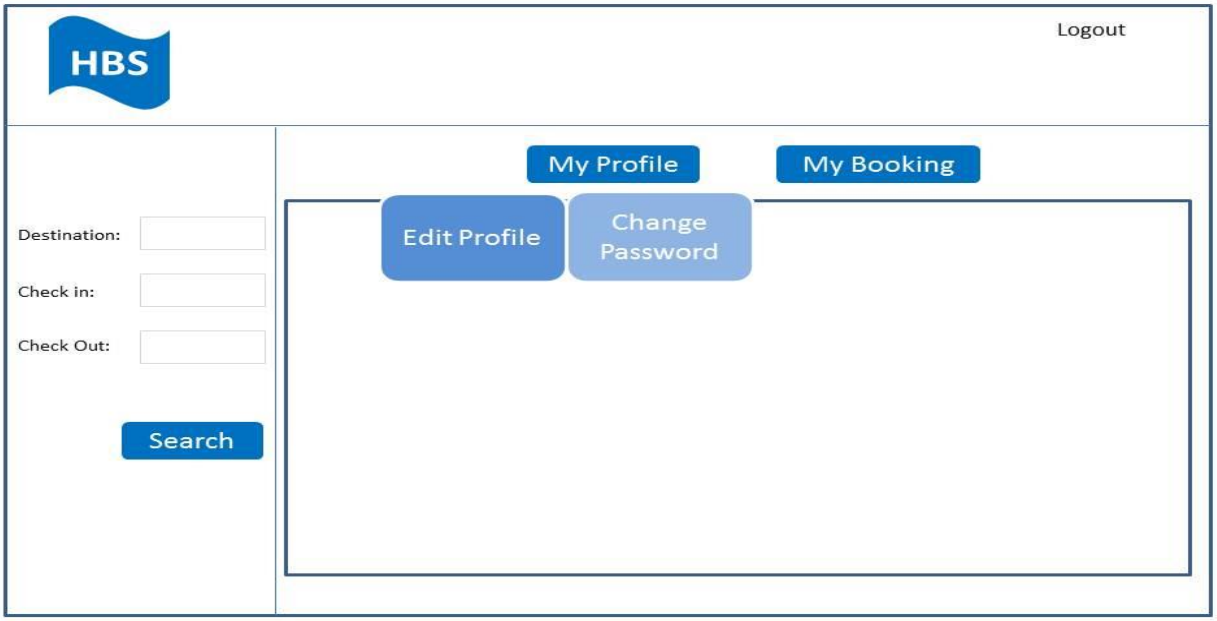
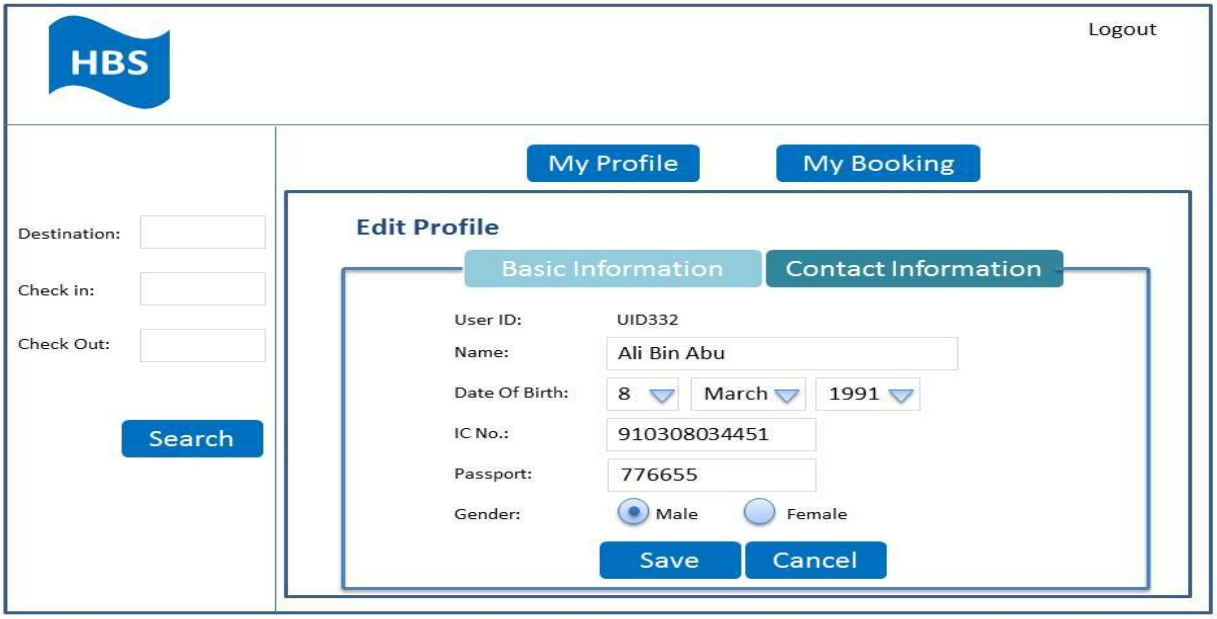


Figure 3.3.2

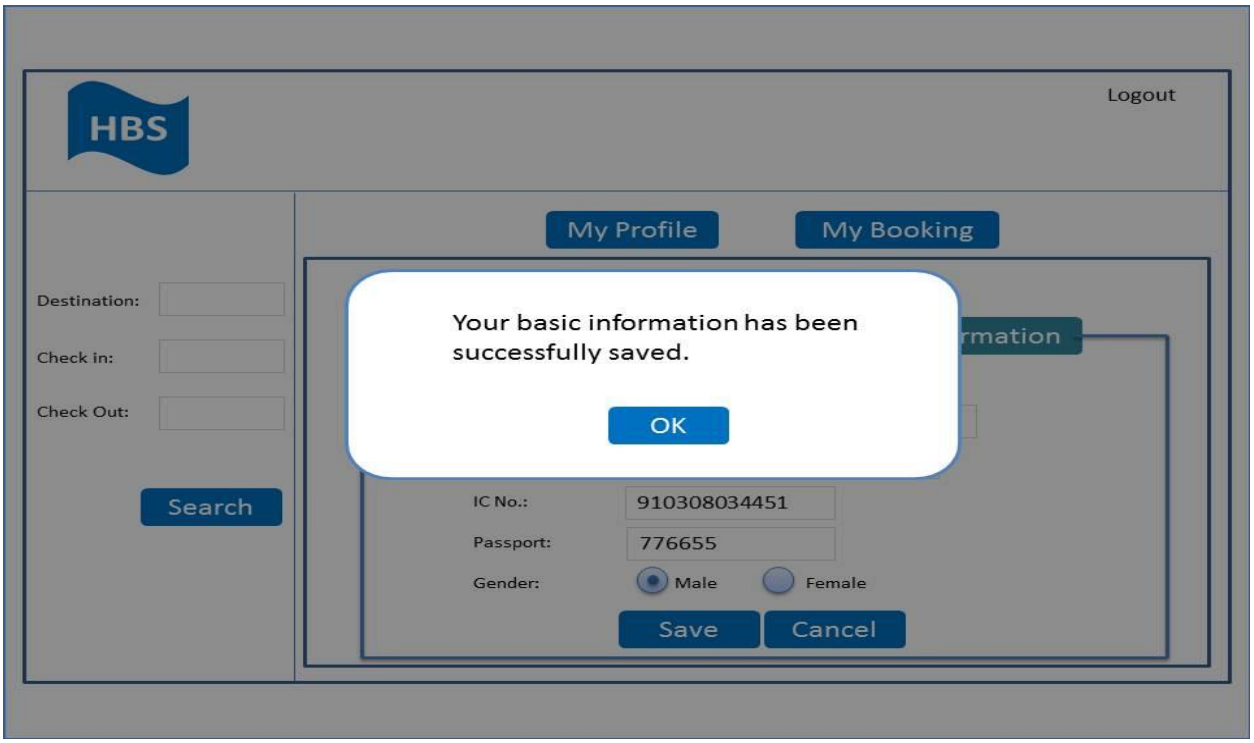
c. Graphic User Interface



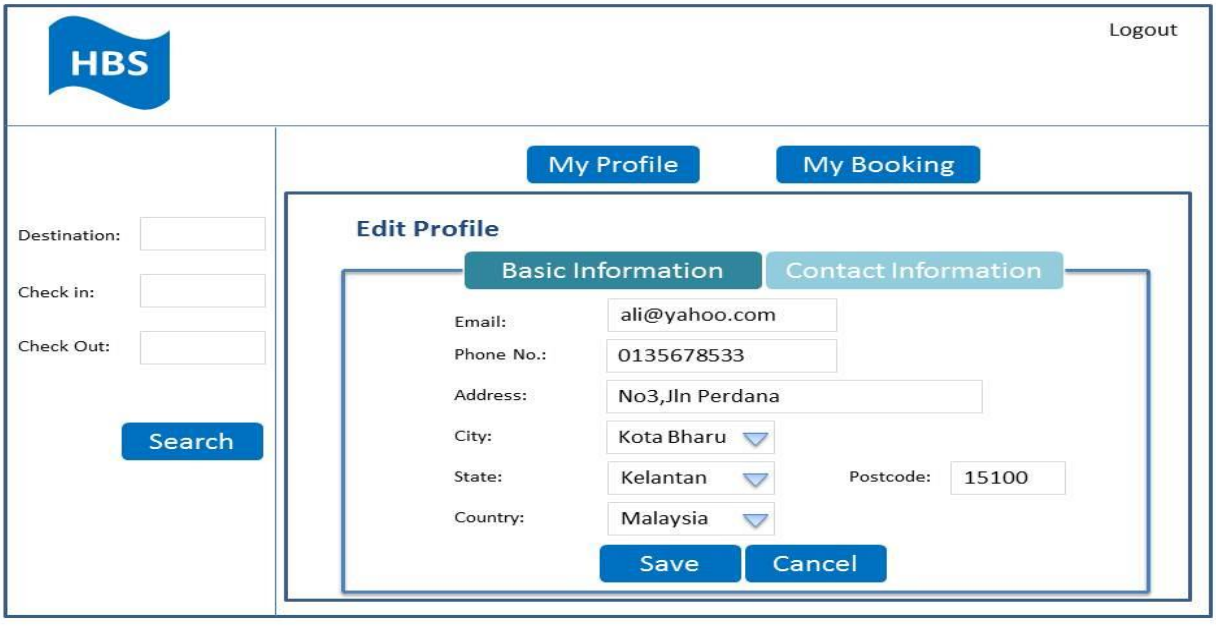
Screen 1



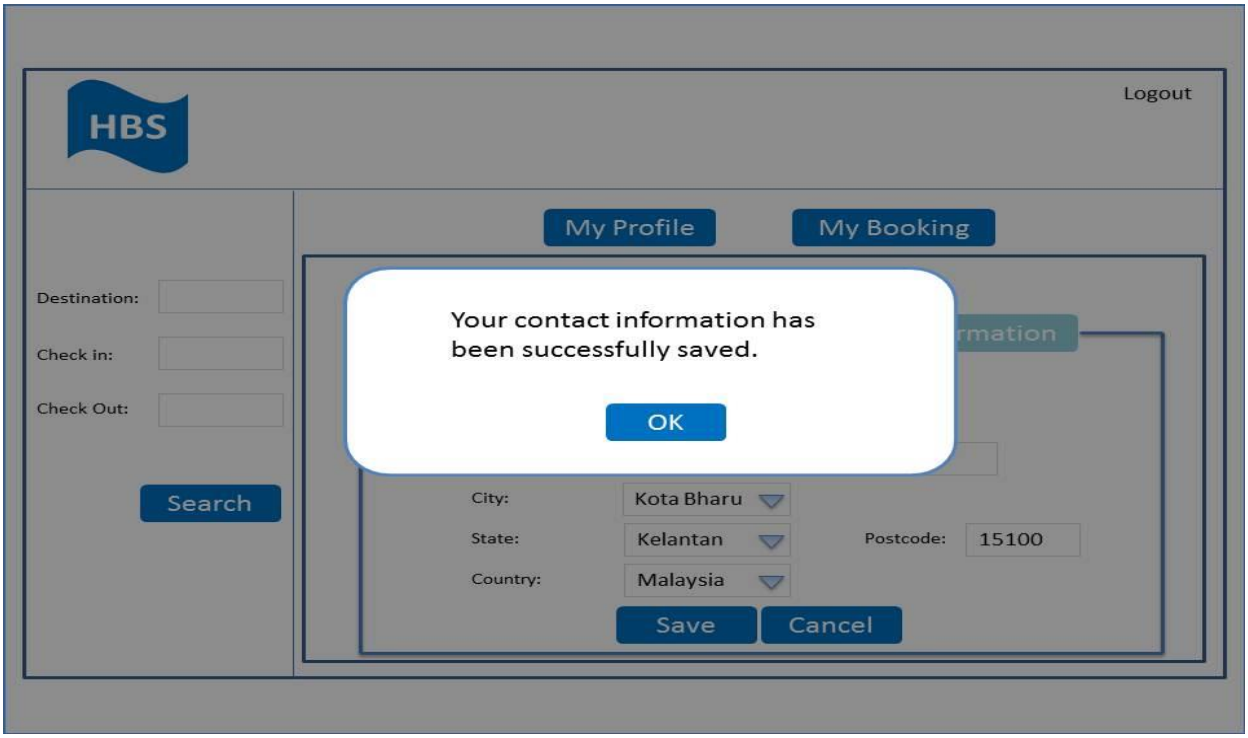
Screen 2



Screen 3



Screen 4


HBS
Logout
My Profile
My Booking
Destination:
Check in:
Check Out:
Search
Your contact information has been successfully saved.
OK
City:
Kota Bharu
State:
Kelantan
Postcode:
15100
Country:
Malaysia
Save
Cancel

Screen 5

3.4 Process 4.0: Manage booking

3.4.1 Process 4.1: check booking record

a. Pseudo code

1. Start
2. Make access “check booking record” page
3. Select intended booking record category
4. Select intended booking record
5. Details of selected booking record is retrieved from database “Booking record”
6. Display the details of booking record
7. End

b. Flow chart

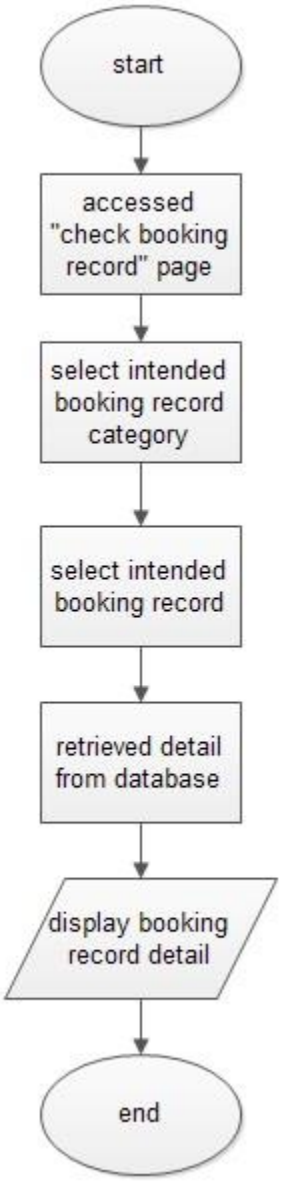
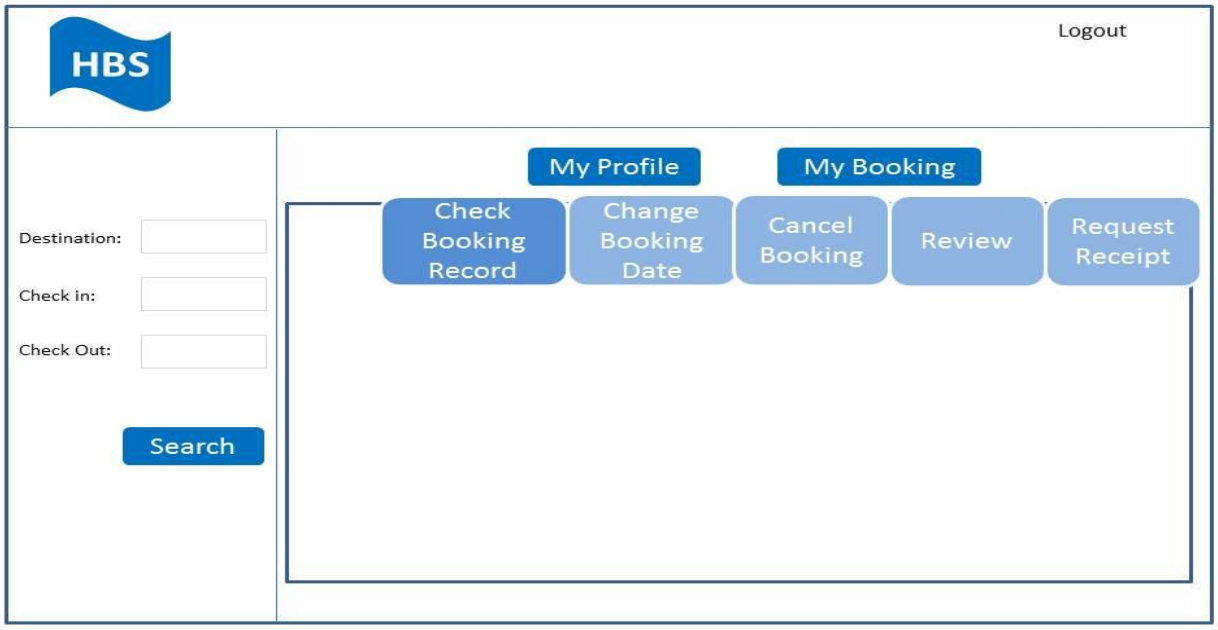


Figure 3.4.1

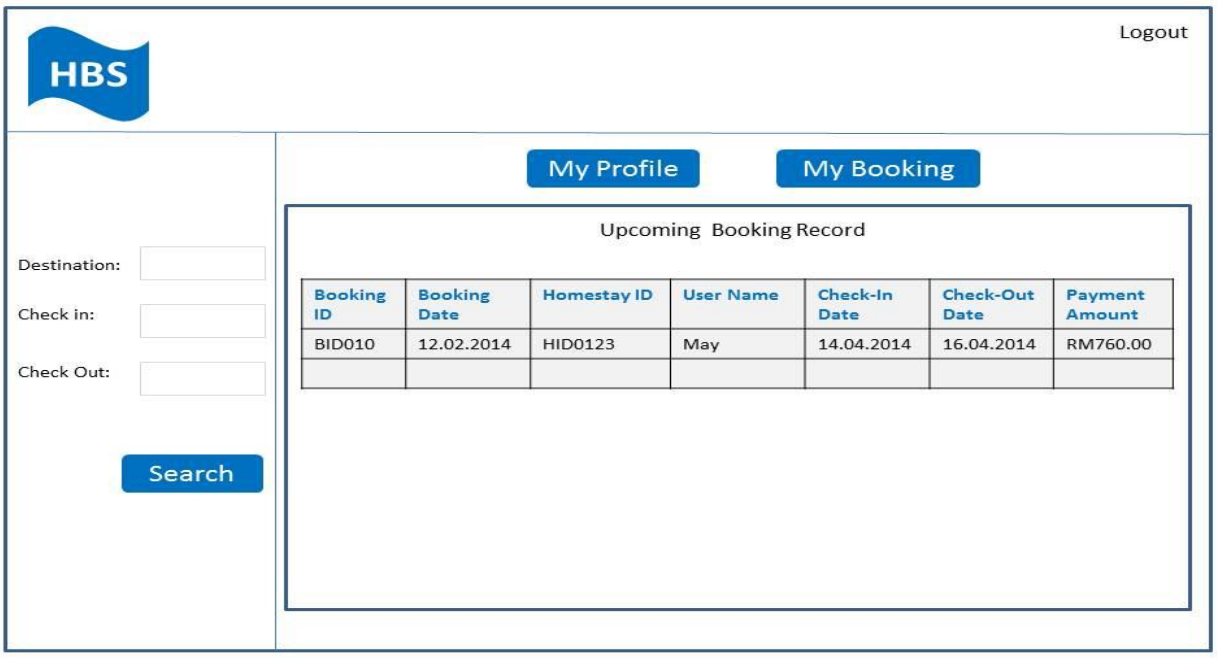
c. Graphic User Interface



Screen 1



Screen 2



Screen 3



Screen 4

3.4.2 Process 4.2: change booking date

a. Pseudo code

1. Start
2. Make access “change booking date” page
3. Select intended booking record to change date
4. Input new date
5. Verify the homestay availability with the database “Booking record”
6. If the new date is able to select
    6.1 The homestay is available for the date
    6.2 The date is successfully changed
    6.3 The changes are saved into database “Booking record”
7. Else
    7.1 The homestay is not available for the date
    7.2 Choose a new date
8. End

b. Flow chart

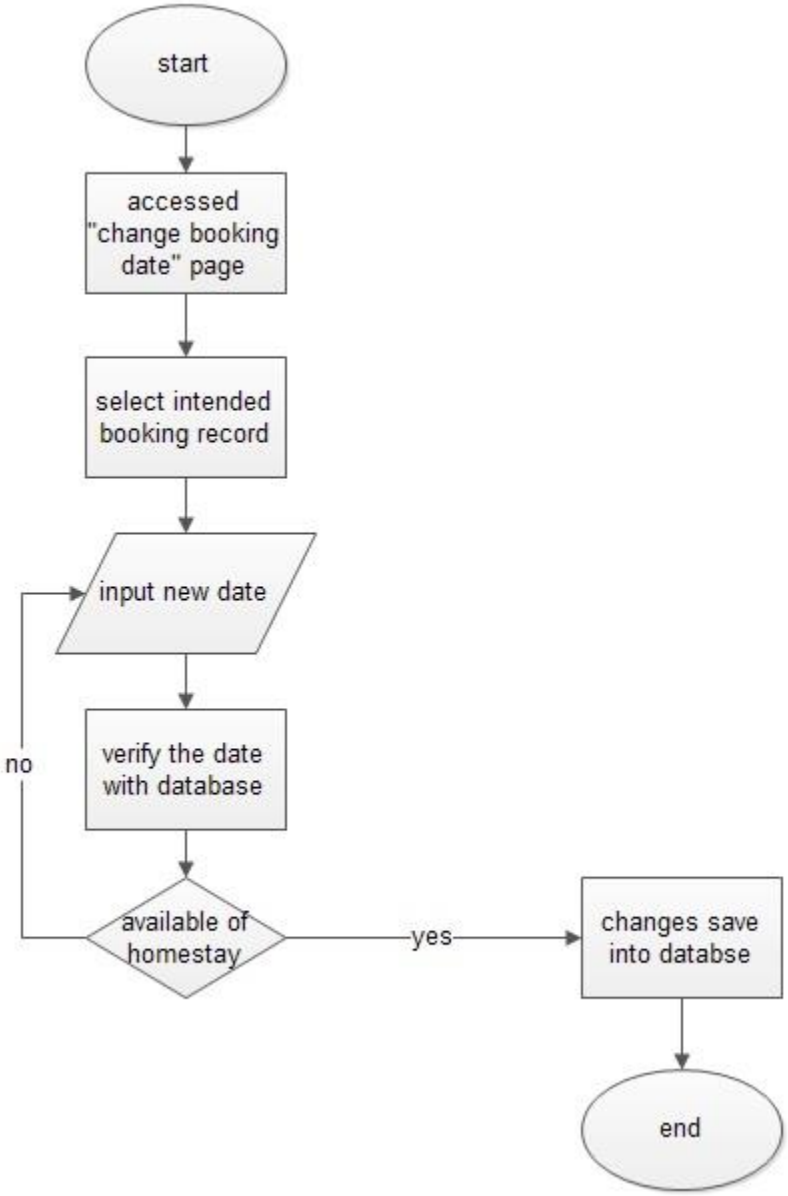
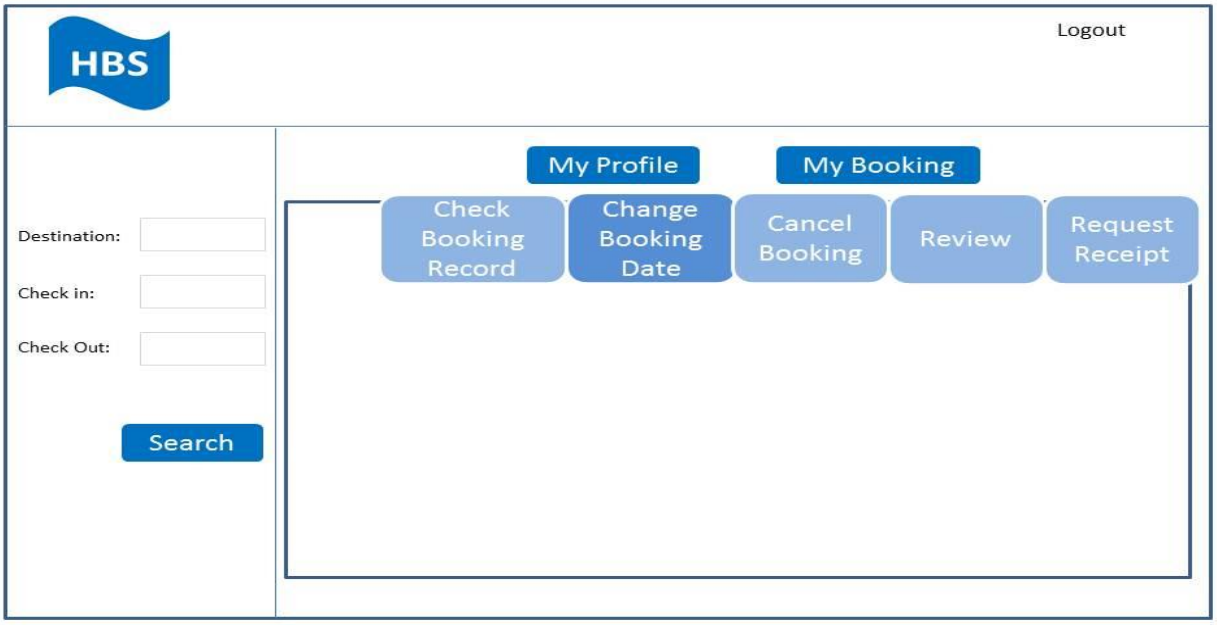
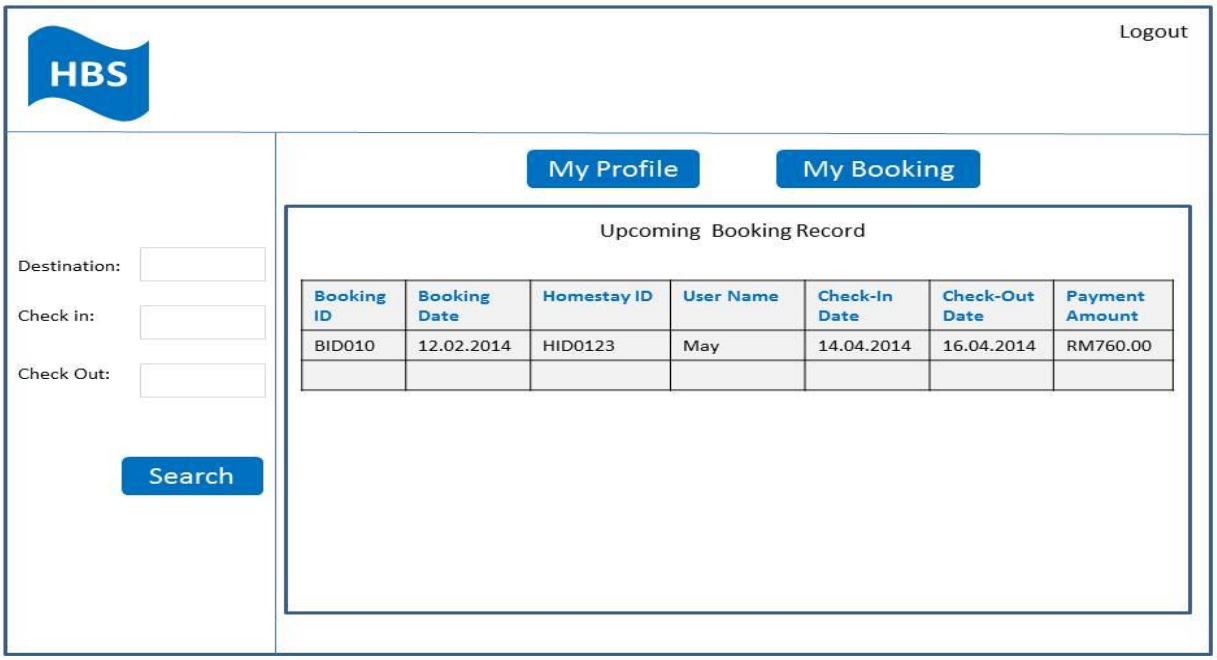


Figure 3.4.2

c. Graphic User Interface



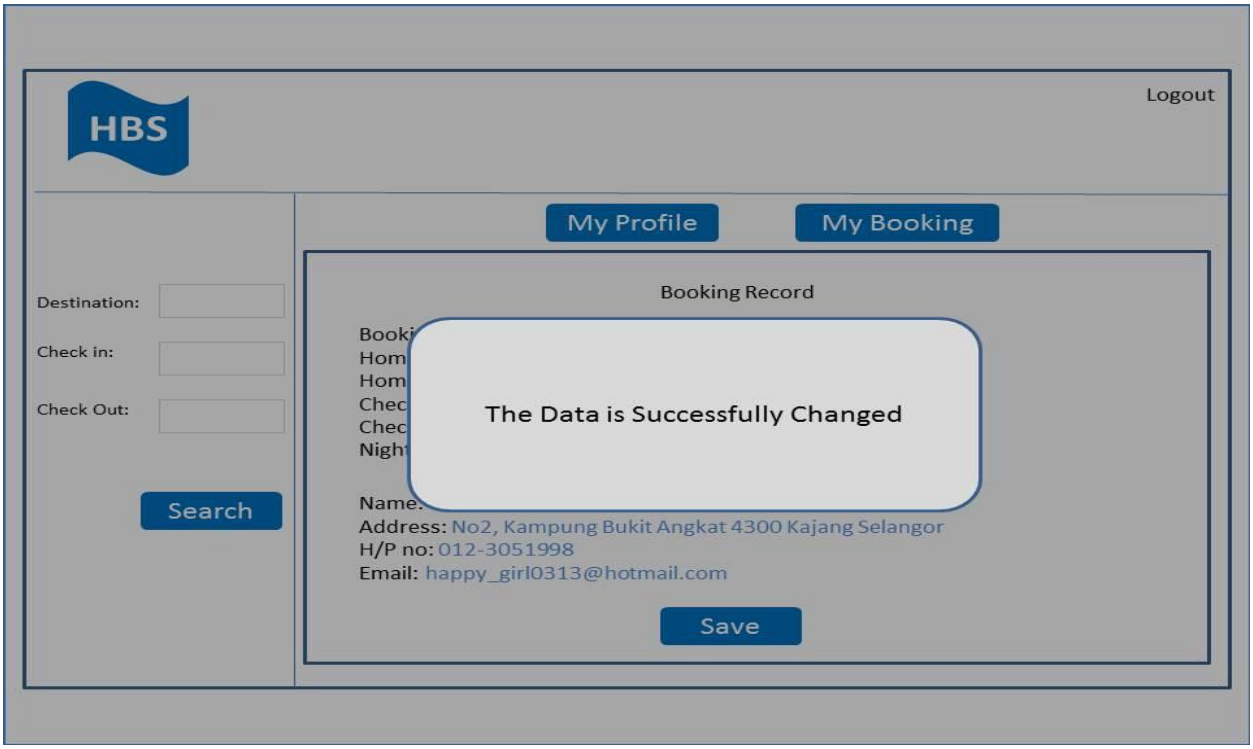
Screen 1



Screen 2



Screen 3



Screen 4

3.4.3 Process 4.3: review

a. Pseudo code

1. Start
2. Make access“review” page
3. Select intended booking record to write comment
4. Input comment
5. Select submit button
6. The comment is saved into database “Homestay record”
7. Display the updated status
8. End

b. Flow chart

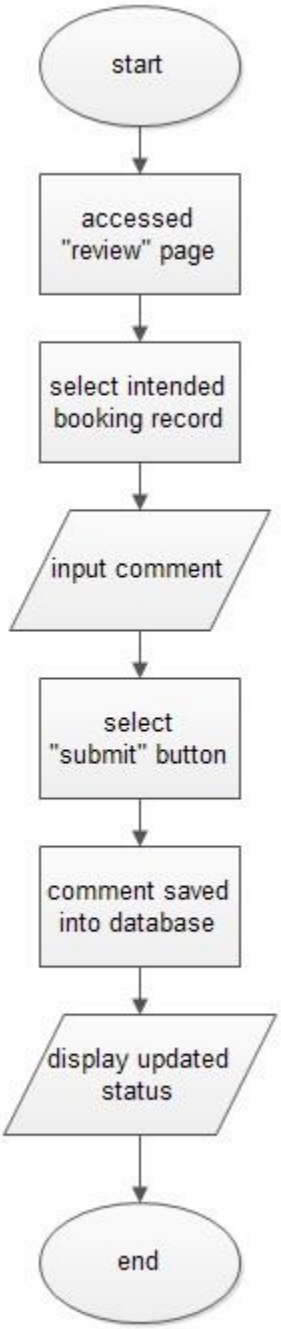
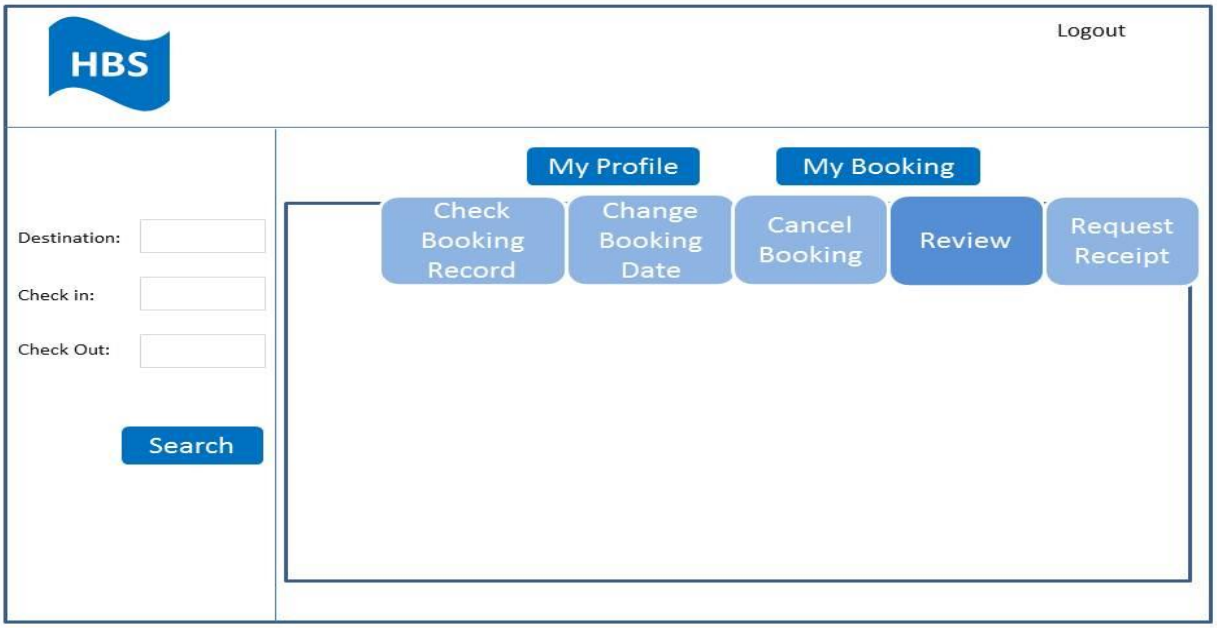


Figure 3.4.3

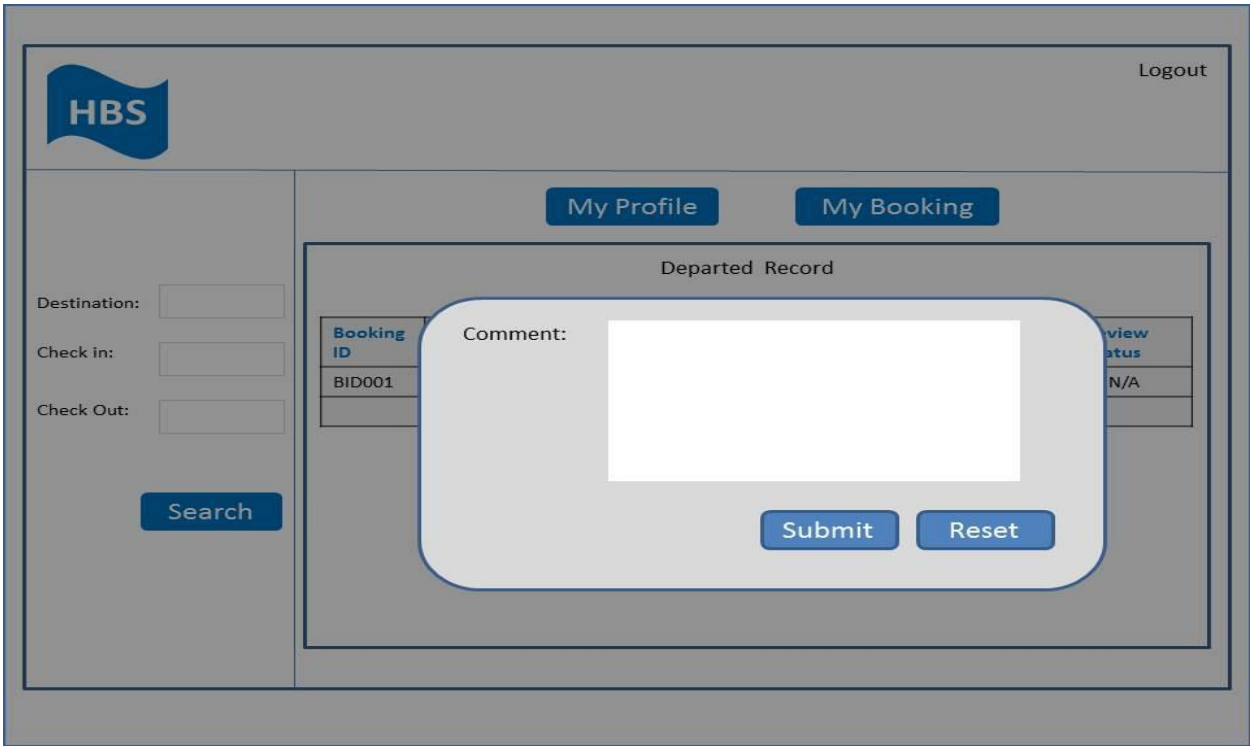
c. Graphic User Interface



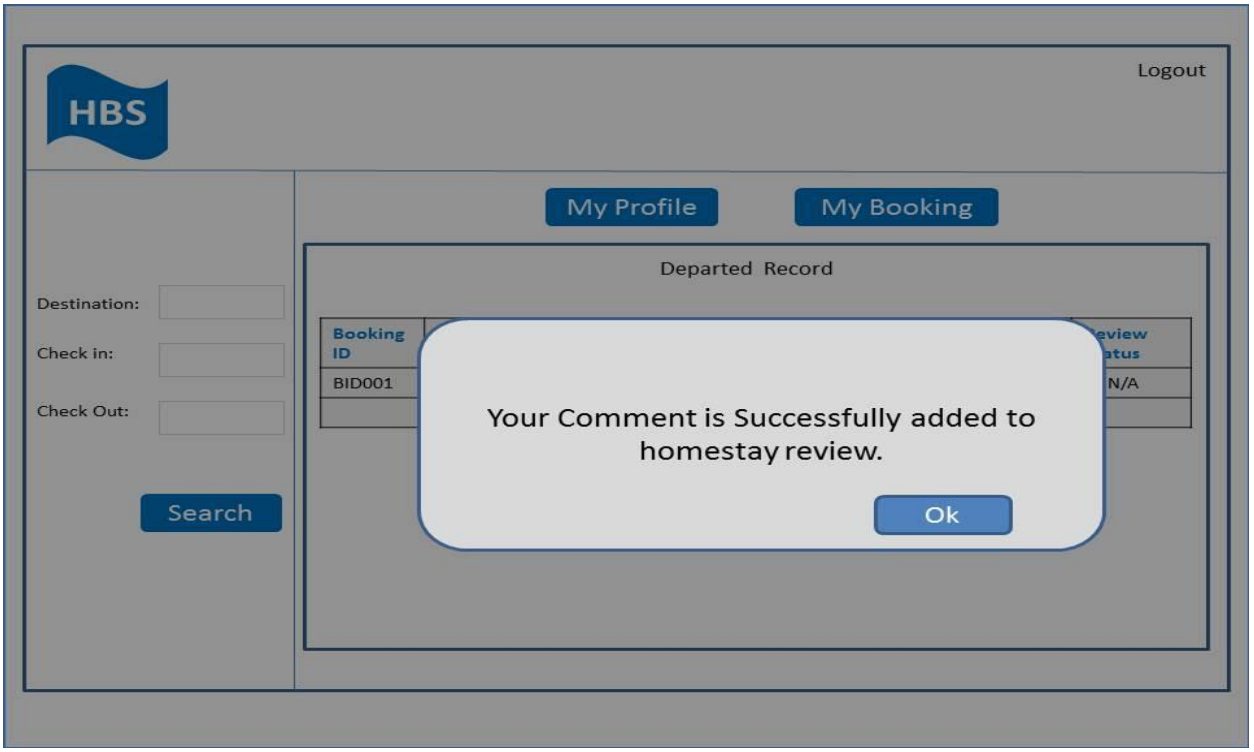
Screen 1



Screen 2



Screen 3



Screen 4

3.4.4 Process 4.4: cancel booking

a. Pseudo code

1. Start
2. Make access “cancel booking” page
3. Select intended booking record to cancel
4. Select the “cancel” button
5. If choose “Yes” button
    5.1 Booking is being cancelled
    5.2 Changes are saved into database “Booking record”
6. Else
    6.1 The booking is remained.
8. End

b. Flow chart

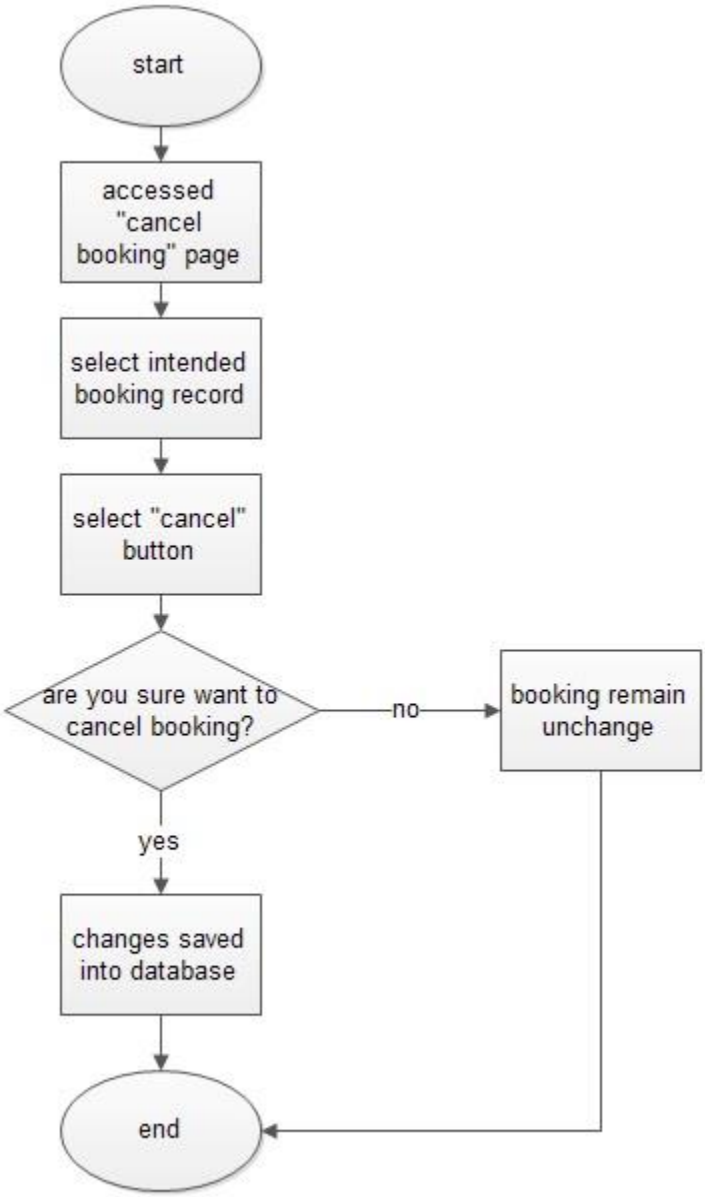
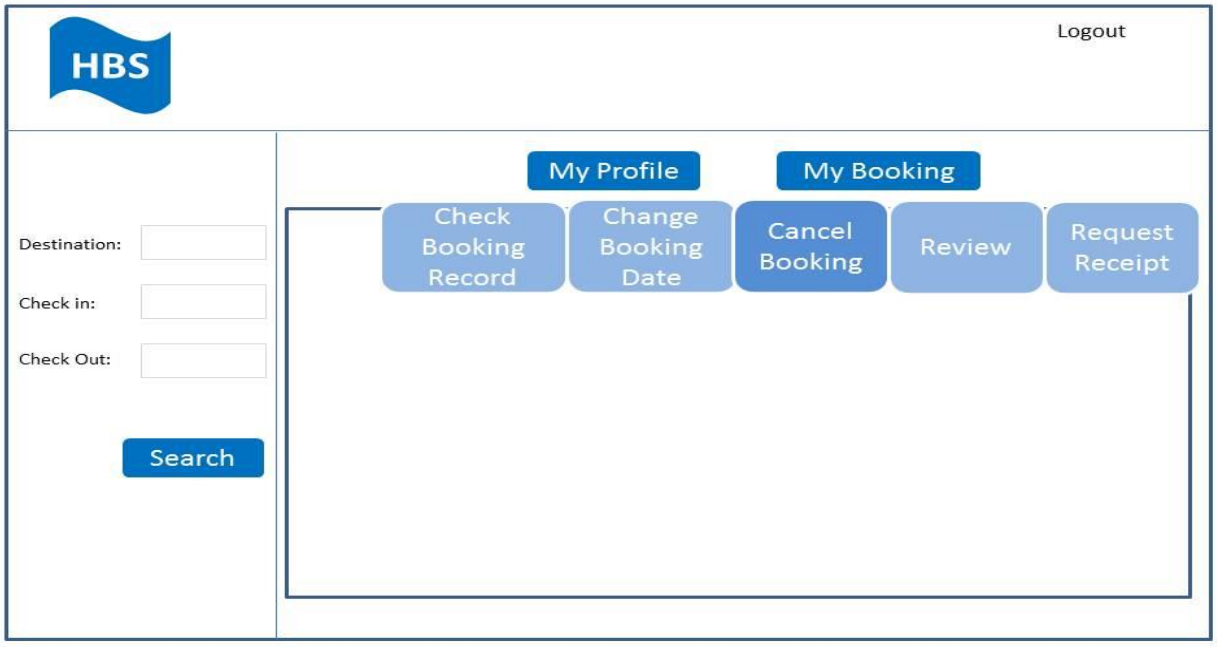
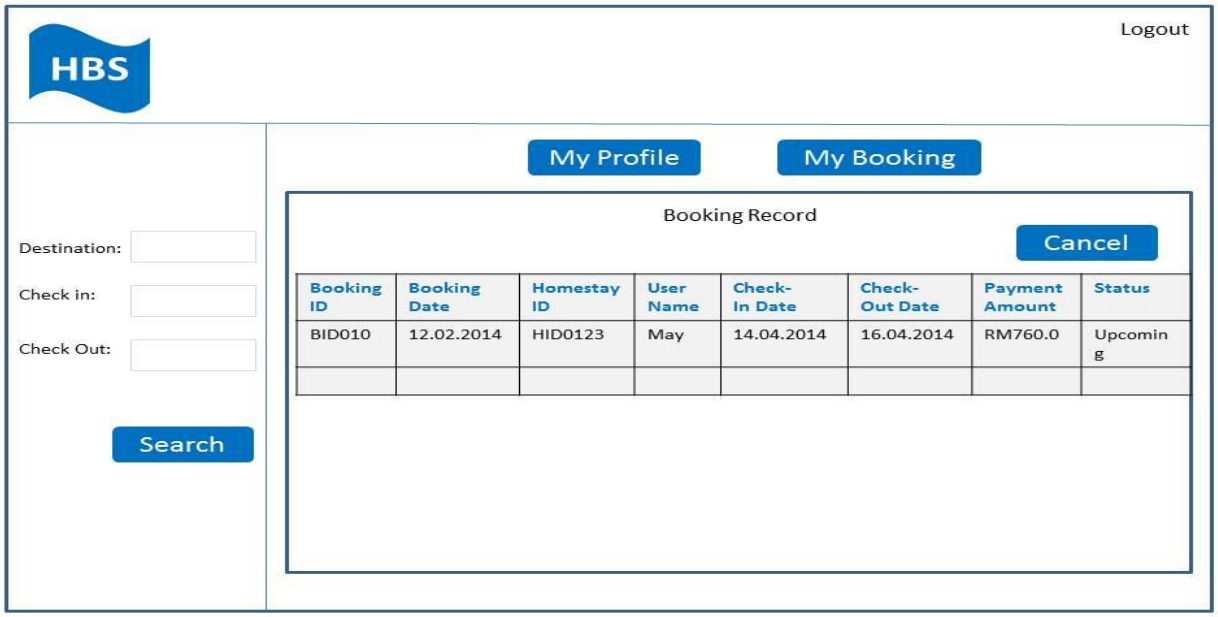


Figure 3.4.4

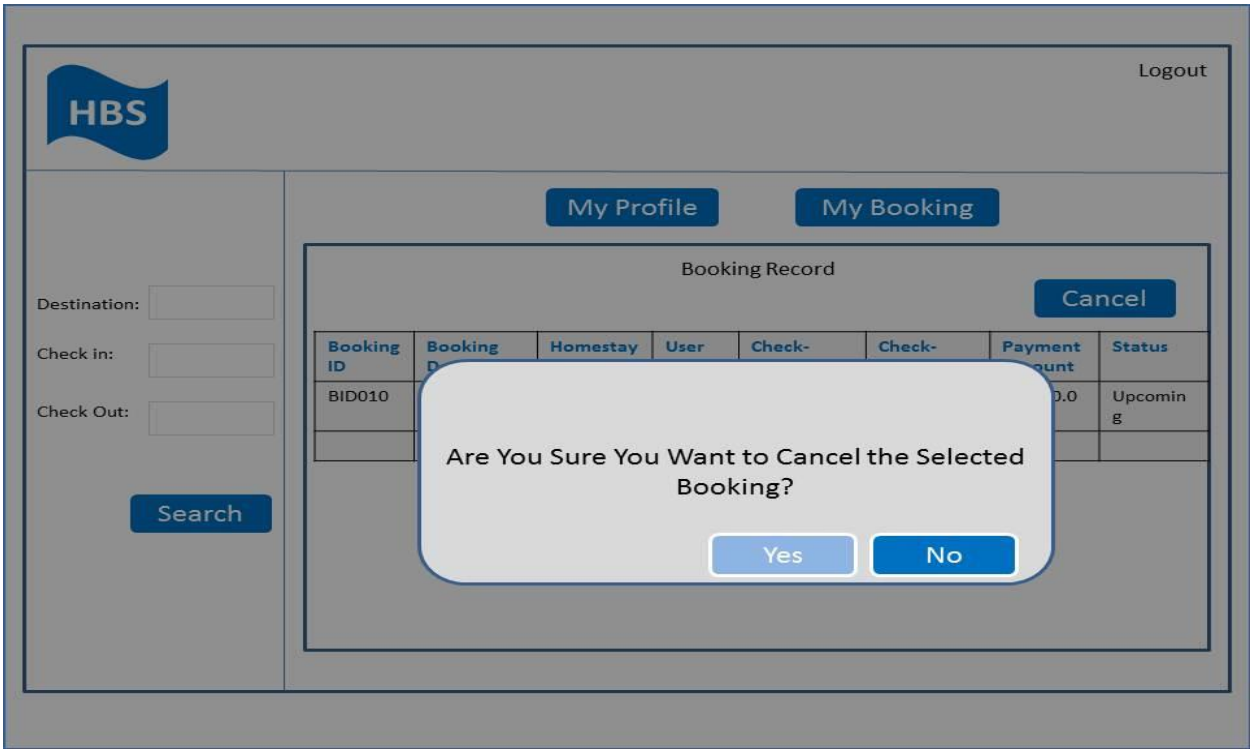
c. Graphic User Interface



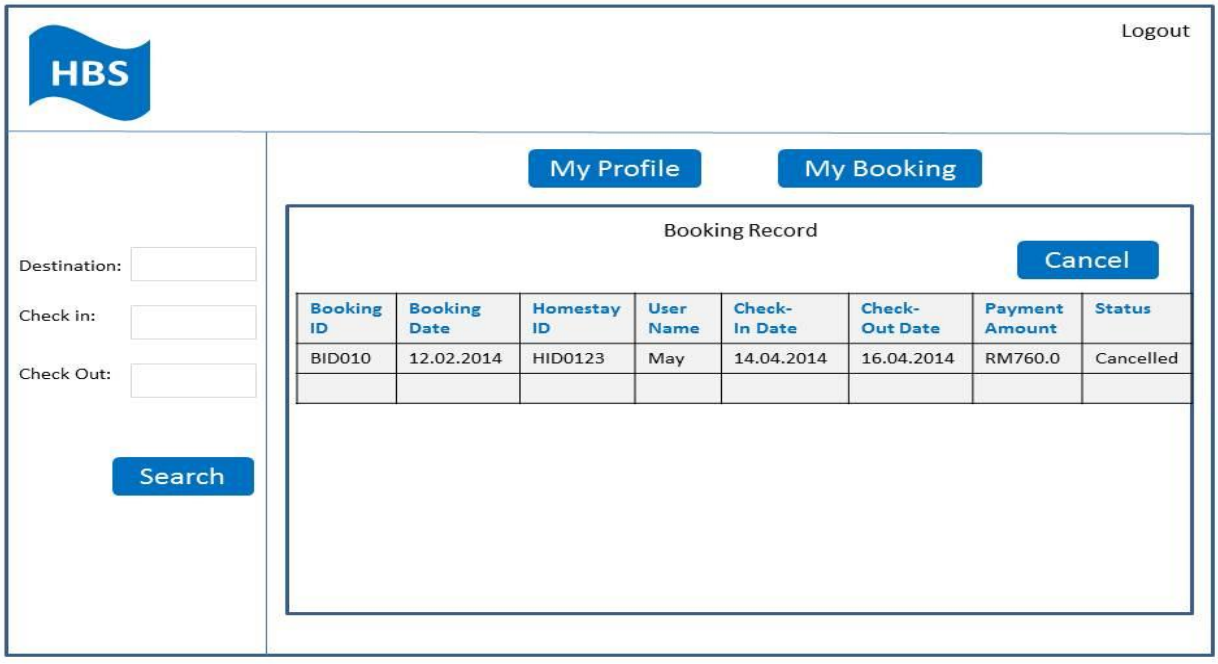
Screen 1



Screen 2



Screen 3



Screen 4

3.5 Process 5.0: request receipt

a. Pseudo code

1. Start
2. Make access “request receipt” page
3. Input booking_id
4. Receipt is retrieved from database “payment”
5. Display the receipt
6. Select “print” button
7. End

b. Flow chart

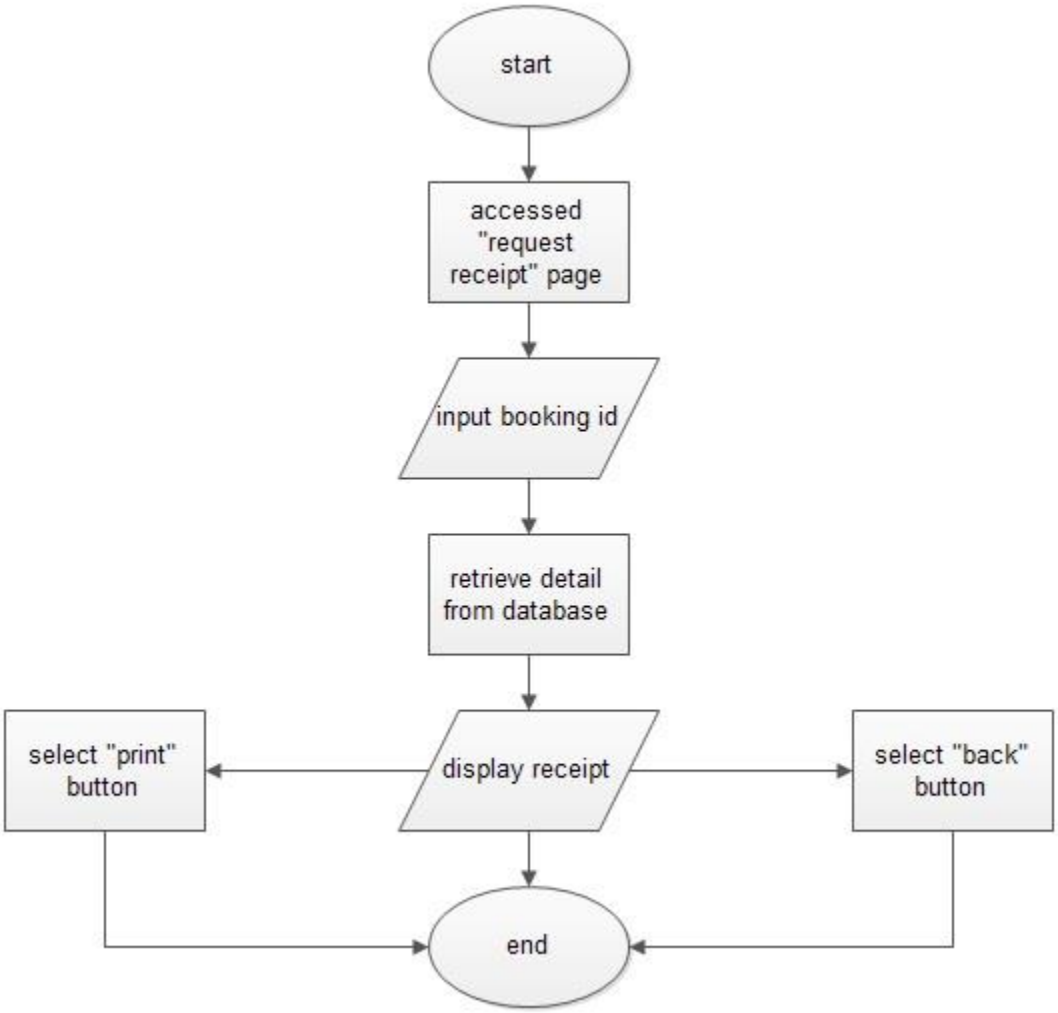
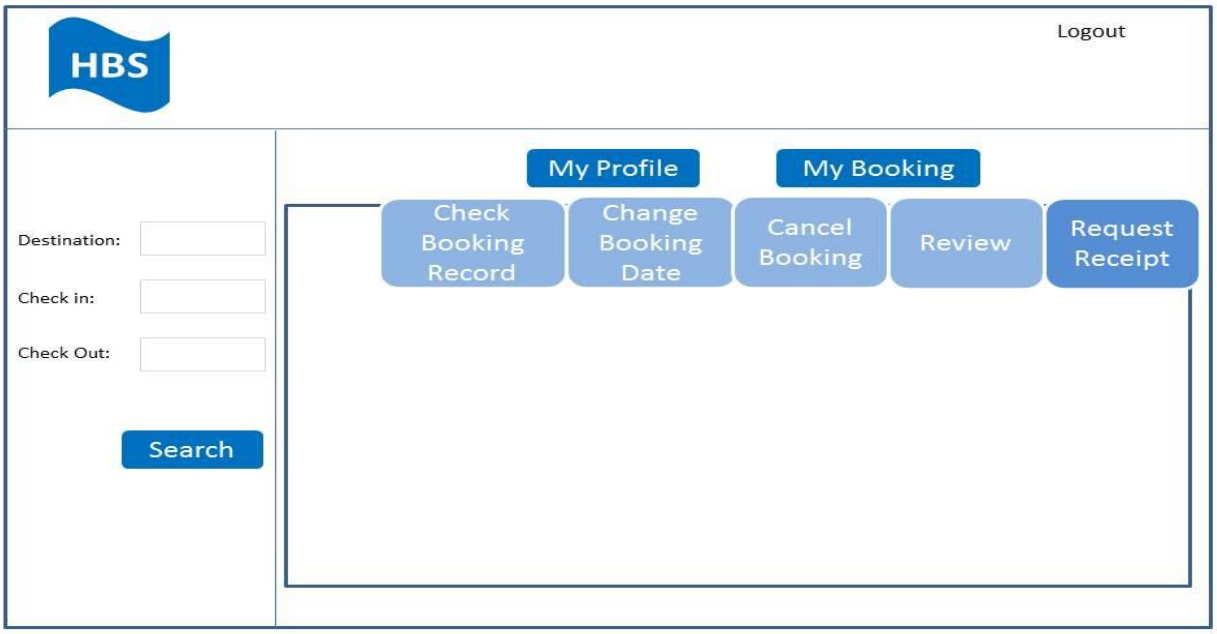
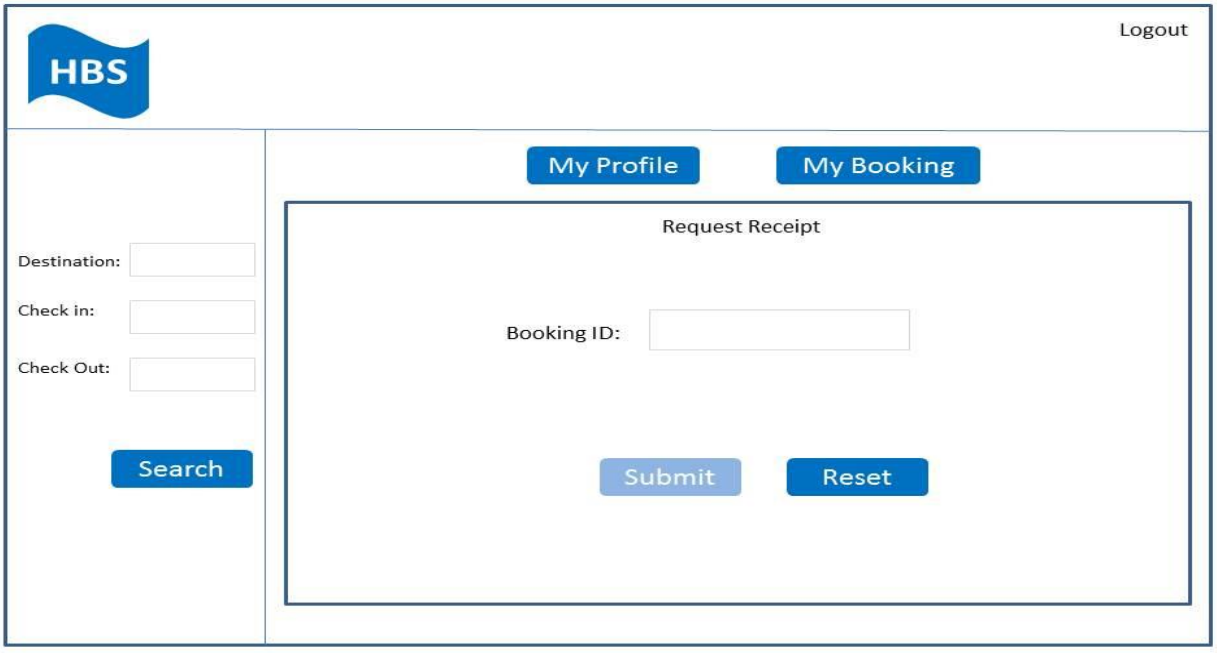


Figure 3.5

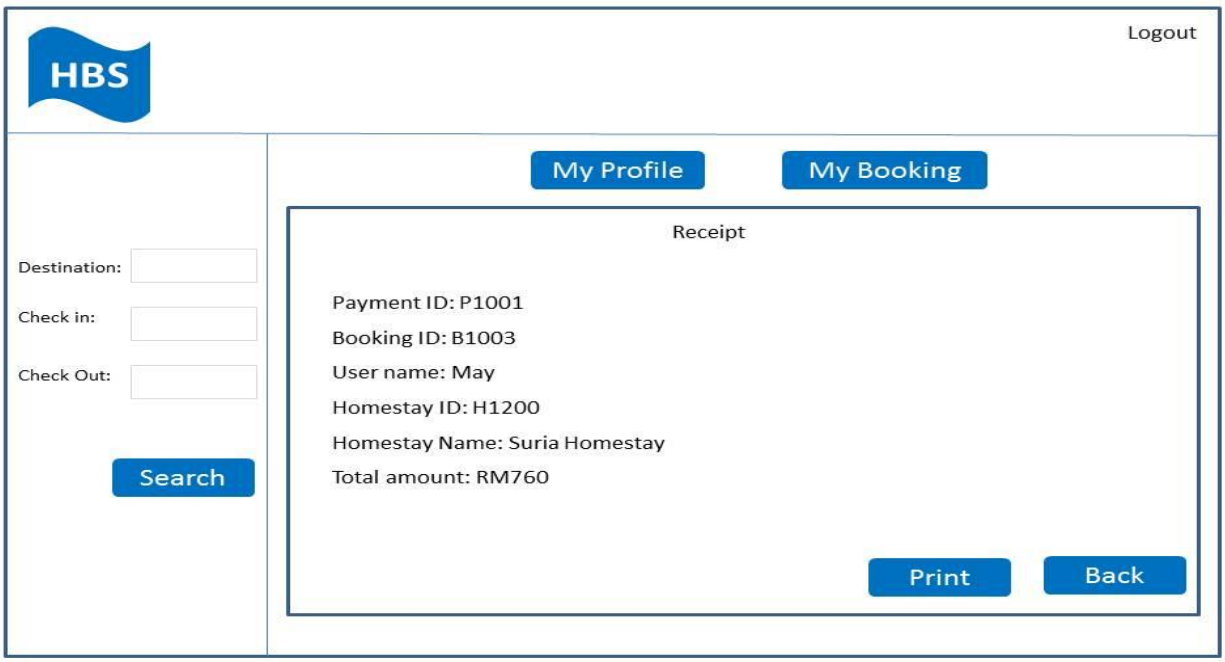
c. Graphic User Interface



Screen 1



Screen 2


HBS
Logout
My Profile
My Booking
Destination:
Check in:
Check Out:
Search
Receipt
Payment ID: P1001
Booking ID: B1003
User name: May
Homestay ID: H1200
Homestay Name: Suria Homestay
Total amount: RM760
Print
Back

Screen 3

PSEUDOCODE ,FLOW CHART AND INTERFACE FOR ADMIN

3.6 Process 1.2.1: Make payment

a. Pseudo code

1. Start
2. Make access “make payment” page
3. Input account number and payment detail
4. Select “submit” button
5. Details saved into database
6. End

b. Flow chart

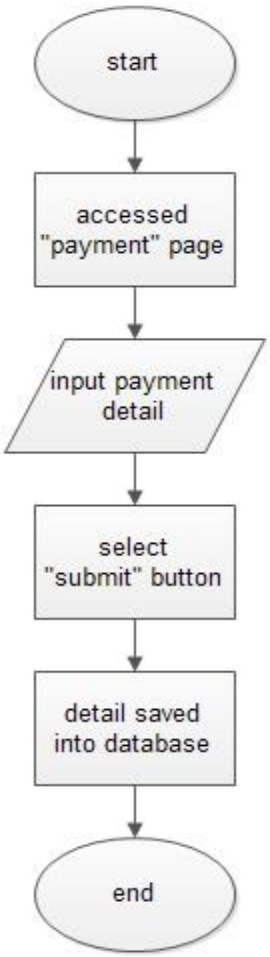
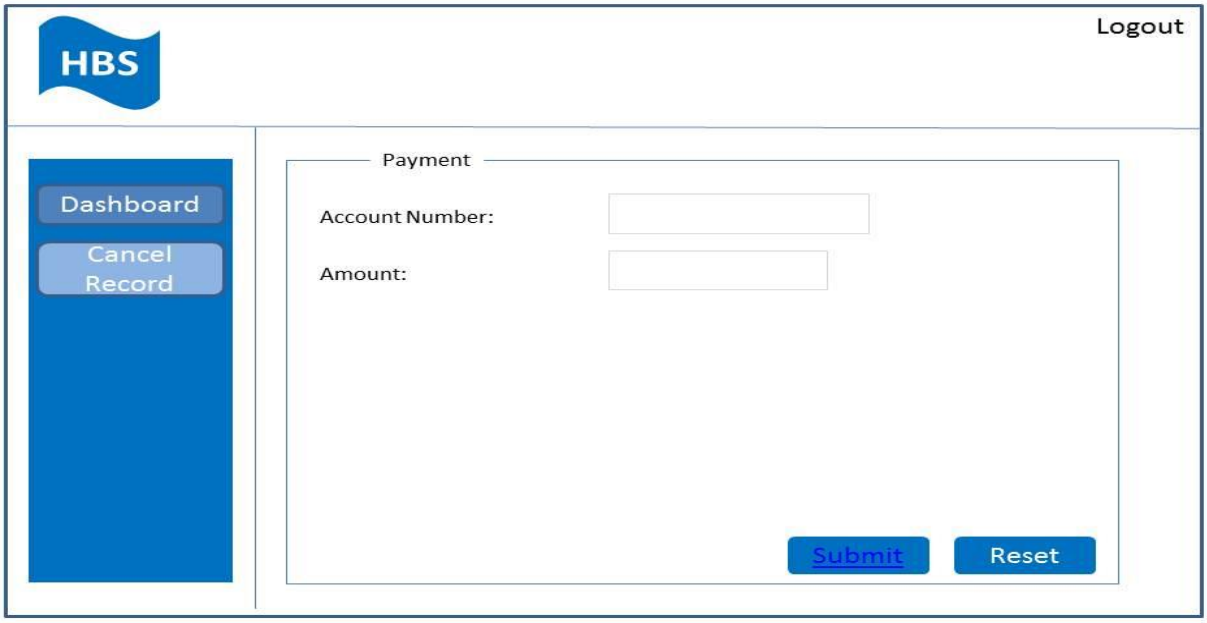
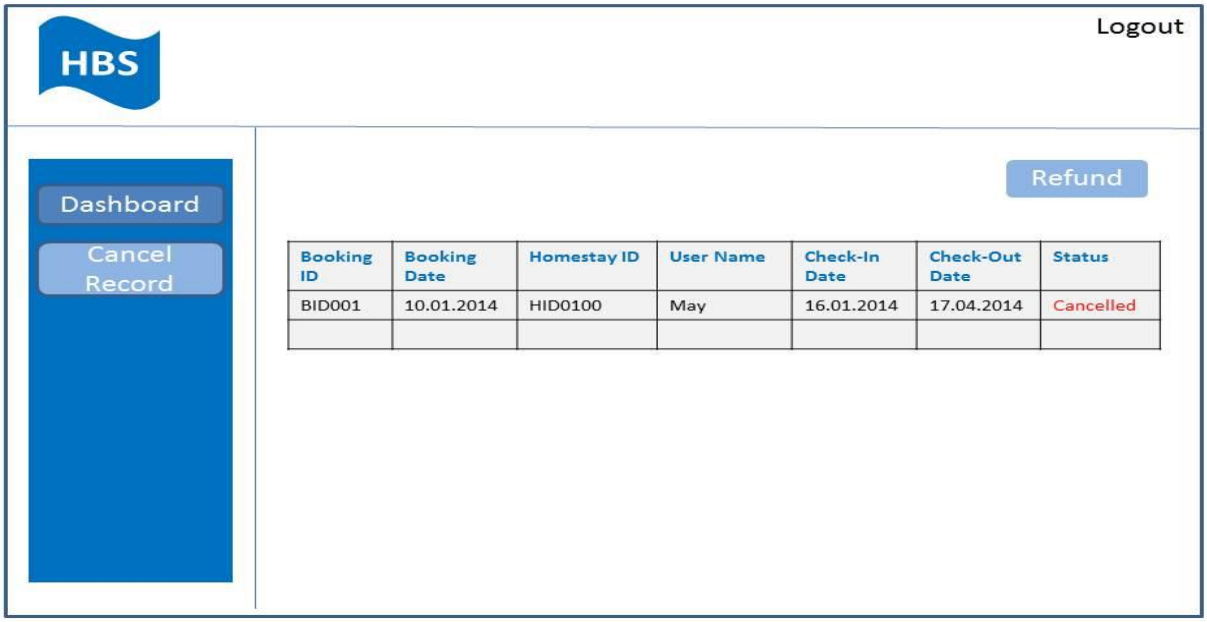


Figure 3.6

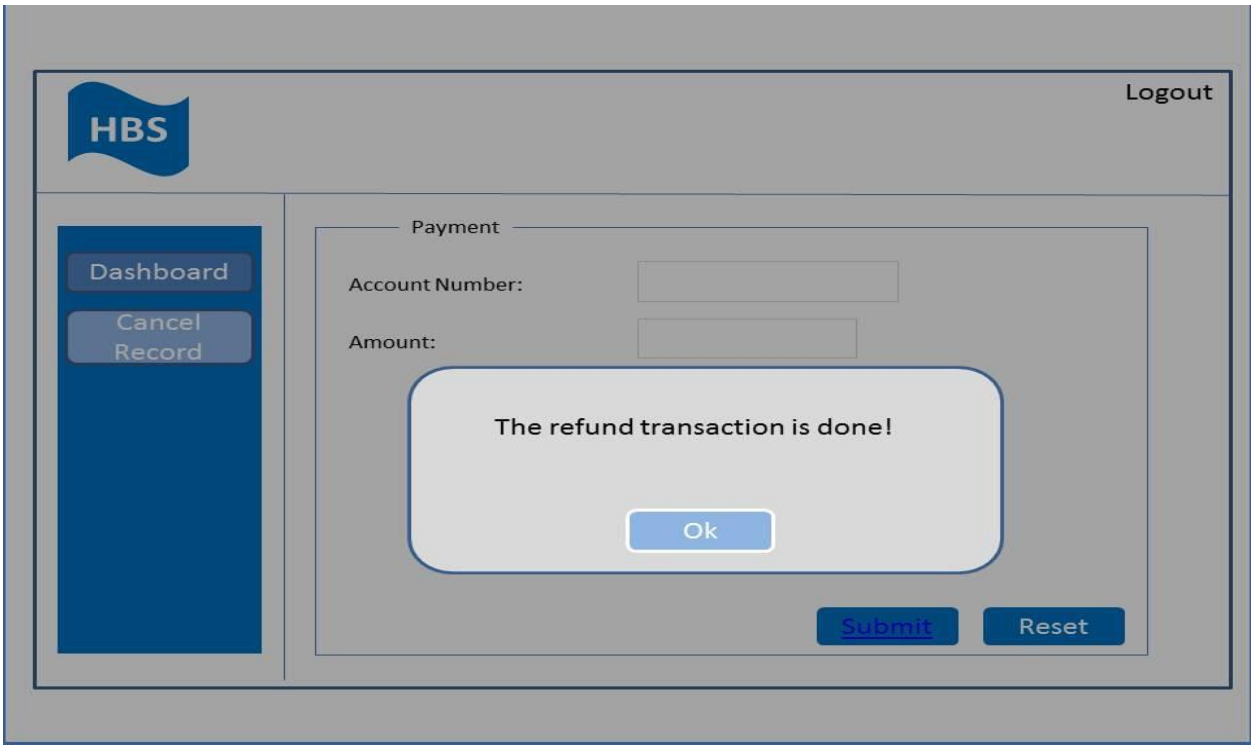
c. Graphic User Interface



Screen 1



Screen 2


HBS
Logout
Dashboard
Cancel Record
Payment
Account Number:
Amount:
The refund transaction is done!
Ok
Submit
Reset

Screen 3

3.7 Process 4.1: Check booking record

a. Pseudo code

1. Start
2. Make access “check booking record” page
3. Select intended booking record category
4. Select/search intended booking record
5. Details of selected booking record is retrieved from database “Booking record”
6. Display the details of booking record
7. End

b. Flow chart

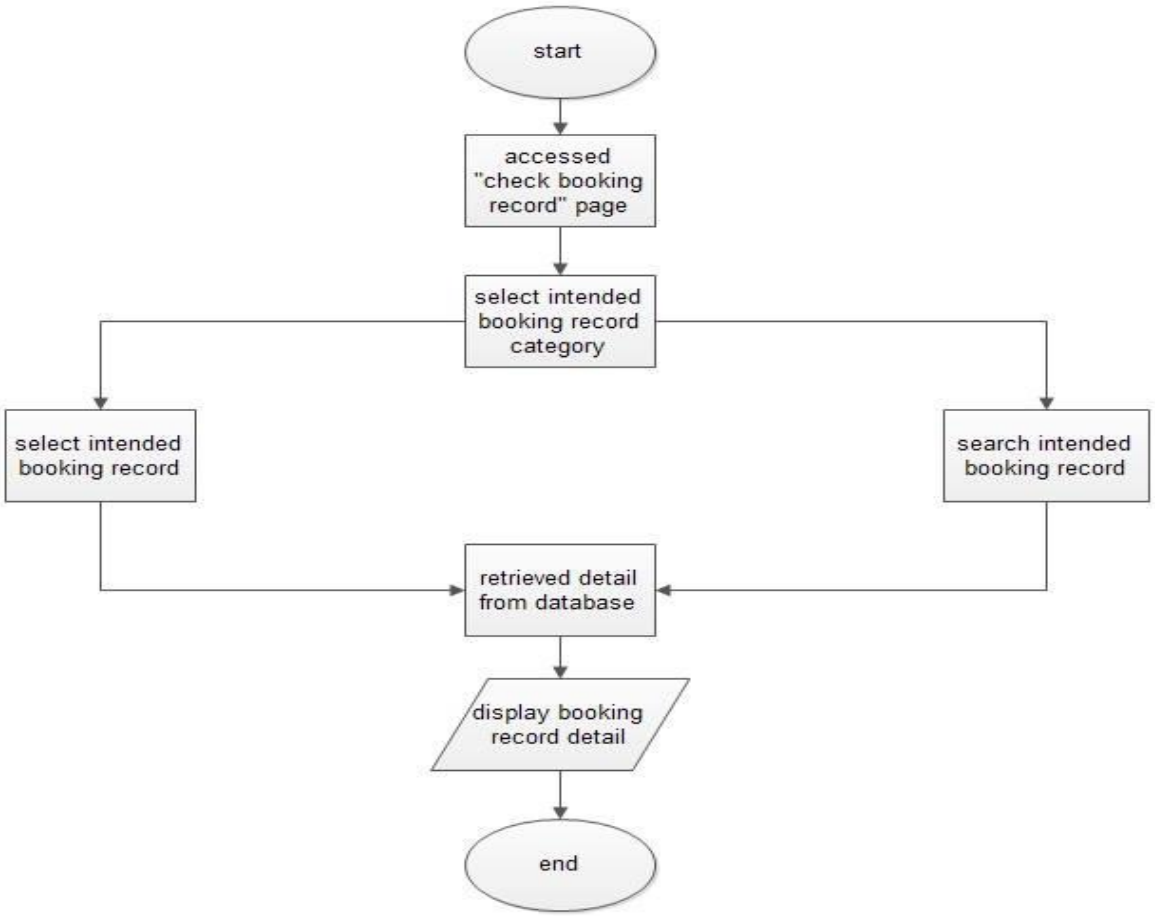
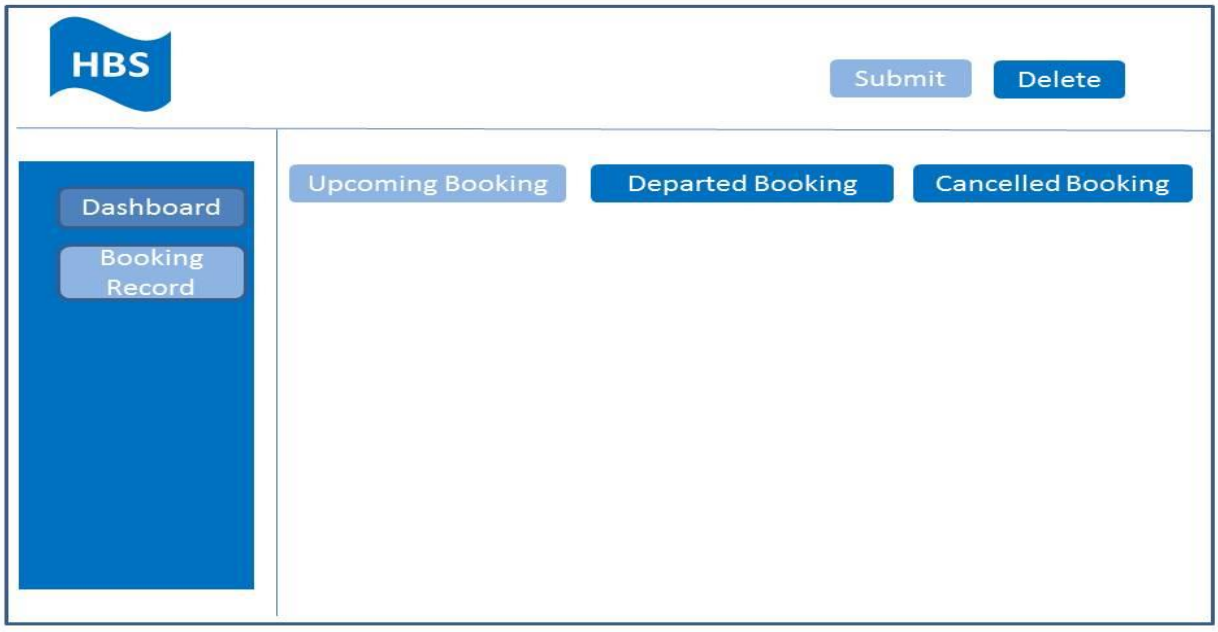
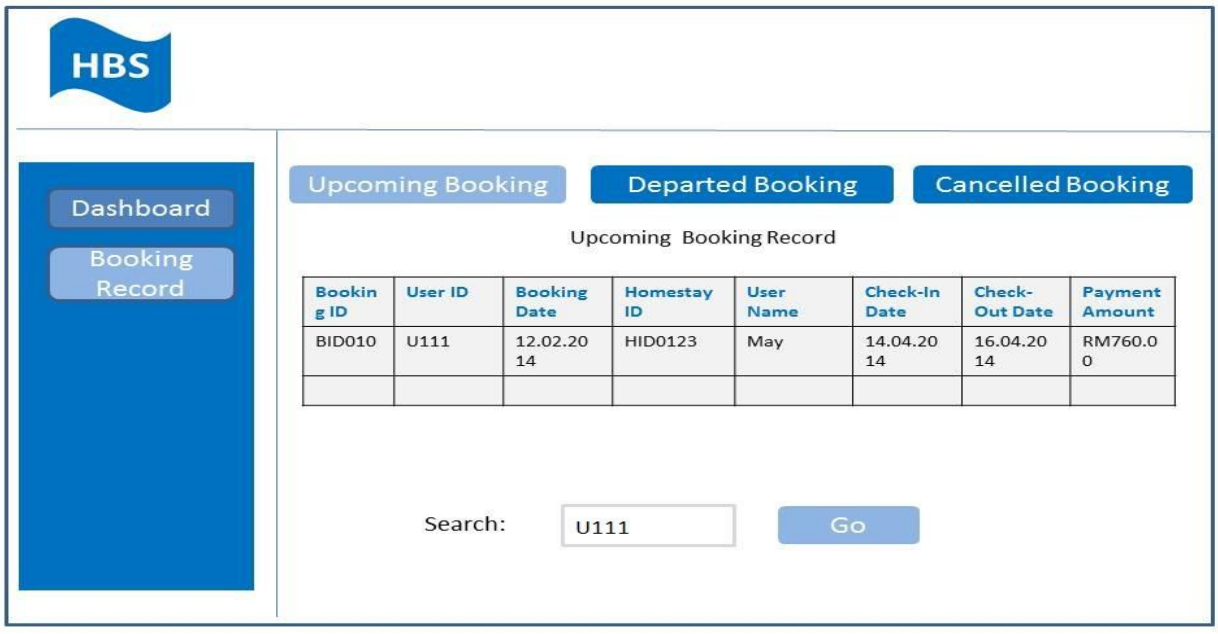


Figure 3.7

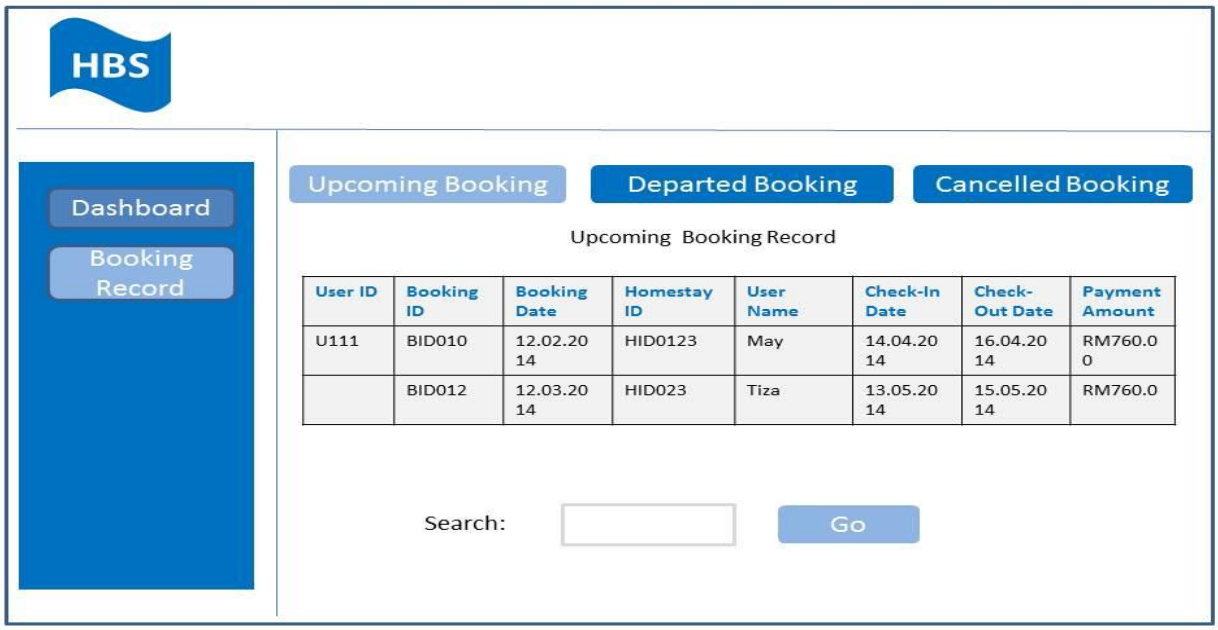
c. Graphic User Interface



Screen 1



Screen 2


HBS
Dashboard
Booking Record
Upcoming Booking
Departed Booking
Cancelled Booking
Upcoming Booking Record
User ID
Booking ID
Booking Date
Homestay ID
User Name
Check-In Date
Check-Out Date
Payment Amount
U111
BID010
12.02.2014
HID0123
May
14.04.2014
16.04.2014
RM760.00
BID012
12.03.2014
HID023
Tiza
13.05.2014
15.05.2014
RM760.0
Search:
Go

Screen 3

3.8 Process 6.0: Check transaction record

a. Pseudo code

1. Start
2. Make access “Transaction Record”
3. Information retrieved from database “TRANSACTION RECORD”
4. Display all payment record
5. End

b. Flow chart

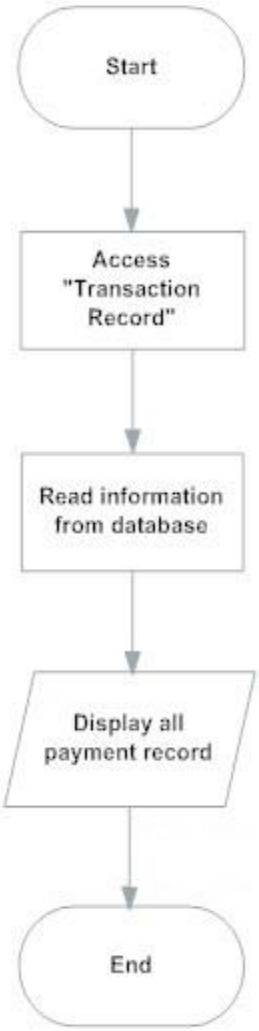
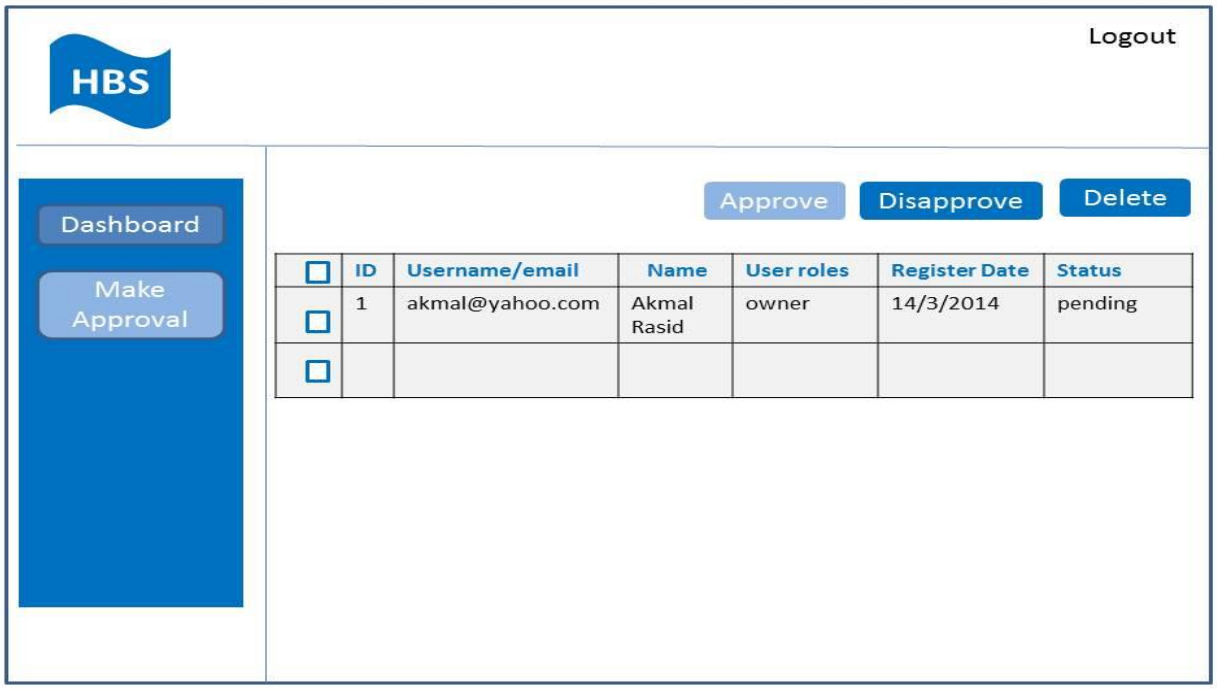
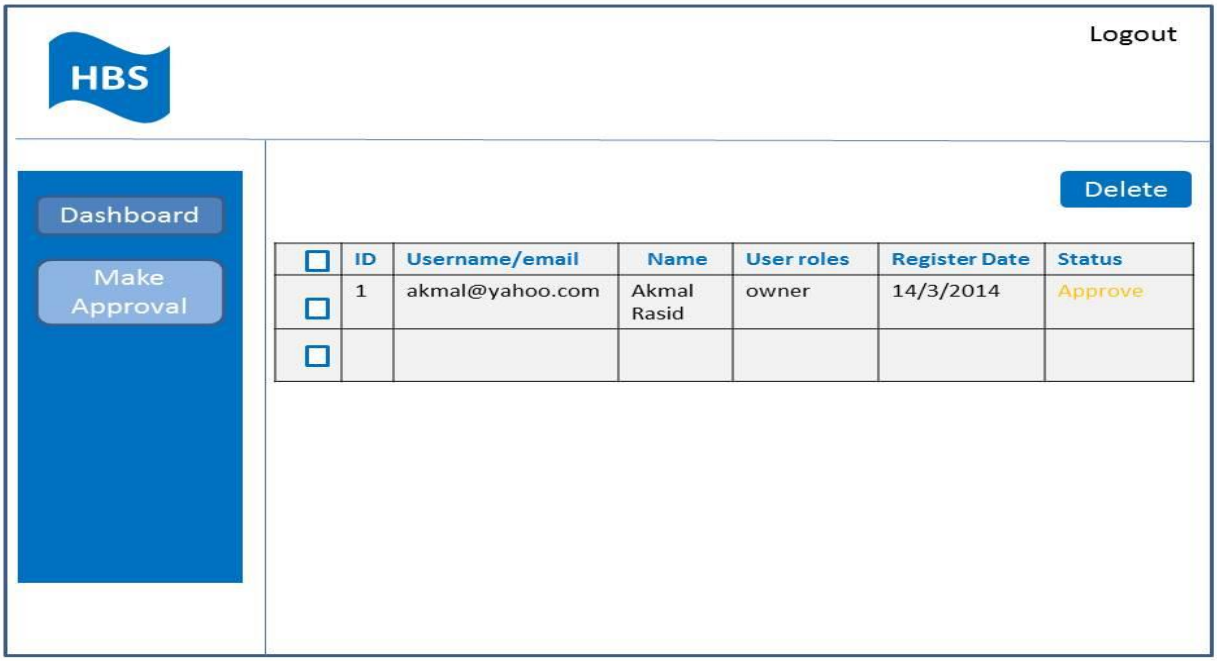


Figure 3.8

c. Graphic User Interface



Screen 1



Screen 2

3.9 Process 7.0: Review account

a. Pseudo code

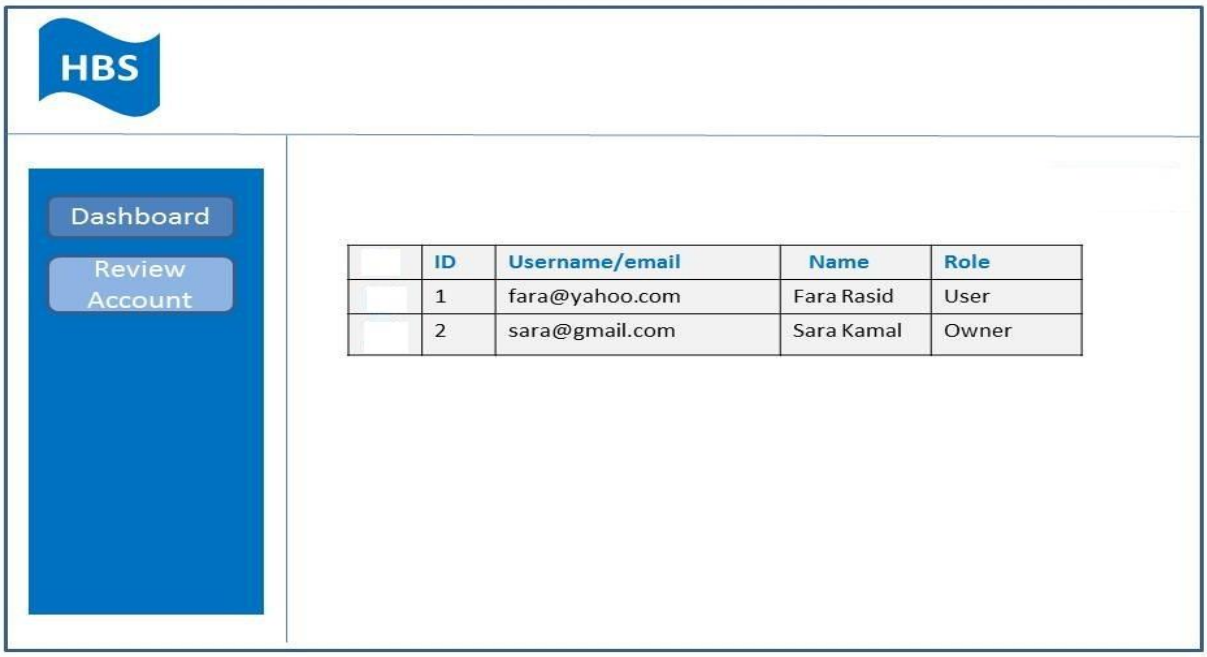
1. Start
2. Make access “Review Account”
3. Display all user and owner information
4. Select user
5. Get Info retrieved from database “USER RECORD”
6. Display user account
7. End

b. Flow chart



Figure 3.9

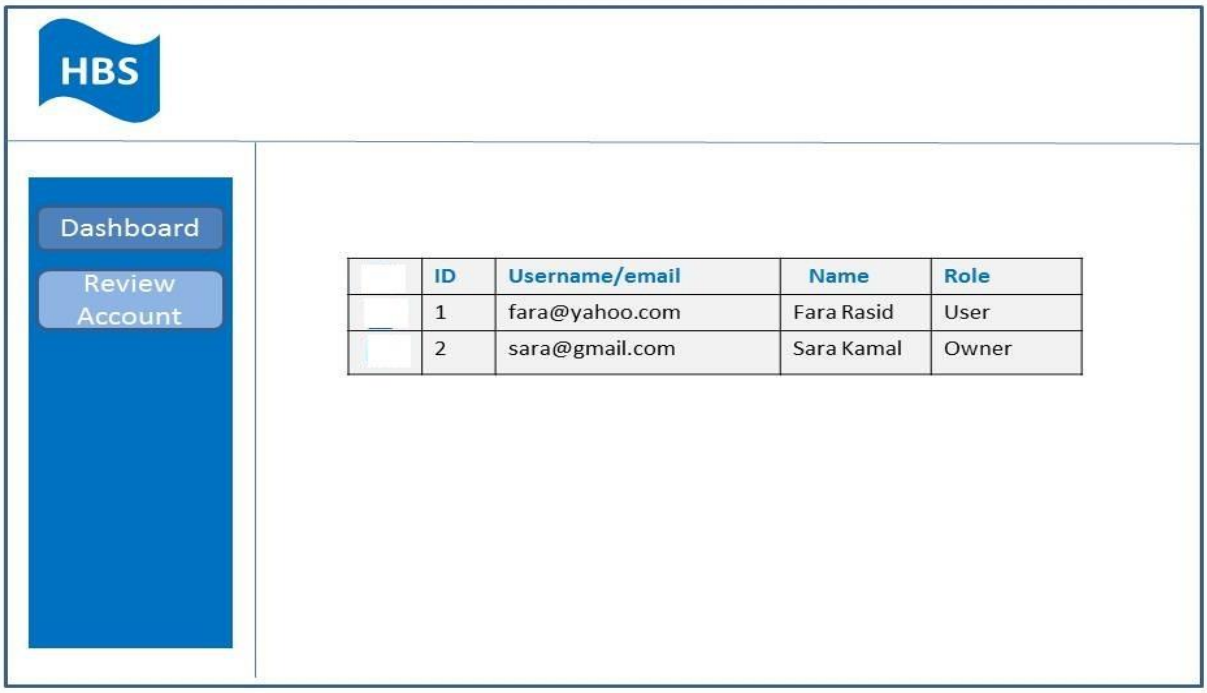
c. Graphic User Interface



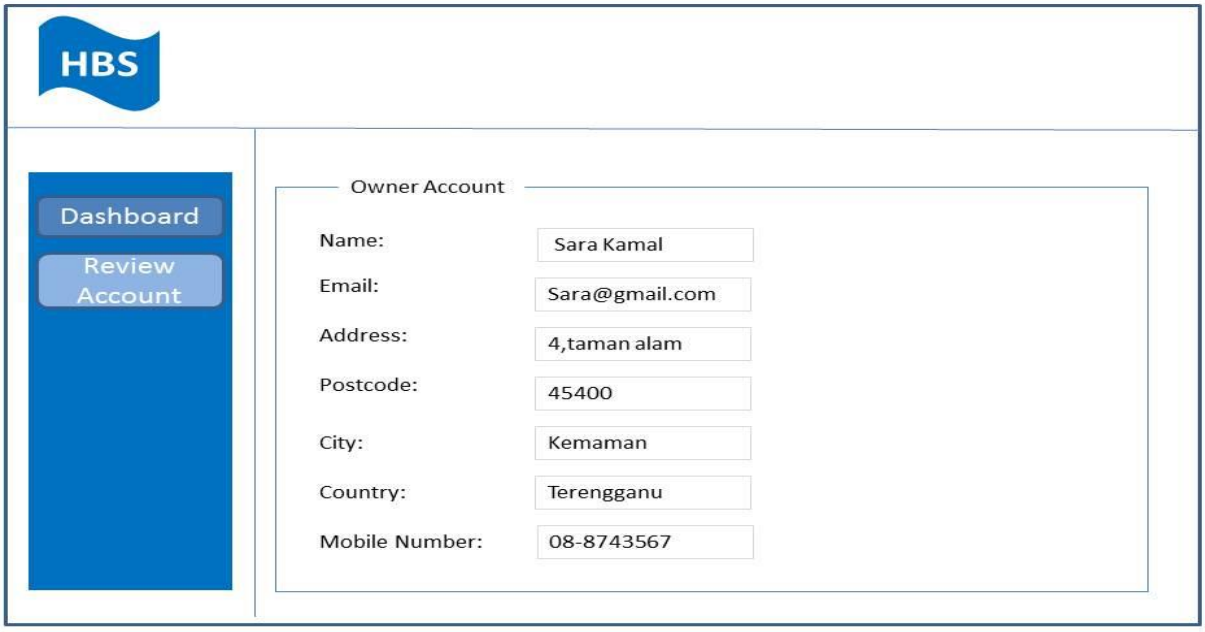
Screen 1



Screen 2



Screen 3



Screen 4

3.10 Process 8.0: Make approval

a. Pseudo code

1. Start
2. Receive notification “pending”
3. Read forms retrieve from database “PARTNERSHIP RECORD”
4. Display filled booking form
5. Review and make decision
6. If (Decision is approve)
   6.1 Application is approved
   6.2 changes saved into database “PARTNERSHIP RECORD”
7. Else
   7.1 Application failed
8. End

b. Flow chart

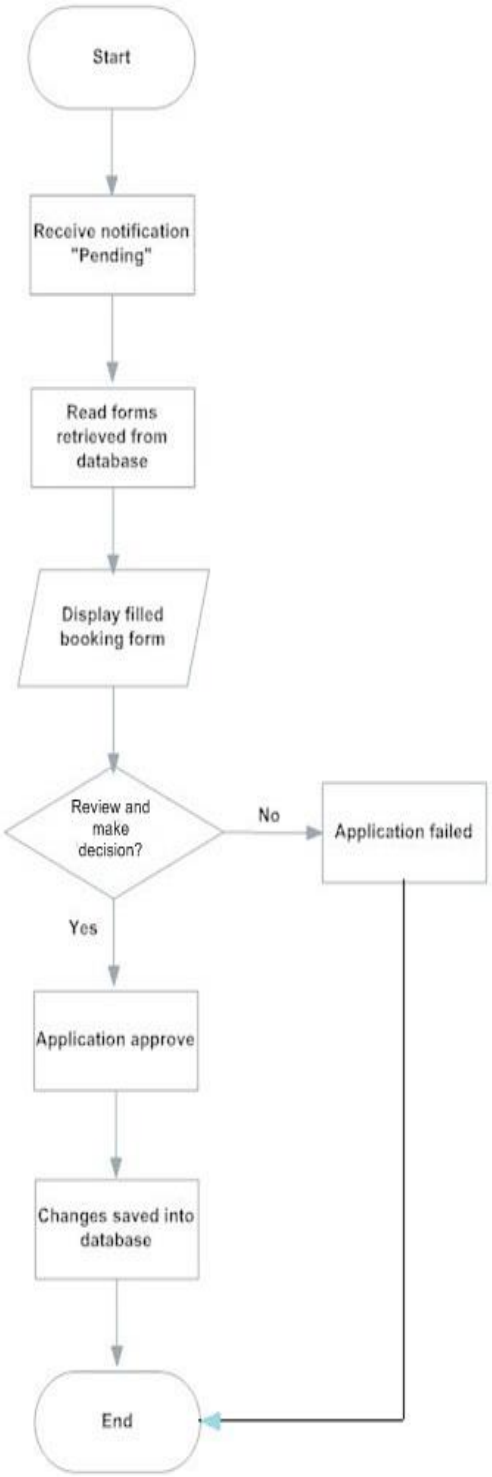
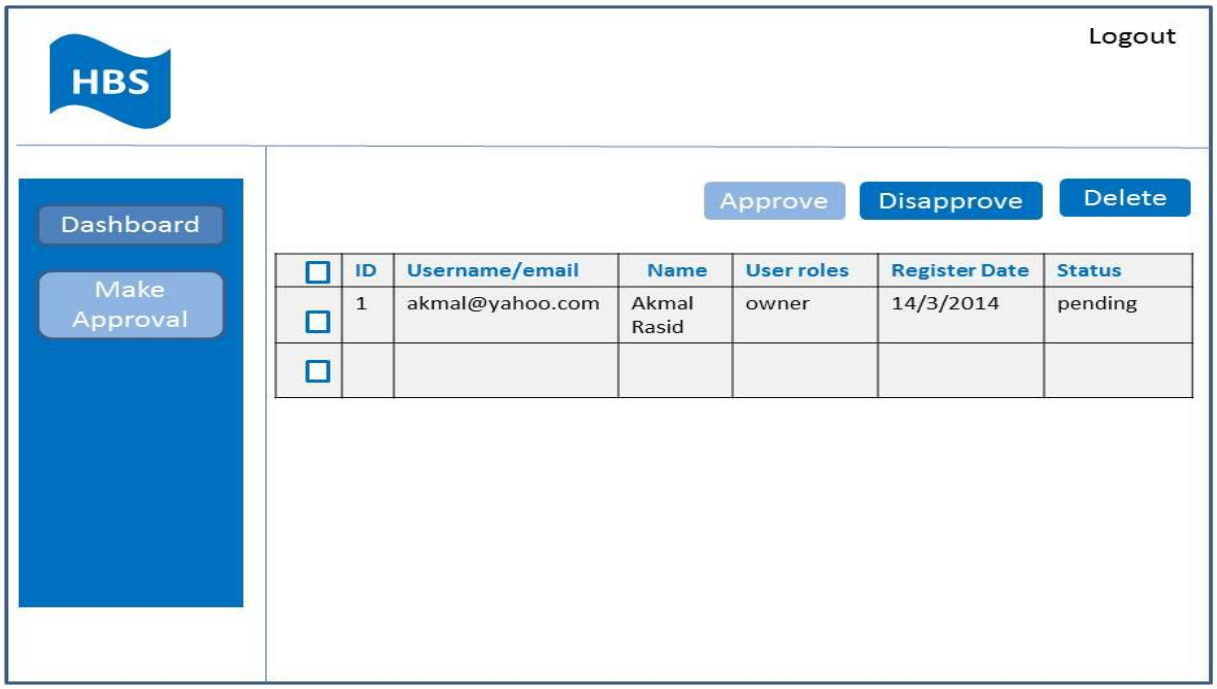
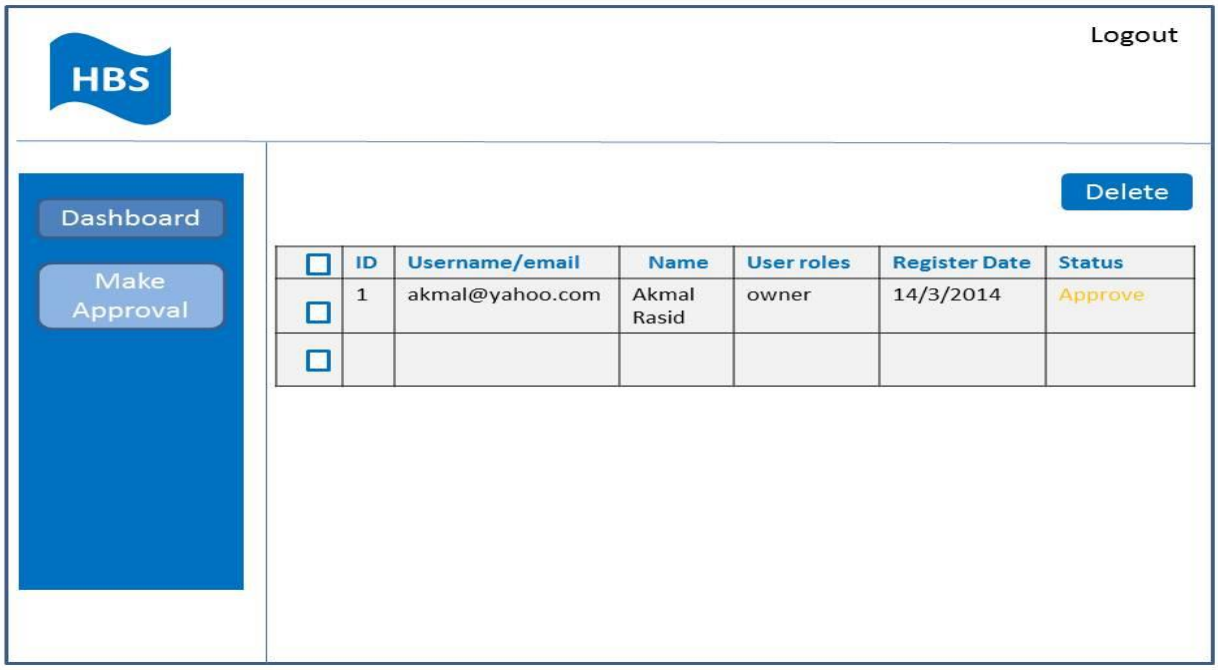


Figure 3.10

c. Graphic User Interface



Screen 1



Screen 2

3.11 Process 9.0: Inactive owner account

a. Pseudo code

1. Start
2. Received notification about “Partnership Termination”
3. Display list of owner that to be terminated
4. Select owner to inactivate
5. Changes saved into database “PARTNERSHIP RECORD”
6. End

b. Flow chart

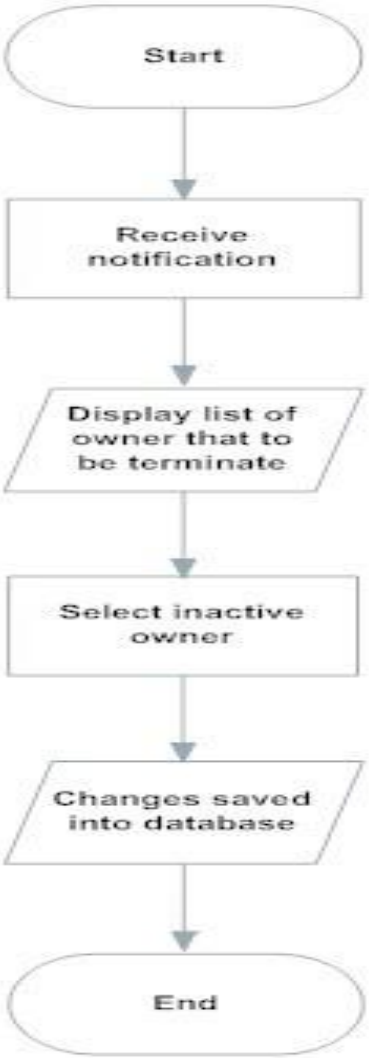
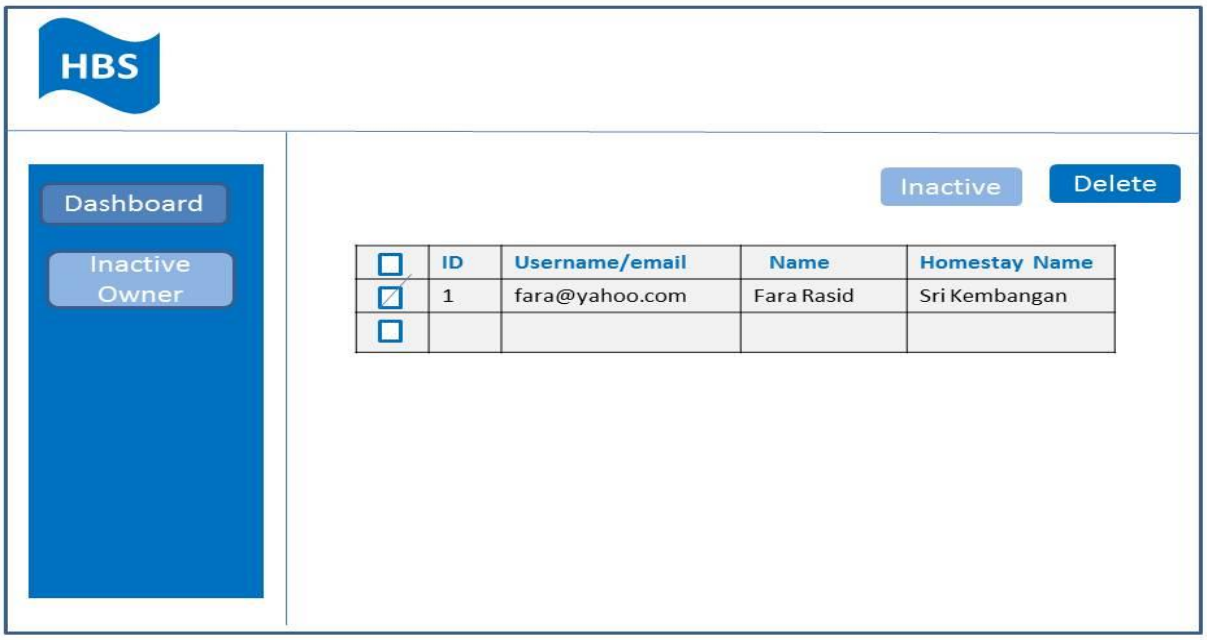
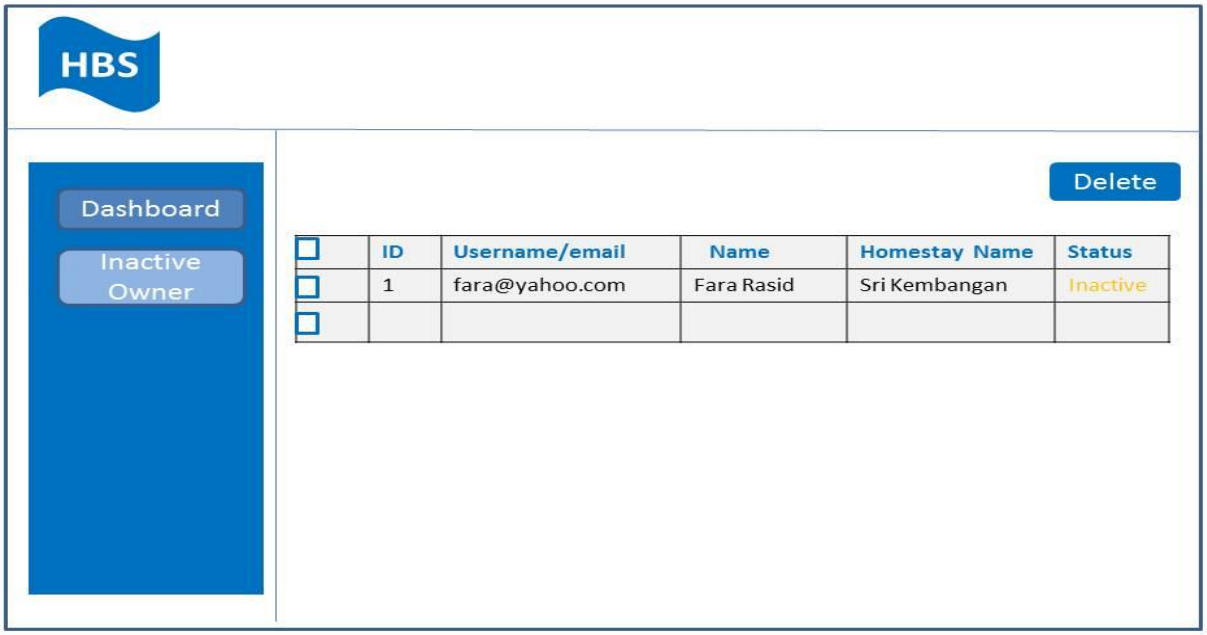


Figure 3.11

c. Graphic User Interface



Screen 1



Screen 2

3.12 Process 3.1: change password

a. Pseudo code

1. Start
2. Make access “My Profile”
3. Select “Change Password”
4. Input change password details
   - 4.1 Input old_password
   - 4.2 Input new_password
   - 4.3 Re-input new _password
5. Verify old_password with database Authorization Record
6. Verify new_password with re-input new_password
7. If (true)
   - 7.1 Update new password into database Authorization Record
   - 7.2 Display message change password successful
8. Else
   - 8.1 Display message change password is cancel
9. End

b. Flow chart

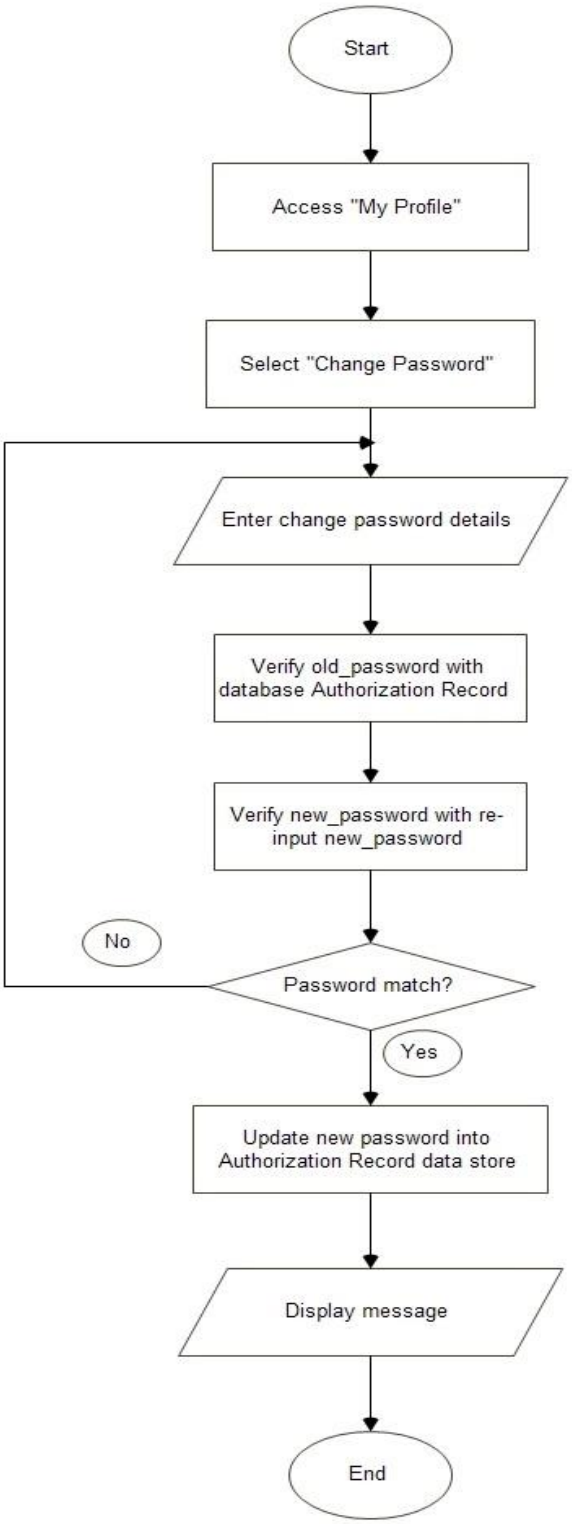
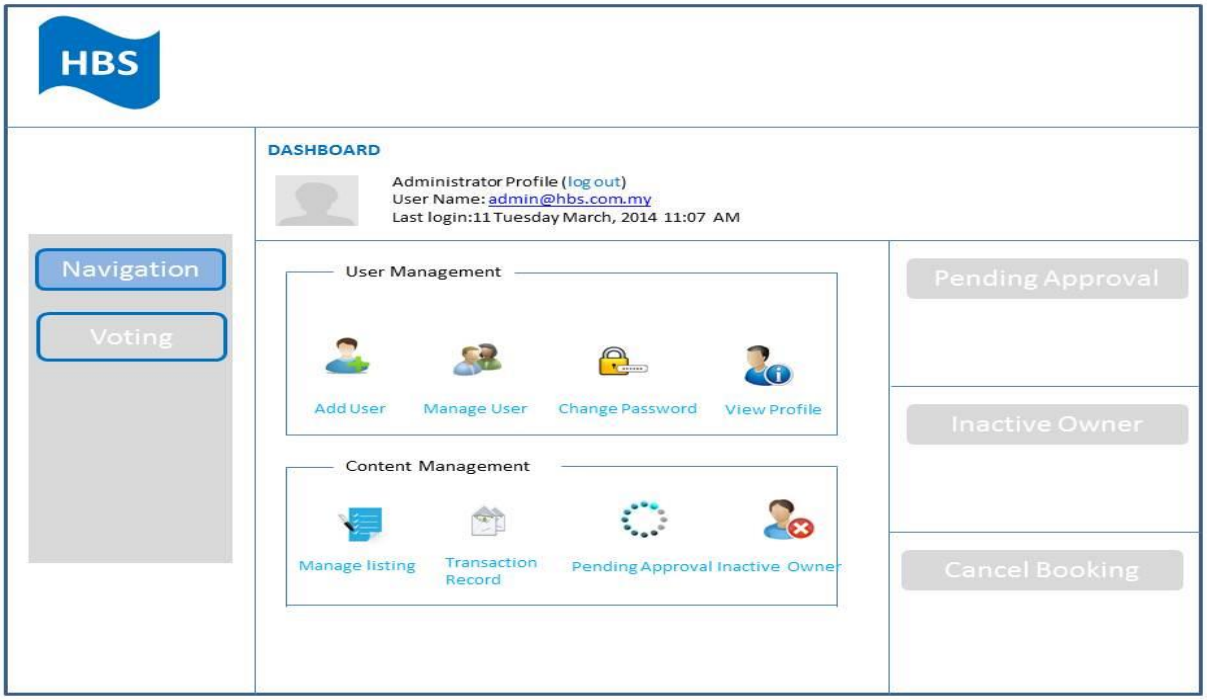
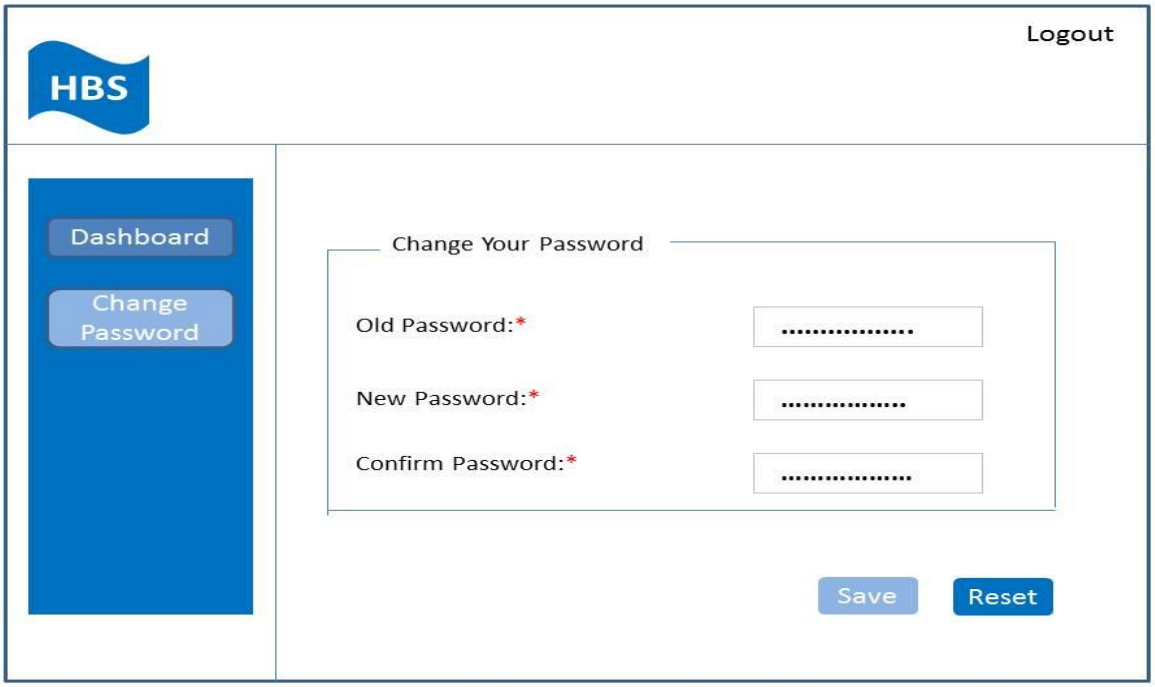


Figure 3.12

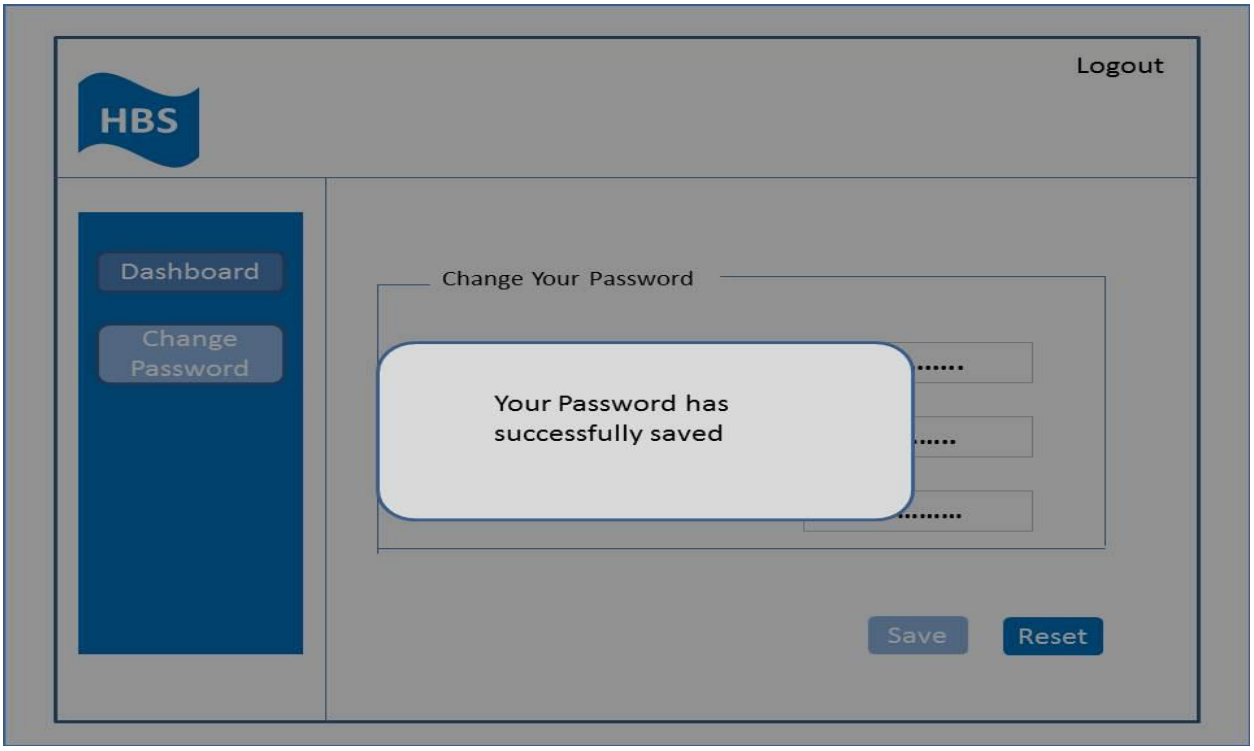
c. Graphic User Interface



Screen 1



Screen 2


Logout
HBS
Dashboard
Change Password
Change Your Password
Your Password has successfully saved
Save
Reset

Screen 2

PSEUDOCODE , FLOW CHART AND INTERFACE FOR OWNER

3.13 Process 3.0 : Manage Profile (For Owner)

3.13.1 Process 3.1 : Change Password

a. Pseudo code

1. Start
2. Make access "My Profile"
3. Select "Change Password"
4. Input change password details
   - 4.1 Input old_password
   - 4.2 Input new_password
   - 4.3 Re-input new _password
5. Verify old_password with database Authorization Record
6. Verify new_password with re-input new_password
7. If (true)
   - 7.1 Update new password into database Authorization Record
   - 7.2 Display message change password successful
8. Else
   - 8.1 Display message change password is cancel
9. End

b. Flow chart

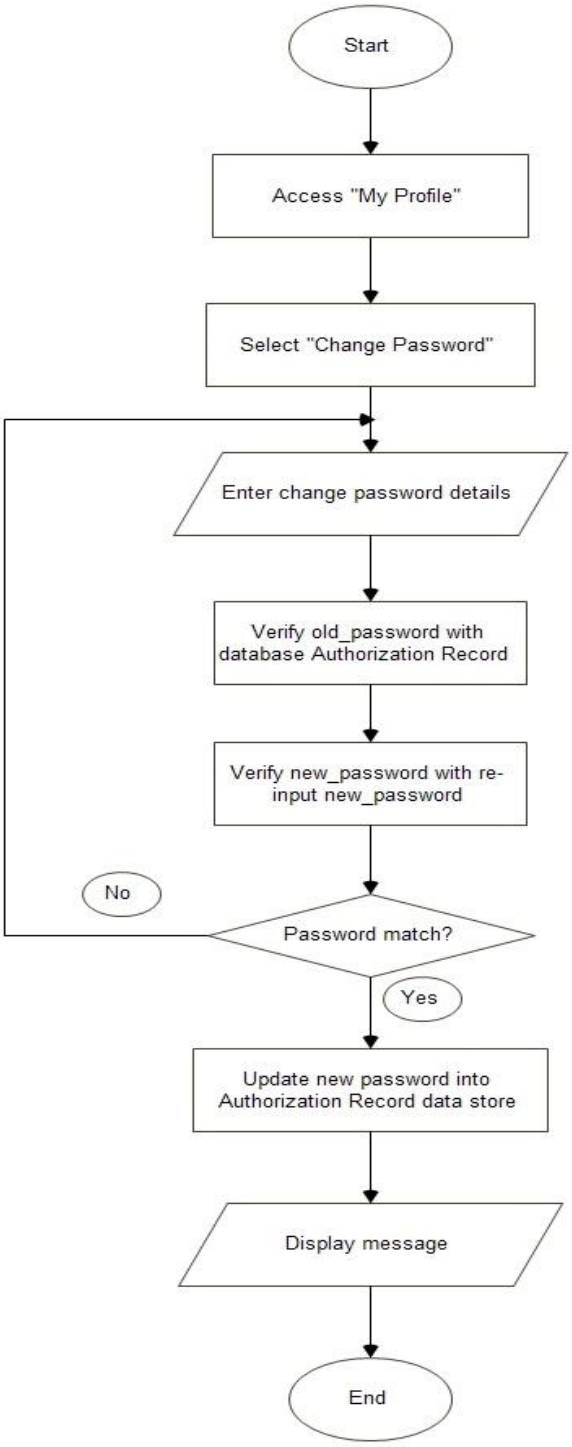
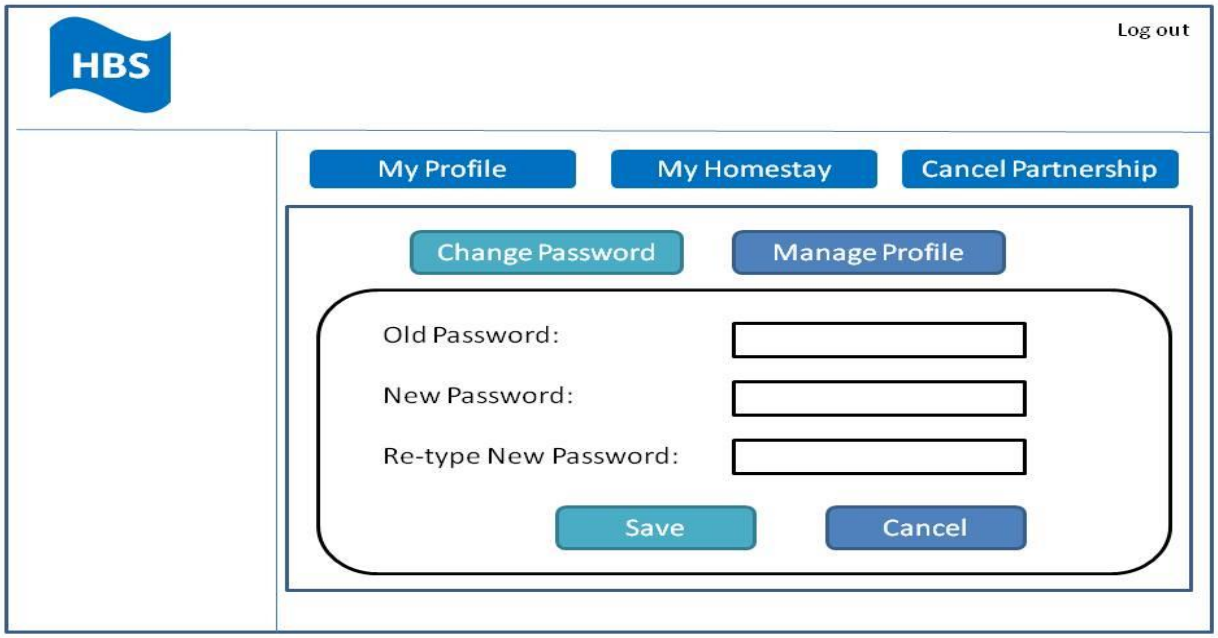


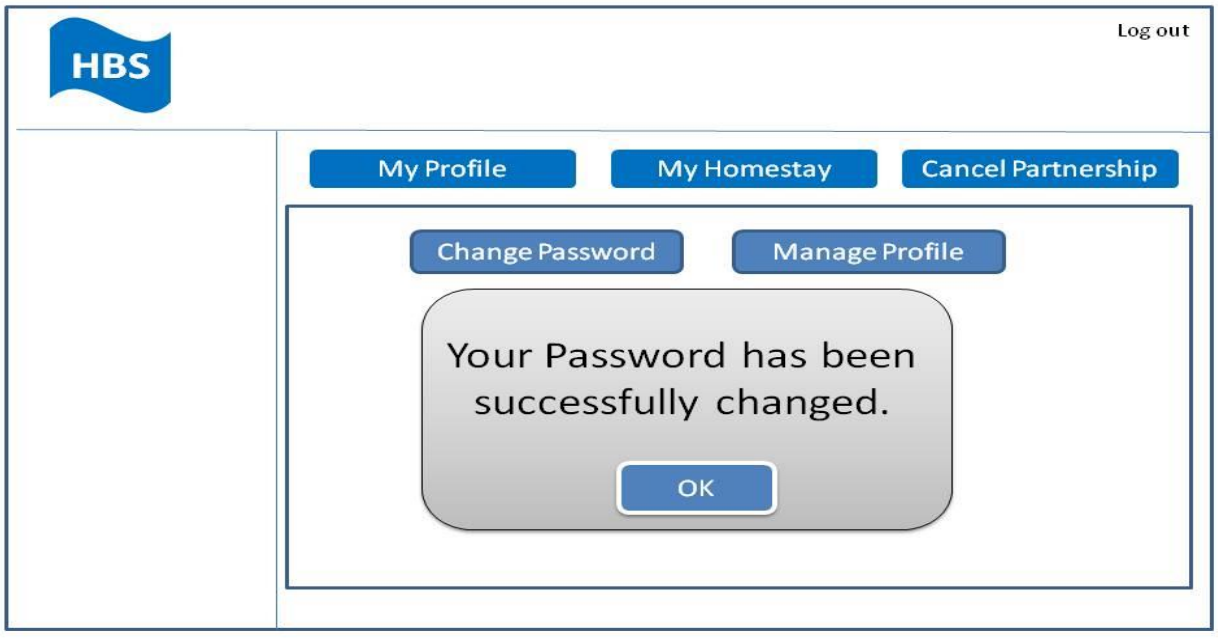
Figure 3.13.1

c. Graphic User Interface

\



Screen 1



Screen 2

3.13.2 Process 3.2 : Edit Profile

a. Pseudo code

1. Start
2. Make access “My Profile”
3. Select “Edit Profile”
4. Load editable profile information form
5. User edit profile information
6. If (select Submit)
    6.1 Save the edited profile information in database Owner Record
    6.2 Display message edit profile is successful
7. Else
    7.1 Cancel edit profile
8. End

b. Flow chart

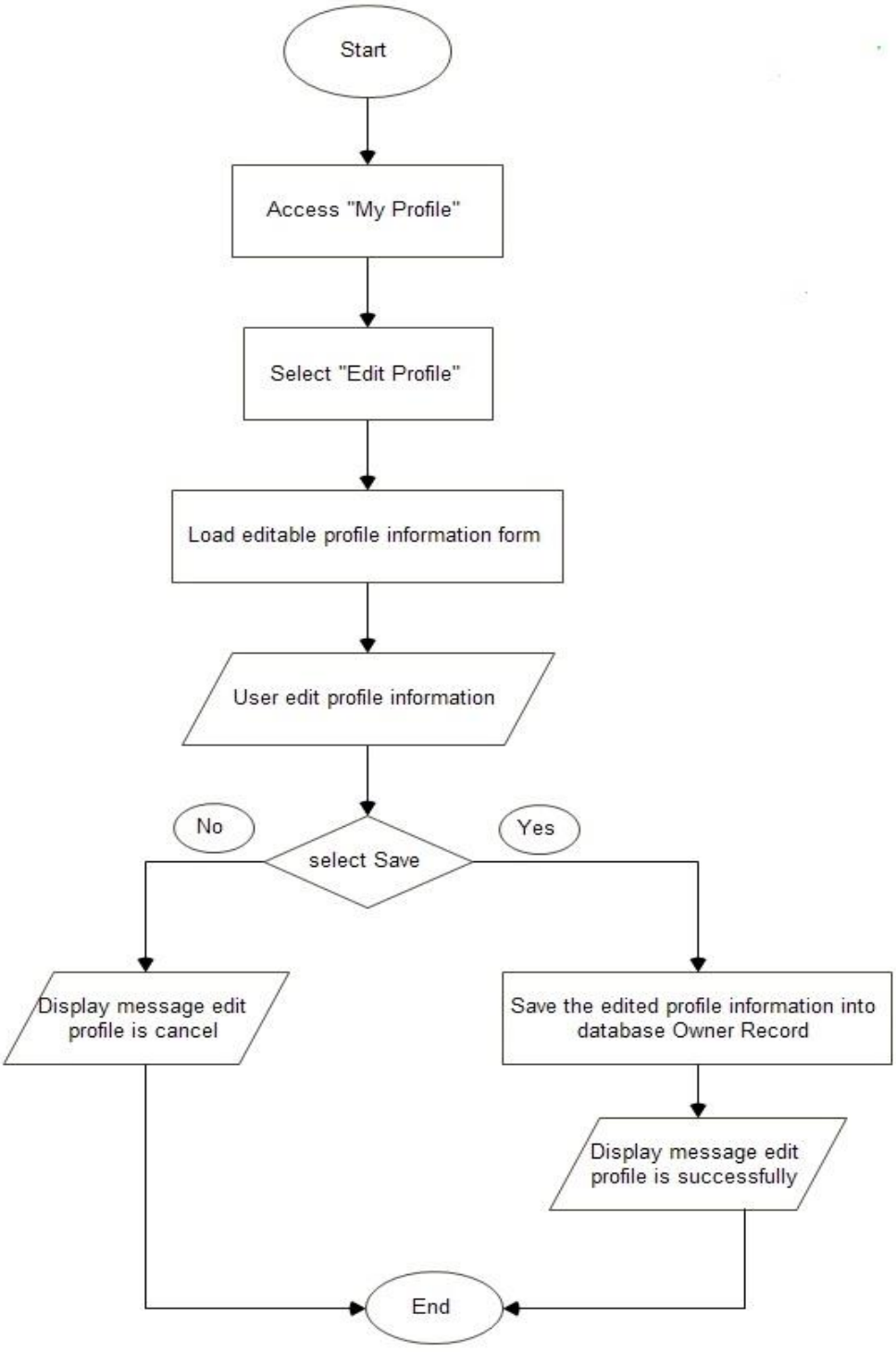
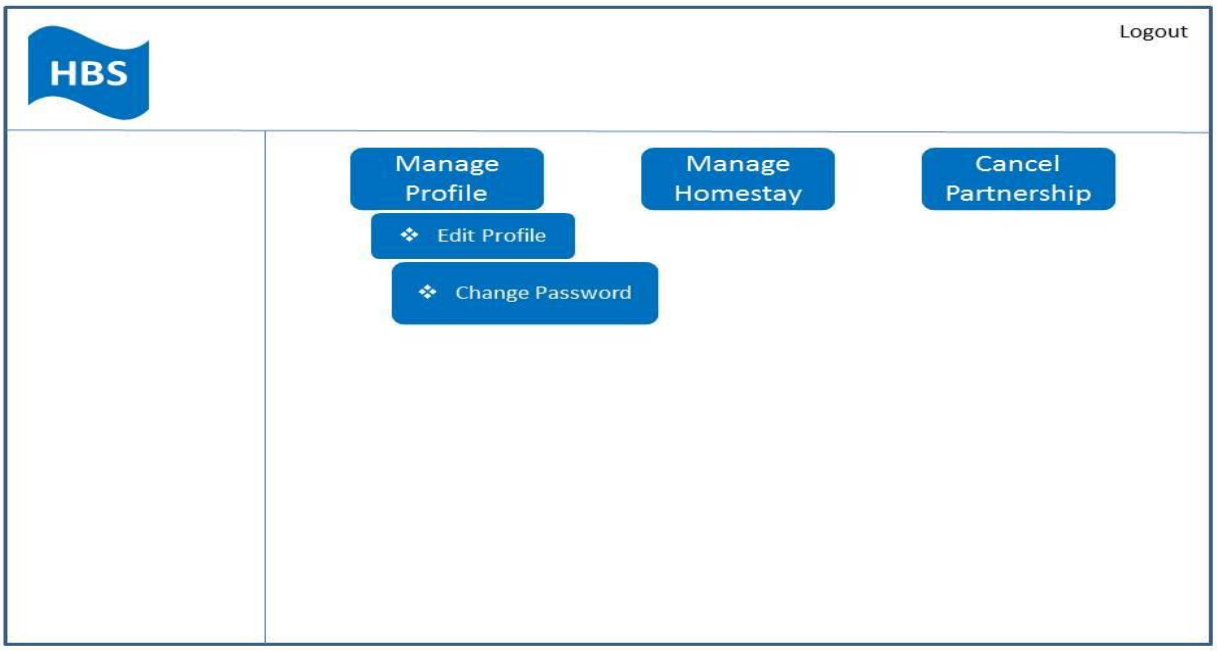
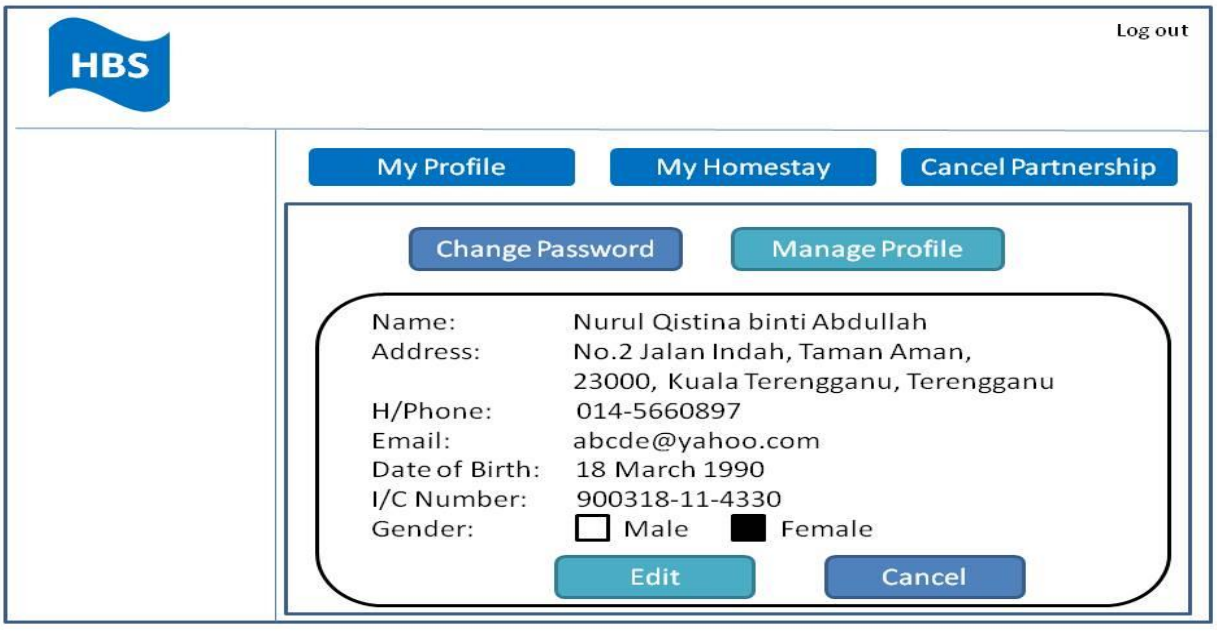


Figure 3.13.2

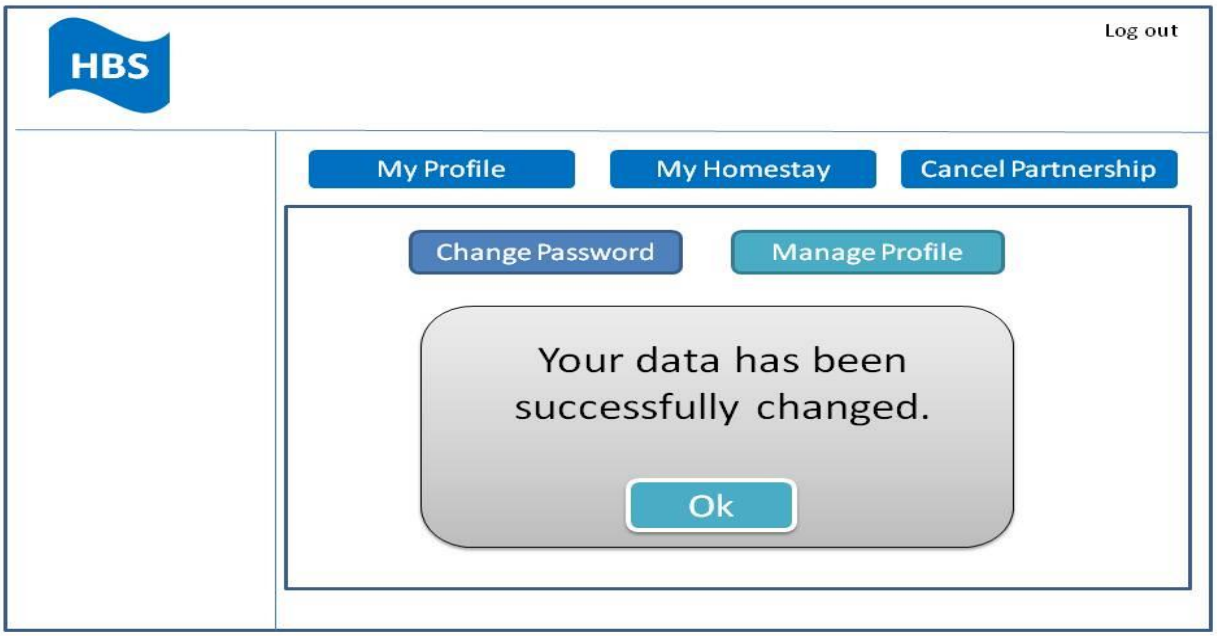
c. Graphic User Interface



Screen 1



Screen 2


HBS
Log out
My Profile
My Homestay
Cancel Partnership
Change Password
Manage Profile
Your data has been successfully changed.
Ok

Screen 3

3.14 Process 11.0 : Apply Partnership

a. Pseudo code

1. Start
2. Make access to “Apply Partnership”
3. Input partnership application details into database
   Partnership Record
4. if (select approve)
   4.1 Owner received PAId, username and password via email
5. Else
   5.1 application failed and owner also notified via email
6. End

b. Flow chart

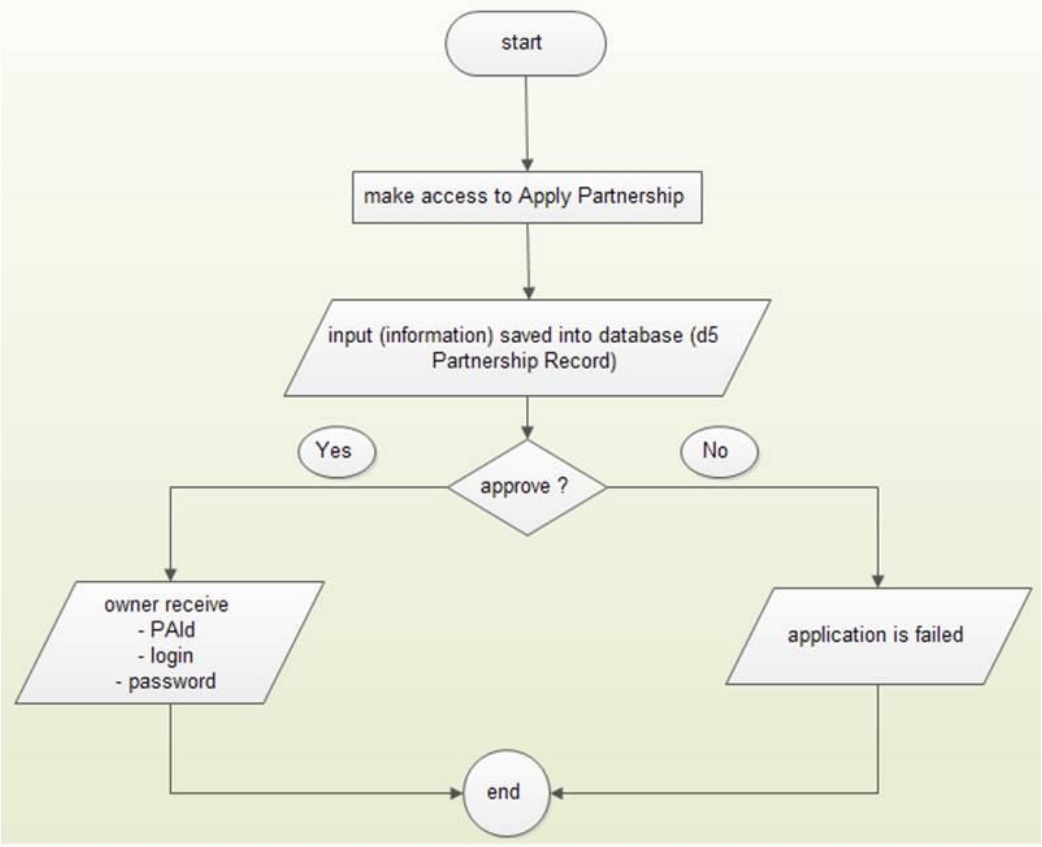
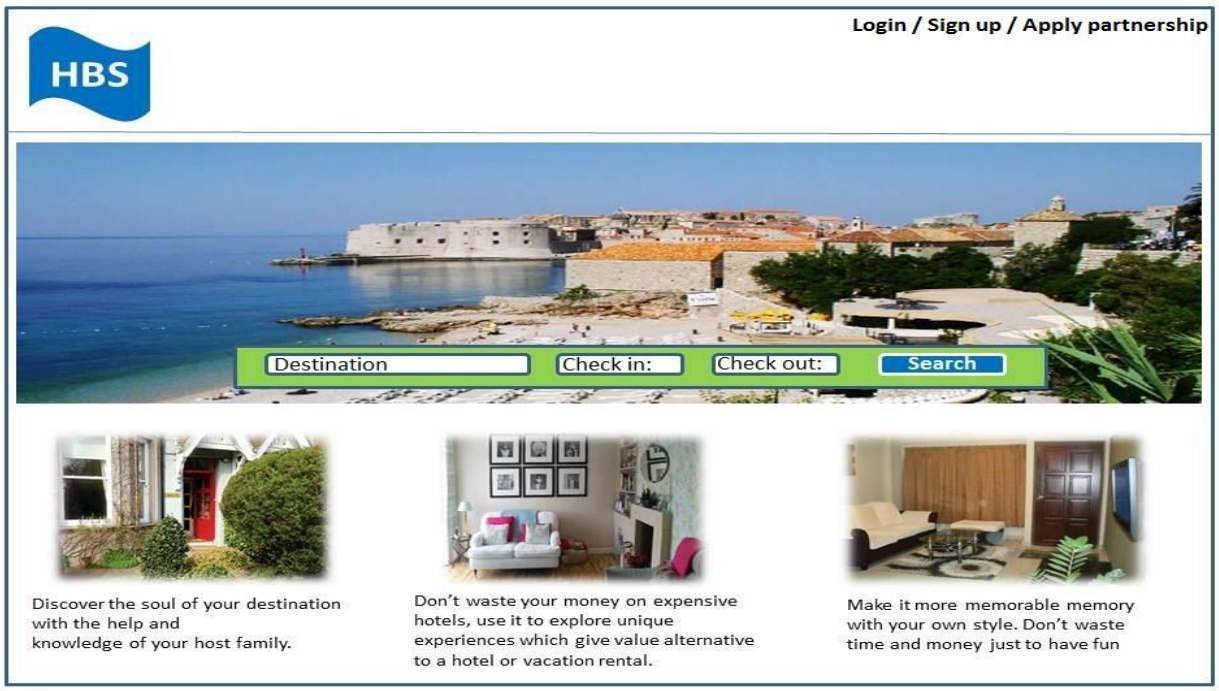
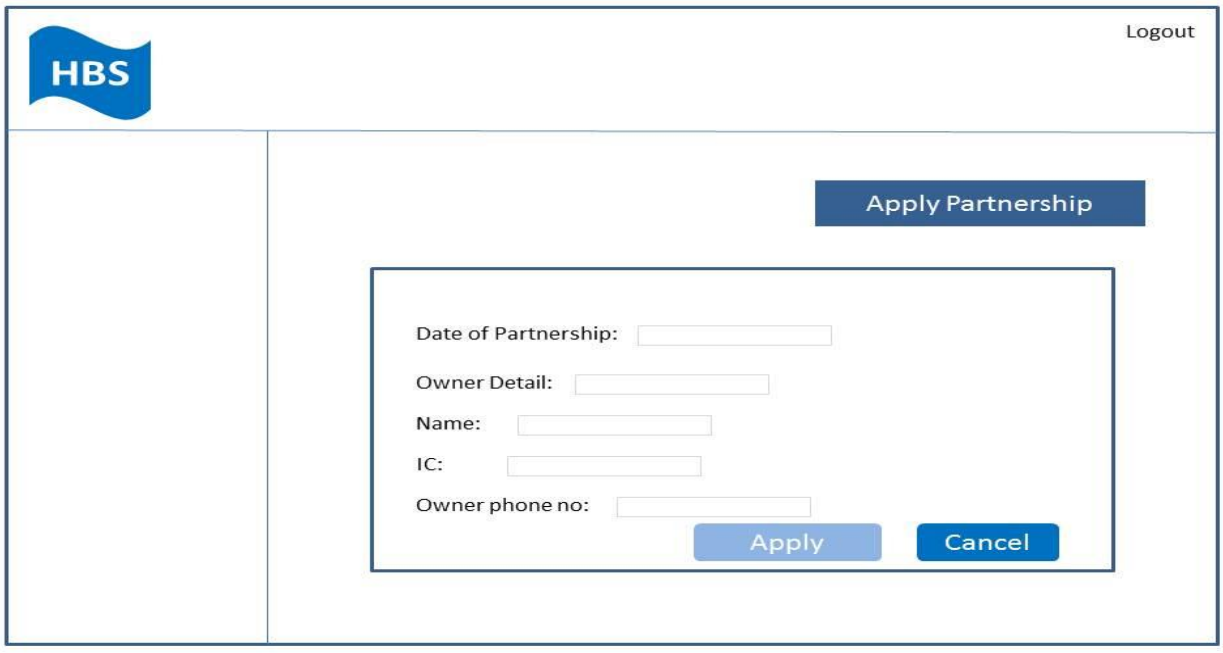


Figure 3.14

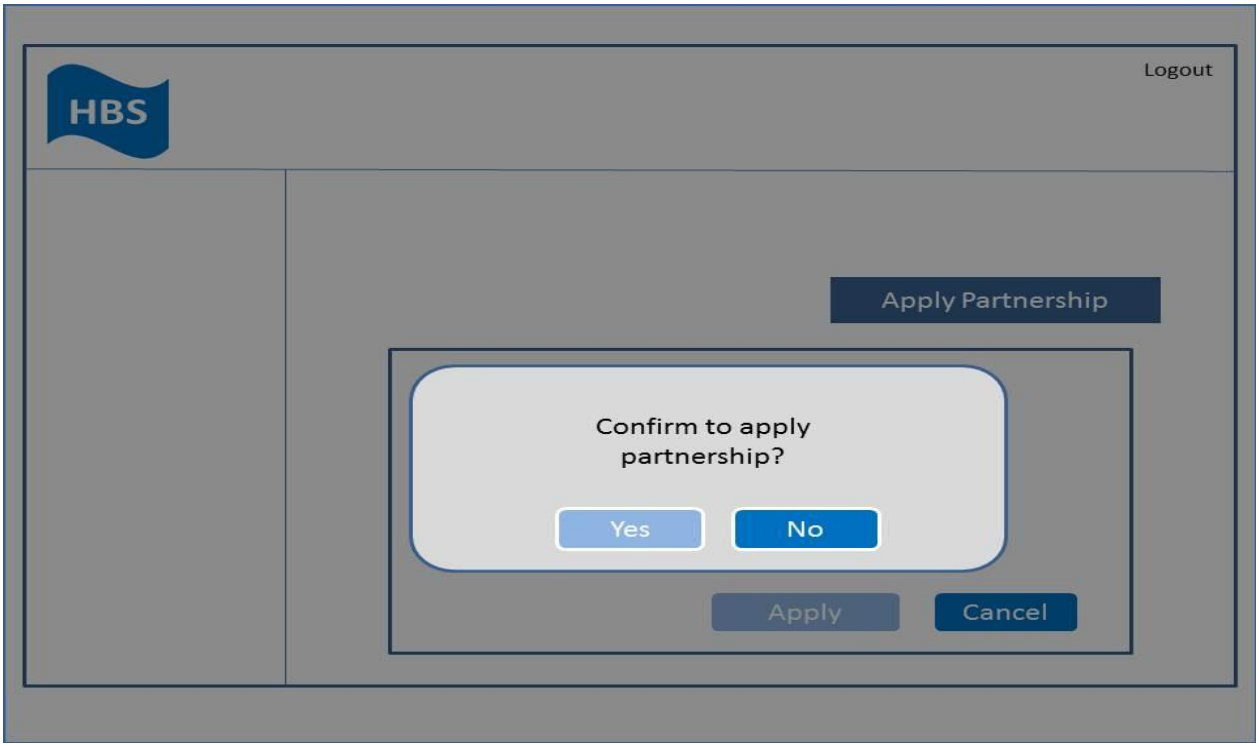
c. Graphic User Interface



Screen 1



Screen 2



Screen 3



Screen 4

3.15 Process 12.0: Manage Homestay

3.15.1 Process 12.1: Add Homestay

a. Pseudo code

1. Start
2. Make access to “Add Homestay”
3. Input add-homestay details
4. if (select Save)
    4.1 input(information) is saved into database Homestay Record
5. Else
    5.1 add-homestay is cancelled
6. End

b. Flow chart

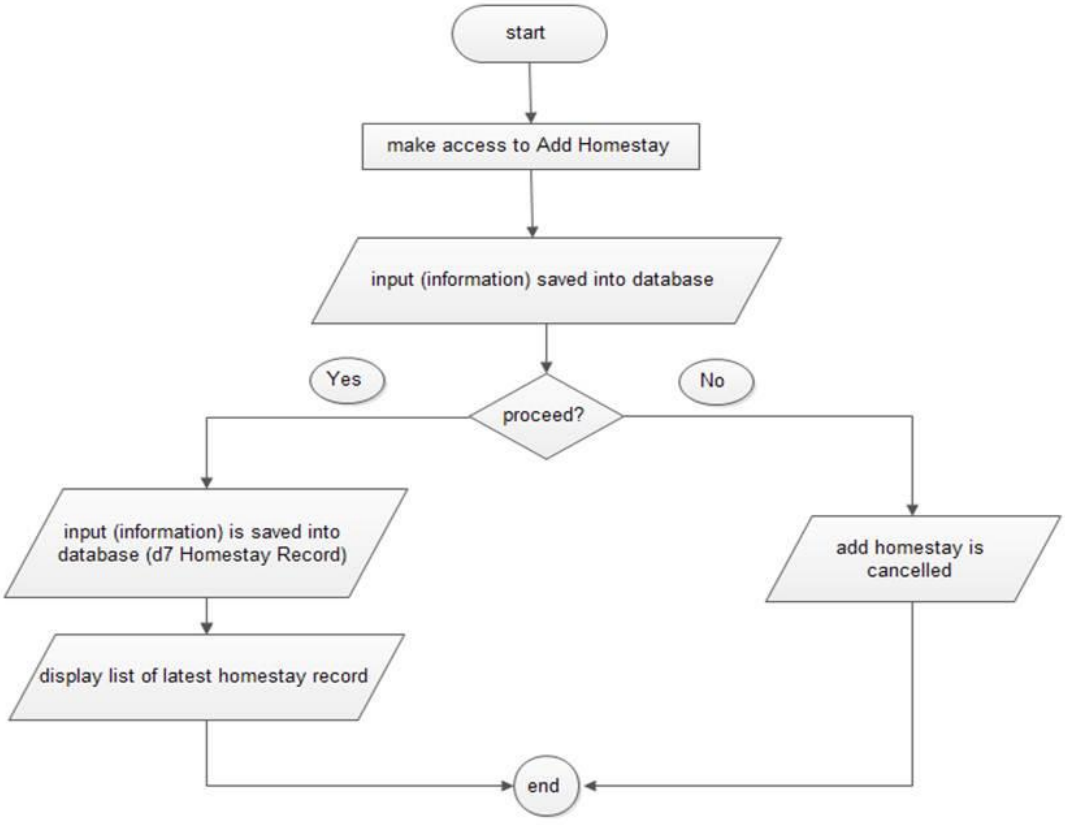
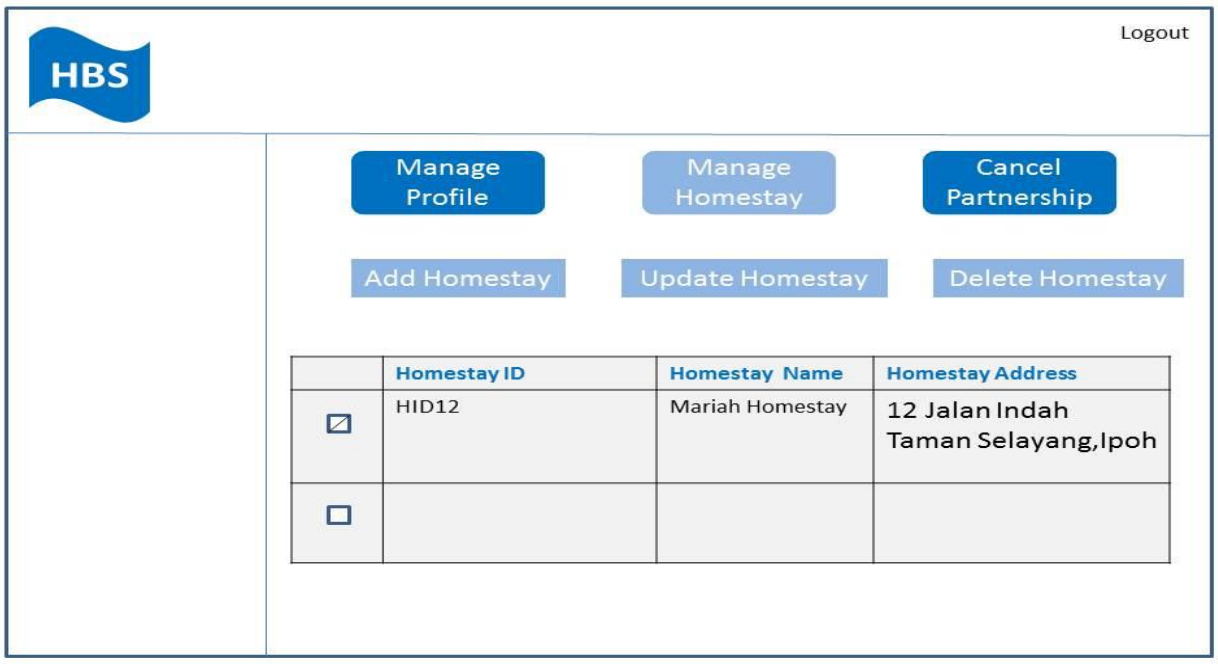
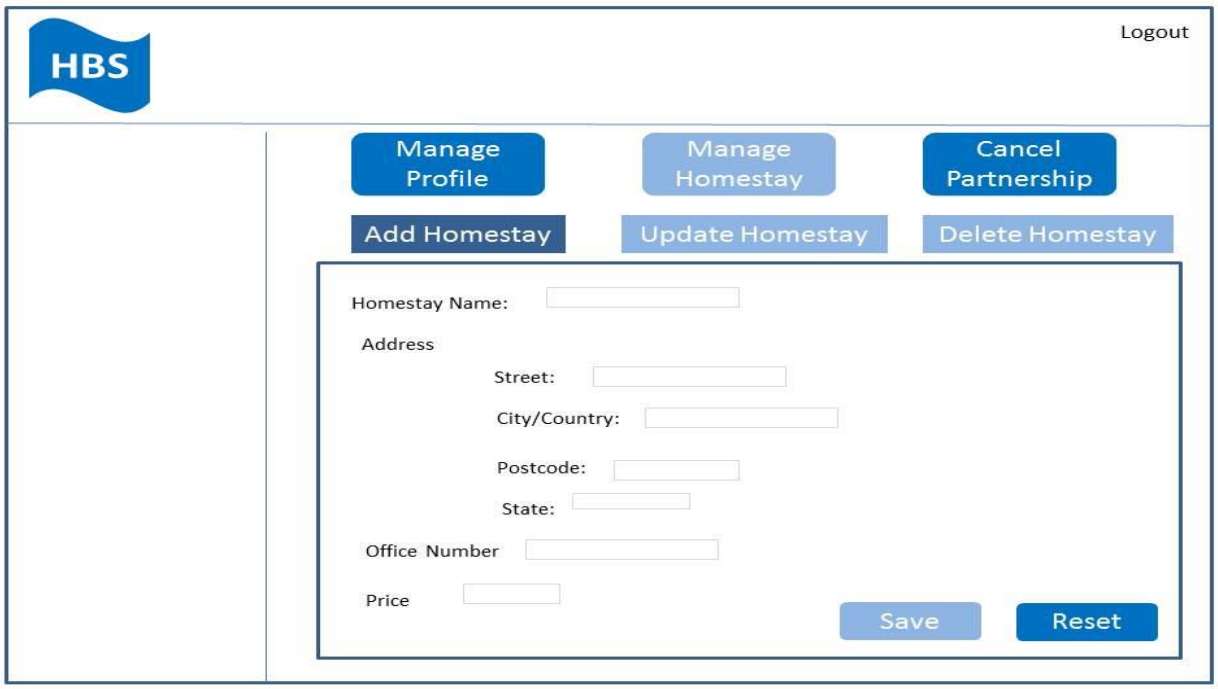


Figure 3.15.1

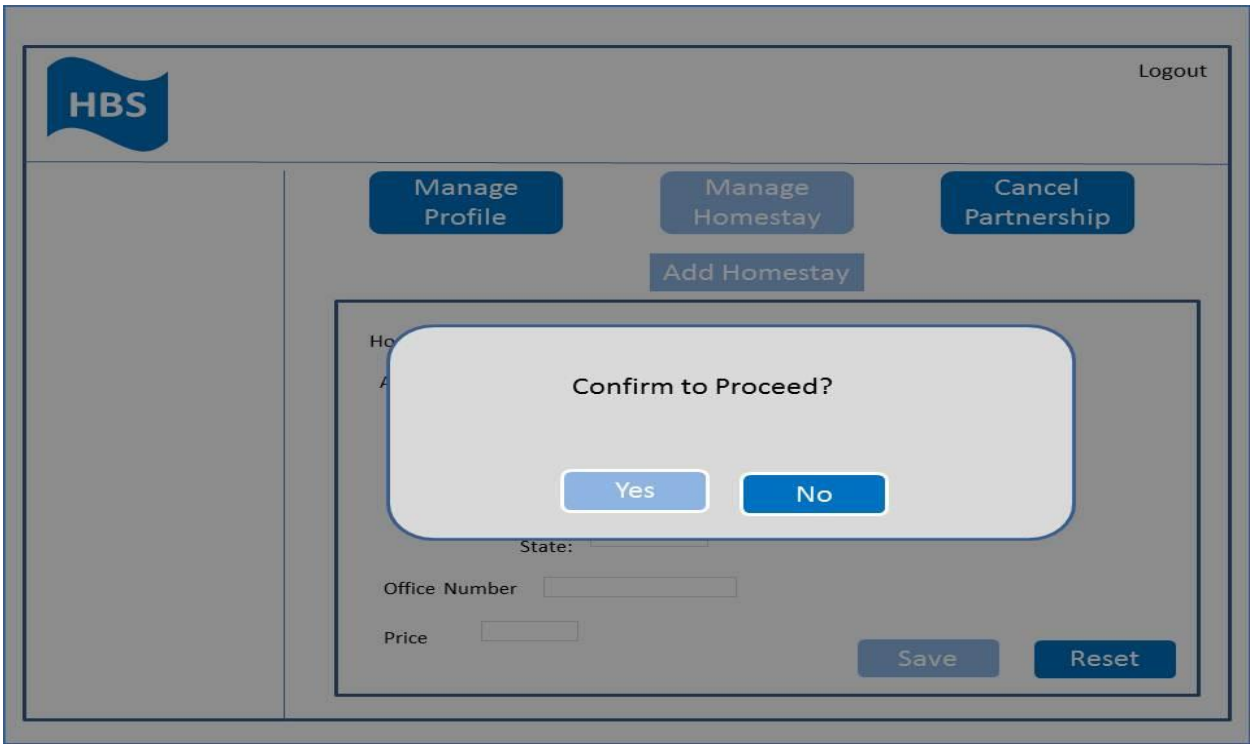
c. Graphic User Interface



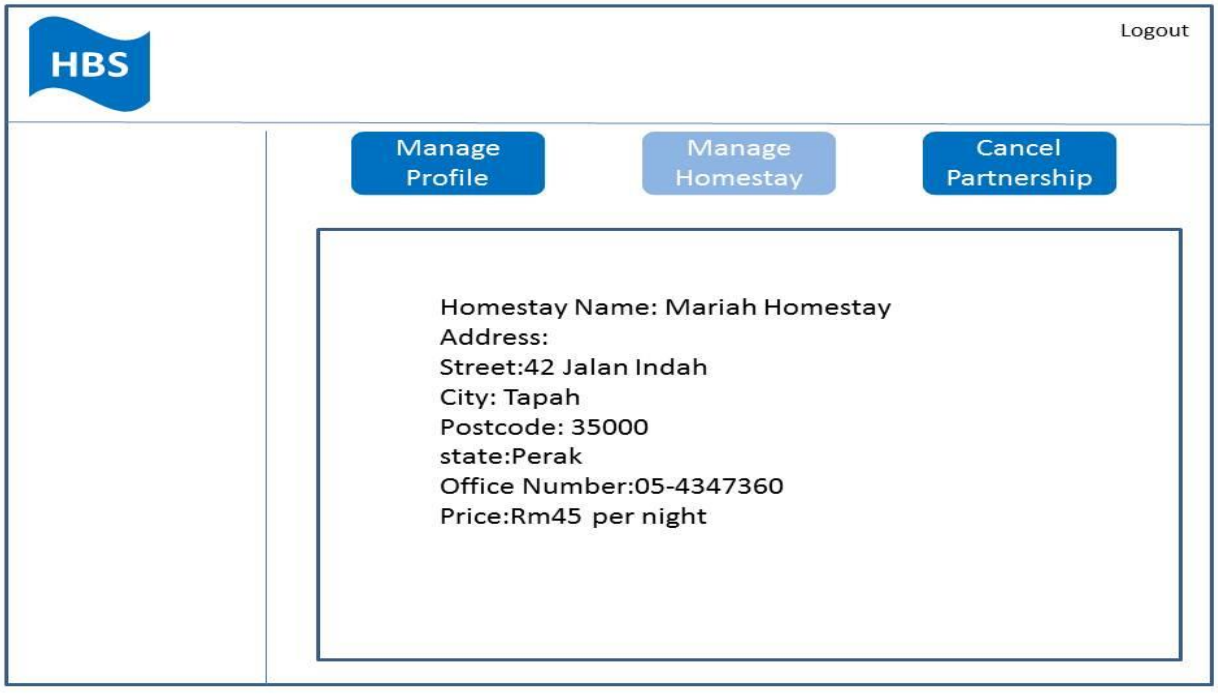
Screen 1



Screen 2



Screen 3



Screen 4

3.15.2 Process 12.2: Delete Homestay

a. Pseudo code

1. Start
2. Make access to “Delete Homestay”
3. Display all homestay list
4. Select intended homestay to delete
5. if (select Yes)
    5.1 the selected homestay is deleted from database Homestay Record
    5.2 display latest homestay list record
6. Else
    6.1 delete-homestay is cancelled
7. End

b. Flow chart

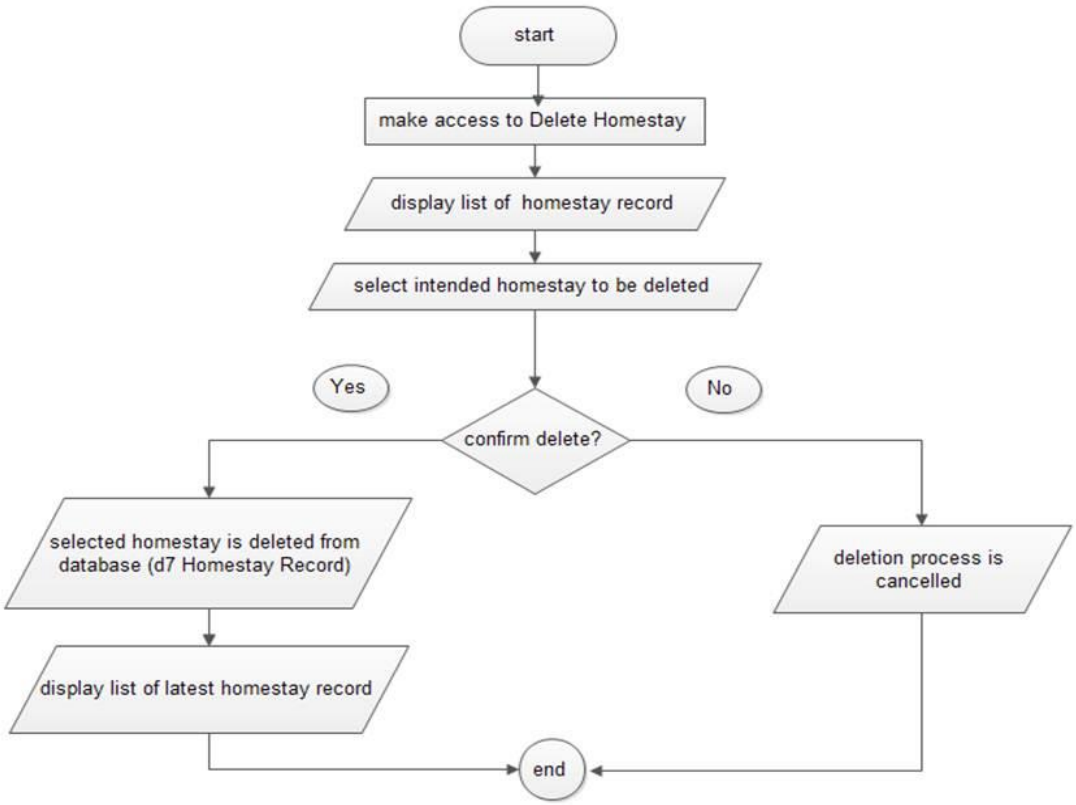
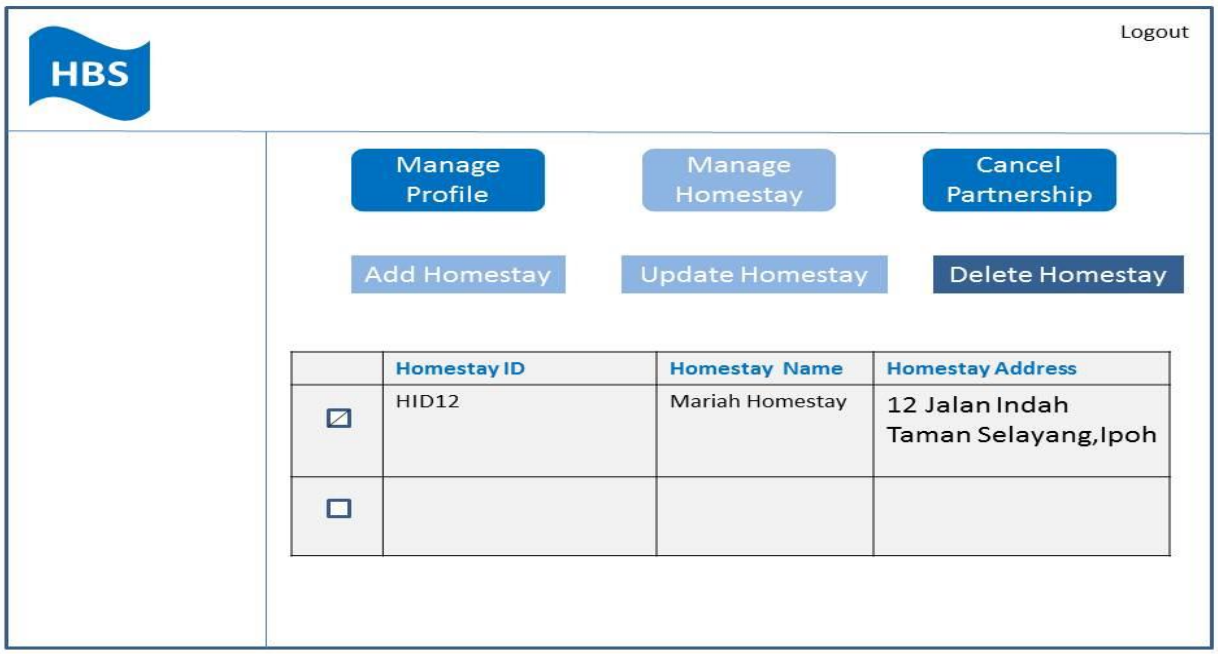


Figure 3.15.2

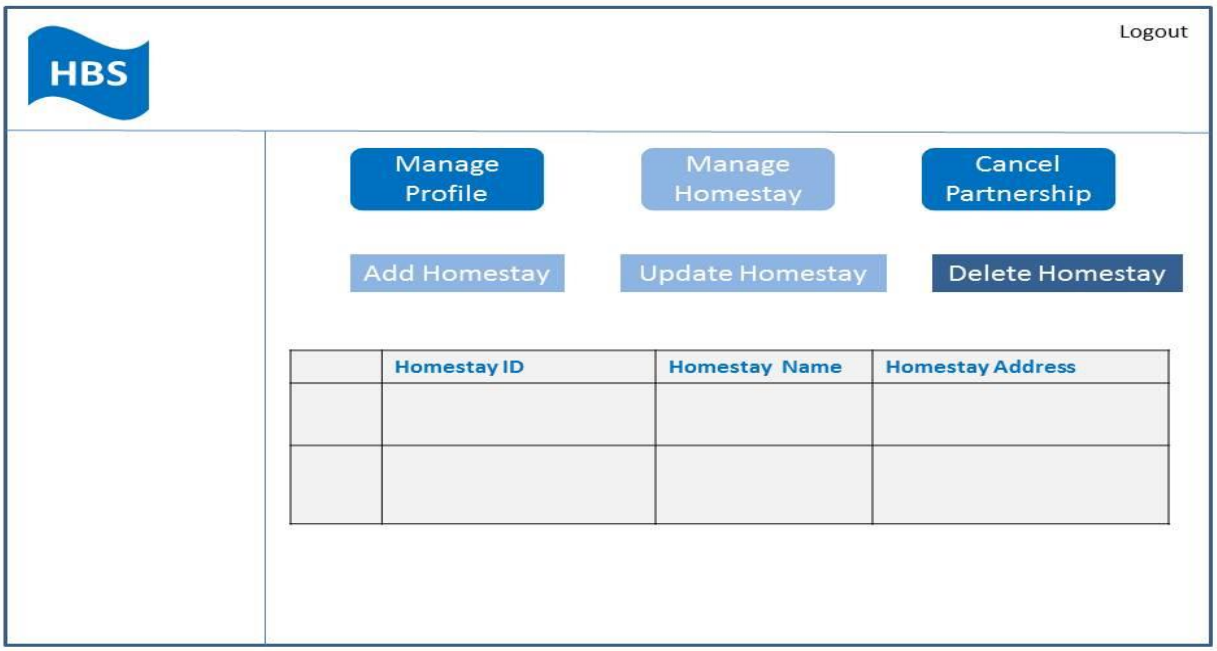
c. Graphic User Interface



Screen 1



Screen 2


HBS
Logout
Manage Profile
Manage Homestay
Cancel Partnership
Add Homestay
Update Homestay
Delete Homestay
Homestay ID
Homestay Name
Homestay Address

Screen 3

3.15.3 Process 12.3: Update Homestay

a. Pseudo code

1. Start
2. Make access to “Update Homestay”
3. Display all the homestay list
4. Select intended homestay to update
5. Update the homestay
6. if (select save)
    6.1 update changes is saved into database Homestay Record
    6.2 display the updated homestay
7. else
    7.2 update changes is cancelled
8. End

b. Flow chart

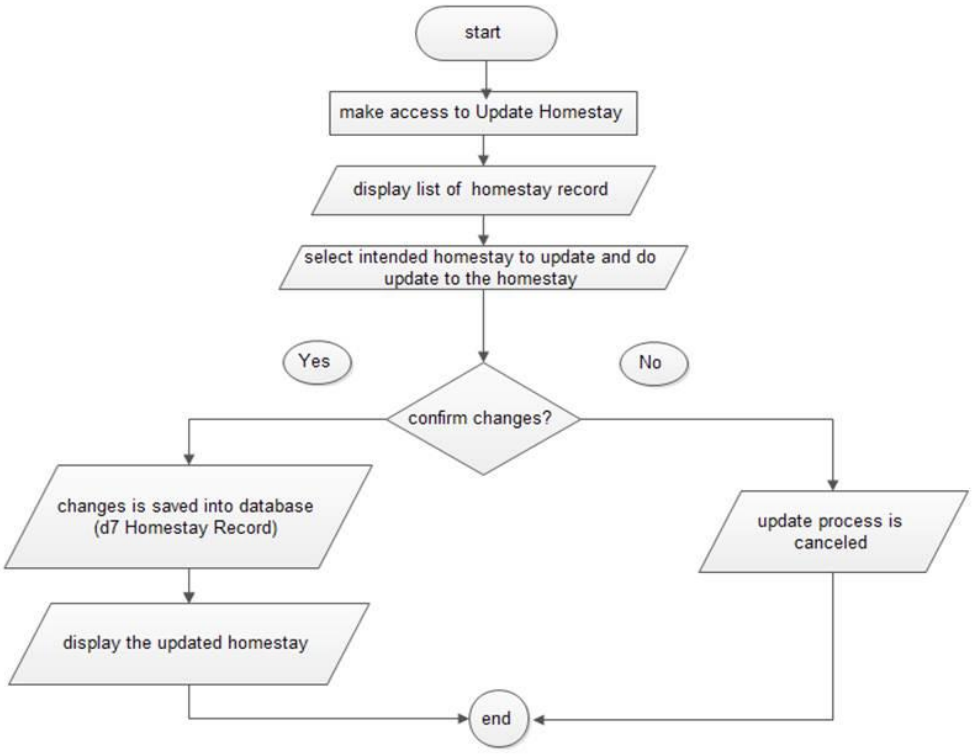
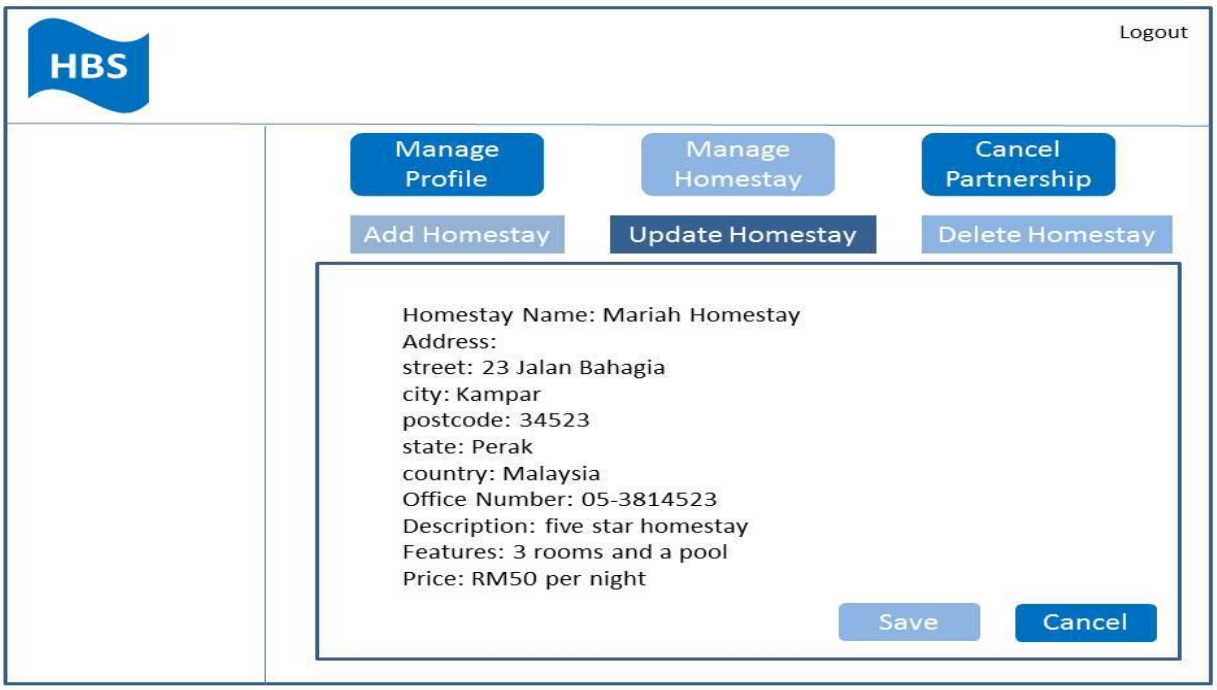
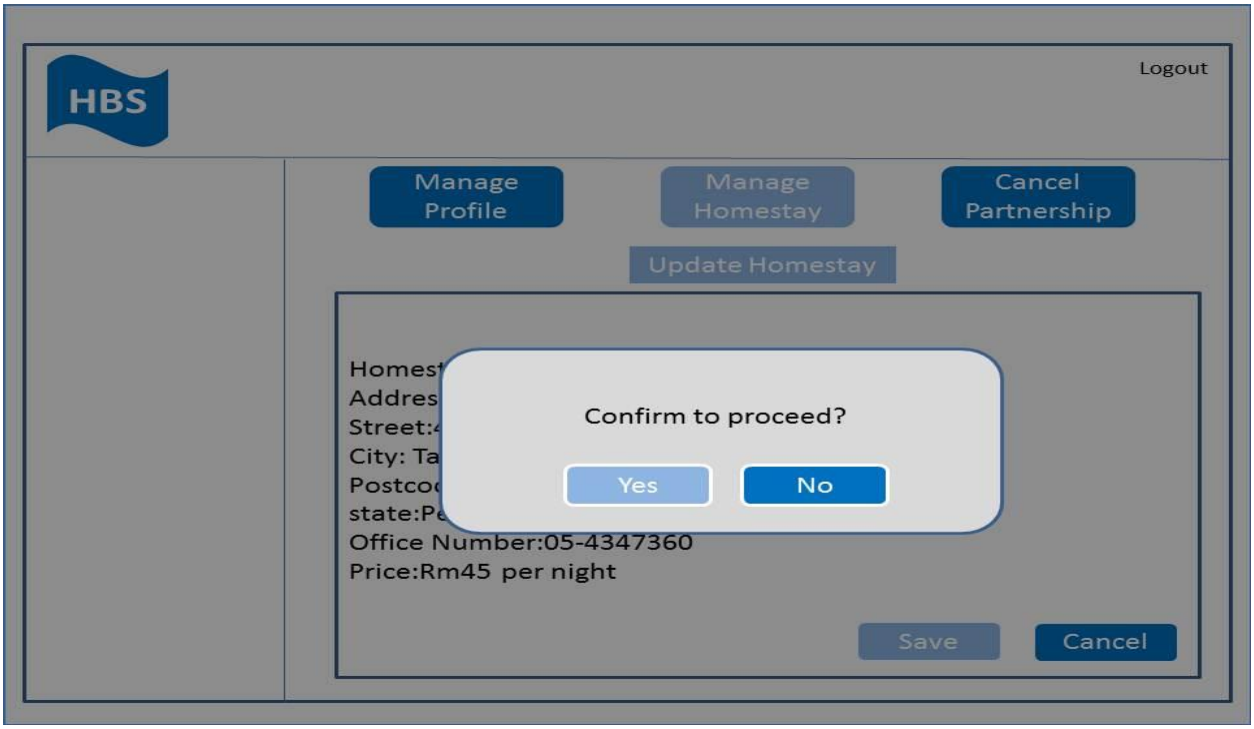


Figure 3.15.3

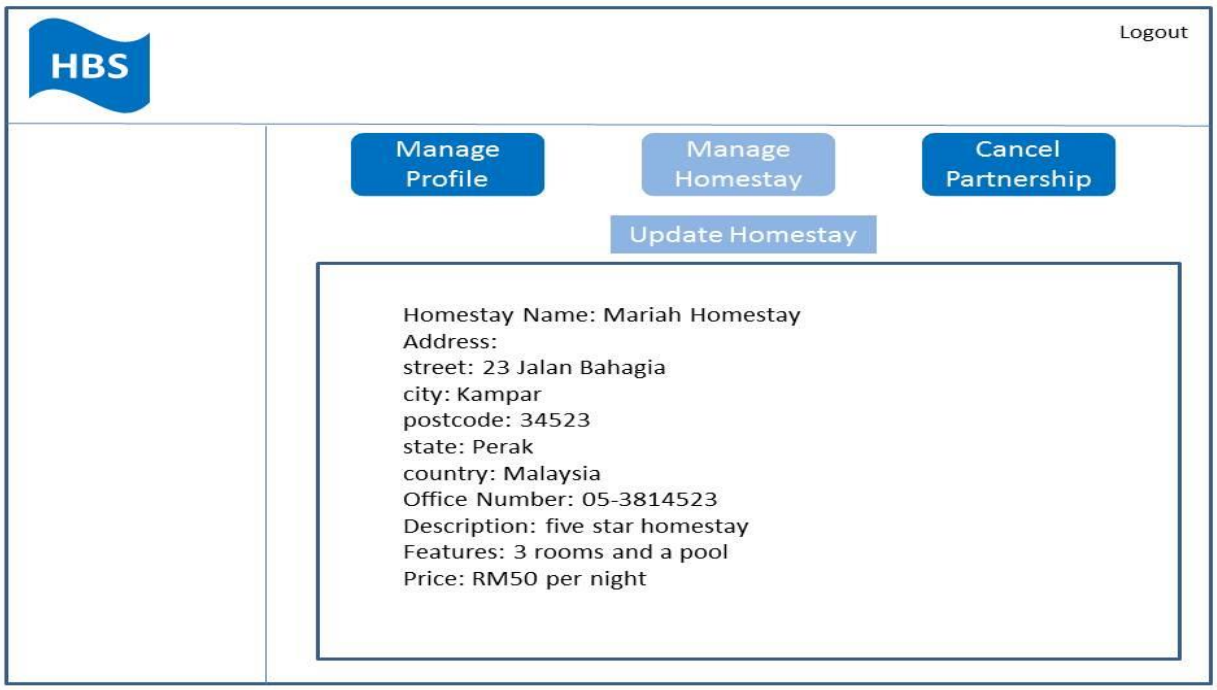
c. Graphic User Interface



Screen 1



Screen 2


HBS
Logout
Manage Profile
Manage Homestay
Cancel Partnership
Update Homestay
Homestay Name: Mariah Homestay
Address:
street: 23 Jalan Bahagia
city: Kampar
postcode: 34523
state: Perak
country: Malaysia
Office Number: 05-3814523
Description: five star homestay
Features: 3 rooms and a pool
Price: RM50 per night

Screen 3

3.16 Process 13.0 : Cancel Partnership

a. Pseudo code

1. Start
2. Make access to “Cancel Partnership”
3. if (select Terminate)
   3.1 changes is saved into database partnership Record
   3.2 owner account is inactivated and notification is sent to Owner’s email
4. Else
   4.1 cancellation is failed
5. End

b. Flow chart

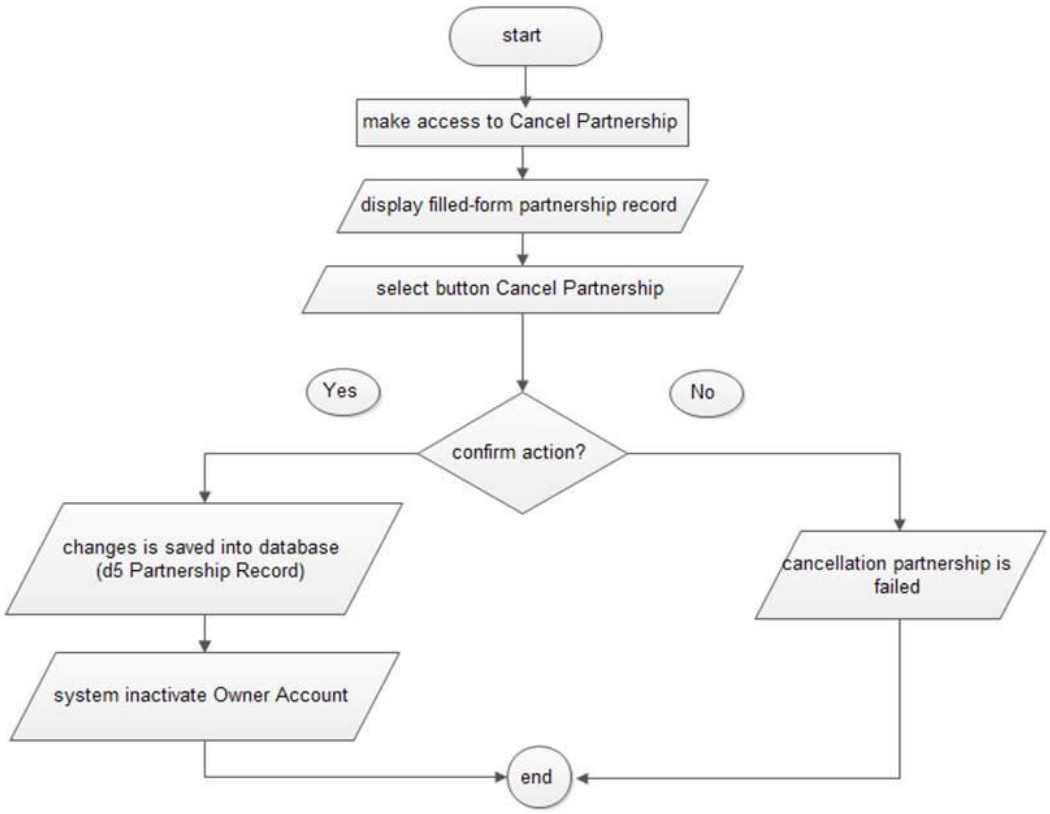
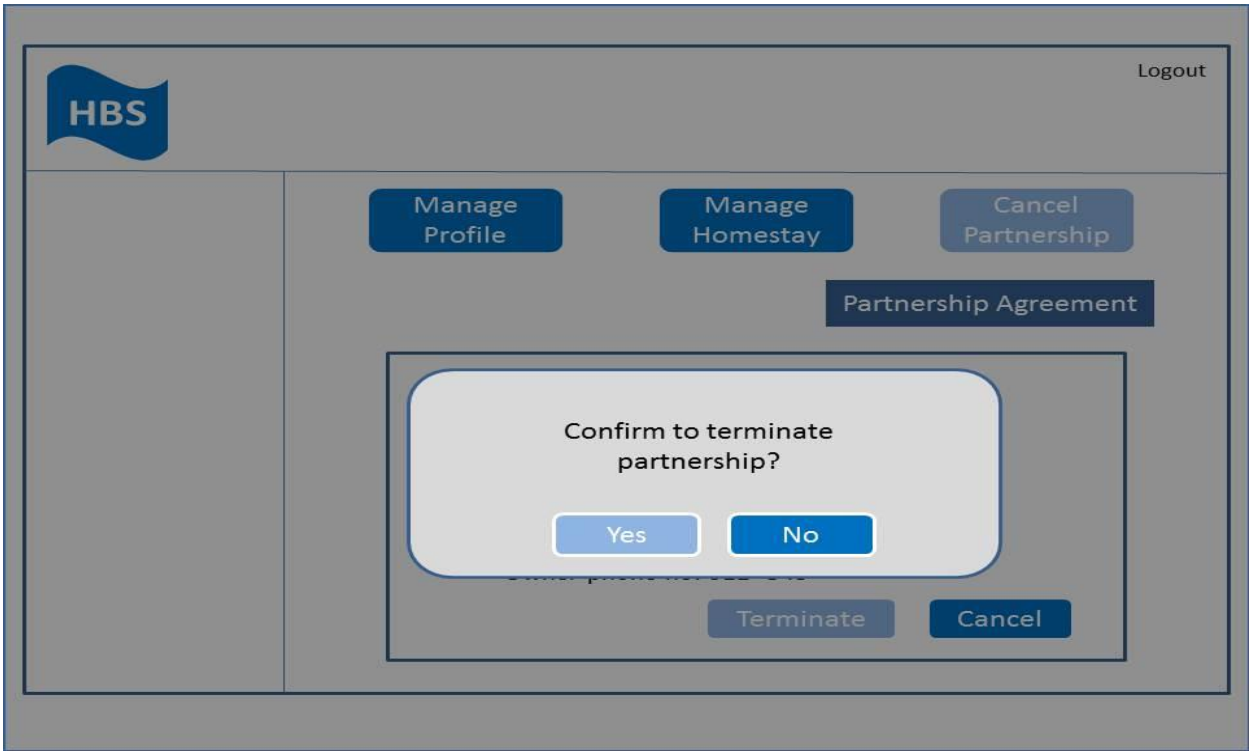


Figure 3.16

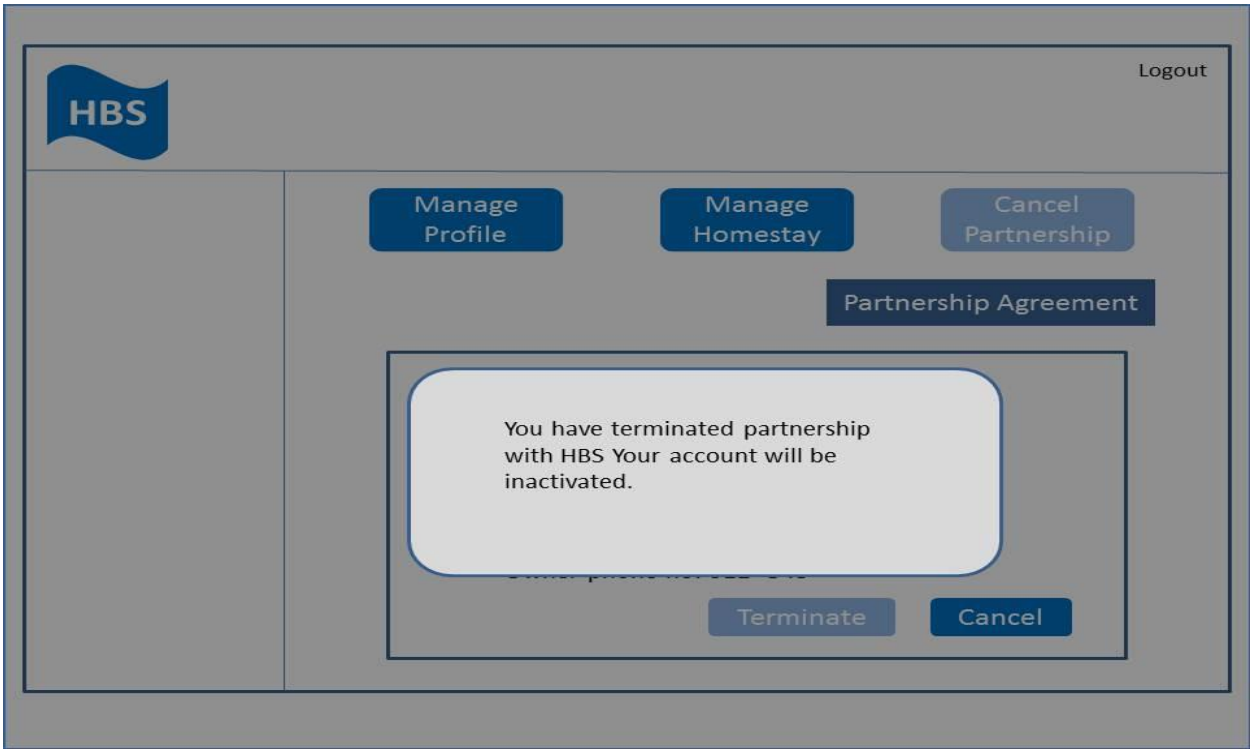
c. Graphic User Interface



Screen 1



Screen 2


HBS
Logout
Manage Profile
Manage Homestay
Cancel Partnership
Partnership Agreement
You have terminated partnership with HBS Your account will be inactivated.
Terminate
Cancel

Screen 3

3.17 Process 14.0: login process (For User , Admin, Owner)

a. Pseudo code

1. Start
2. Enter username
3. Enter password
4. Compare username and password with database.
5. If (true)
    5.1 Log in into system success
6. Else if (false)
    6.1 Re-enter the username or password
7. Else (forget password)
    7.1 Send an email to reset password
8. End

b. Flow chart

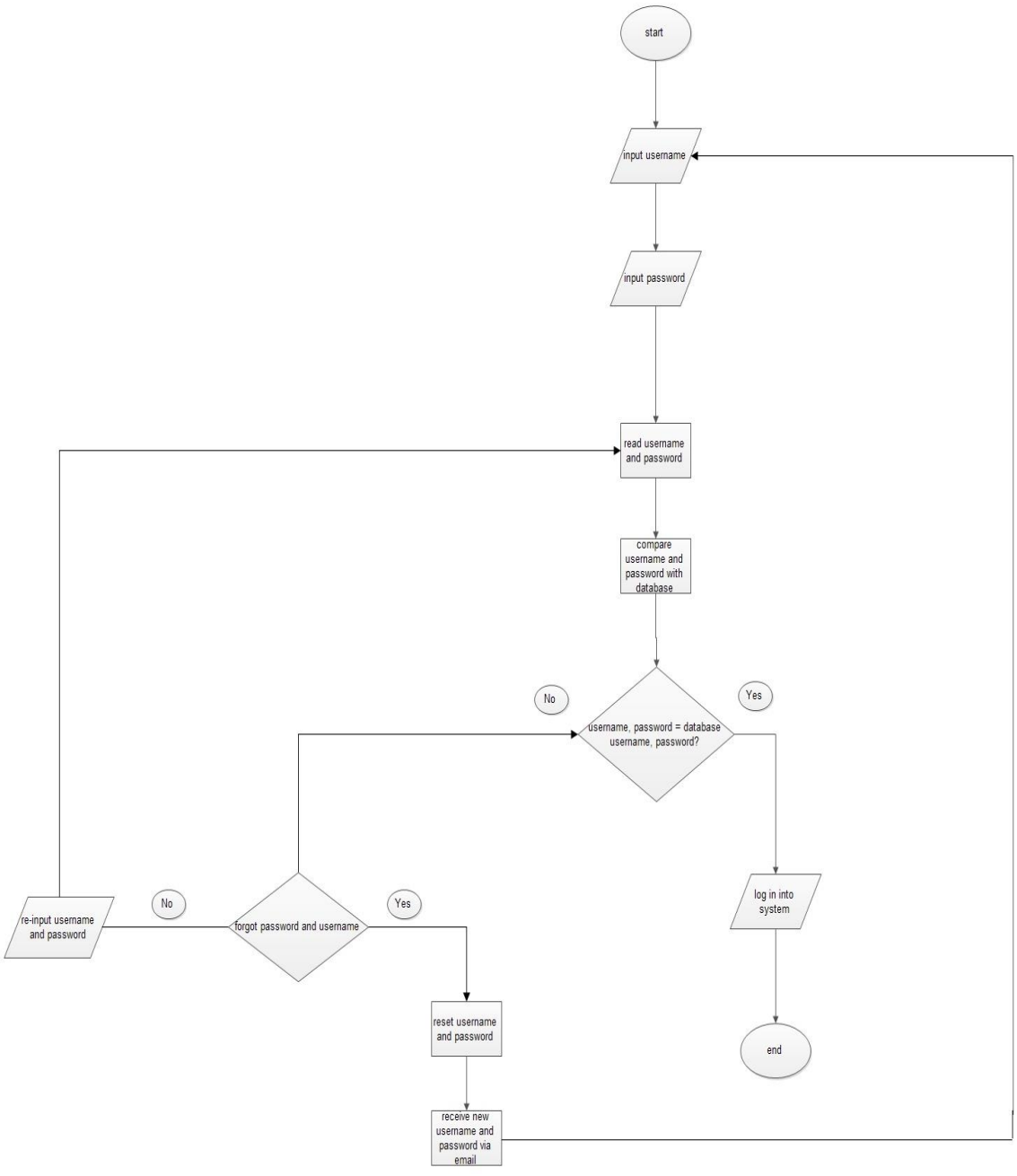
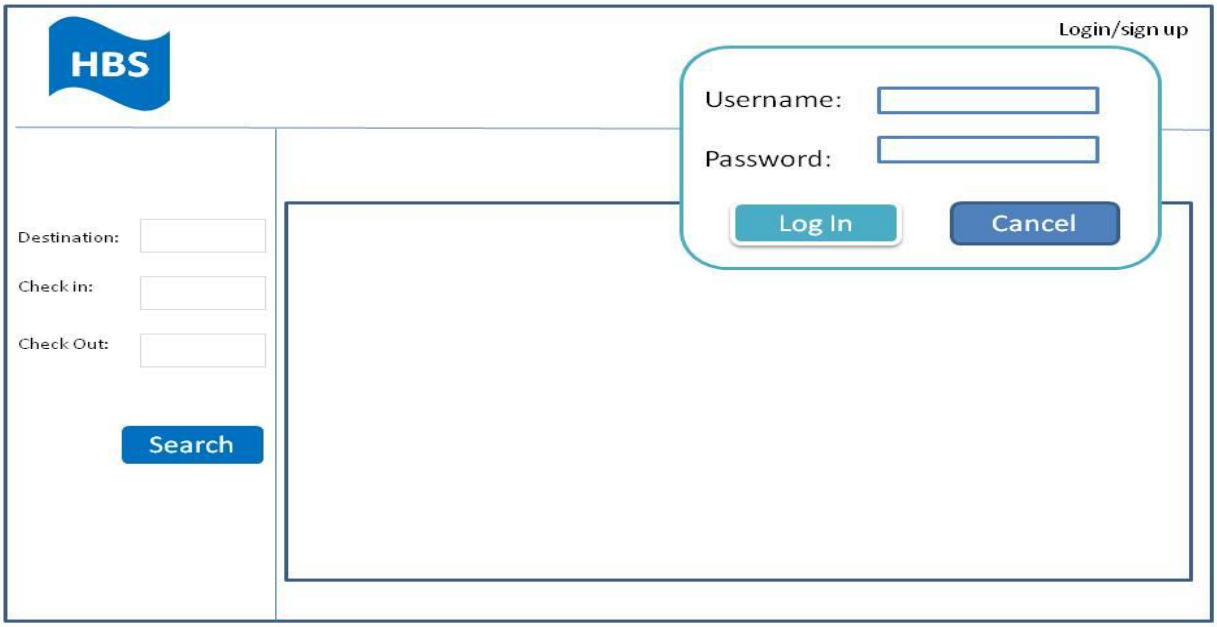
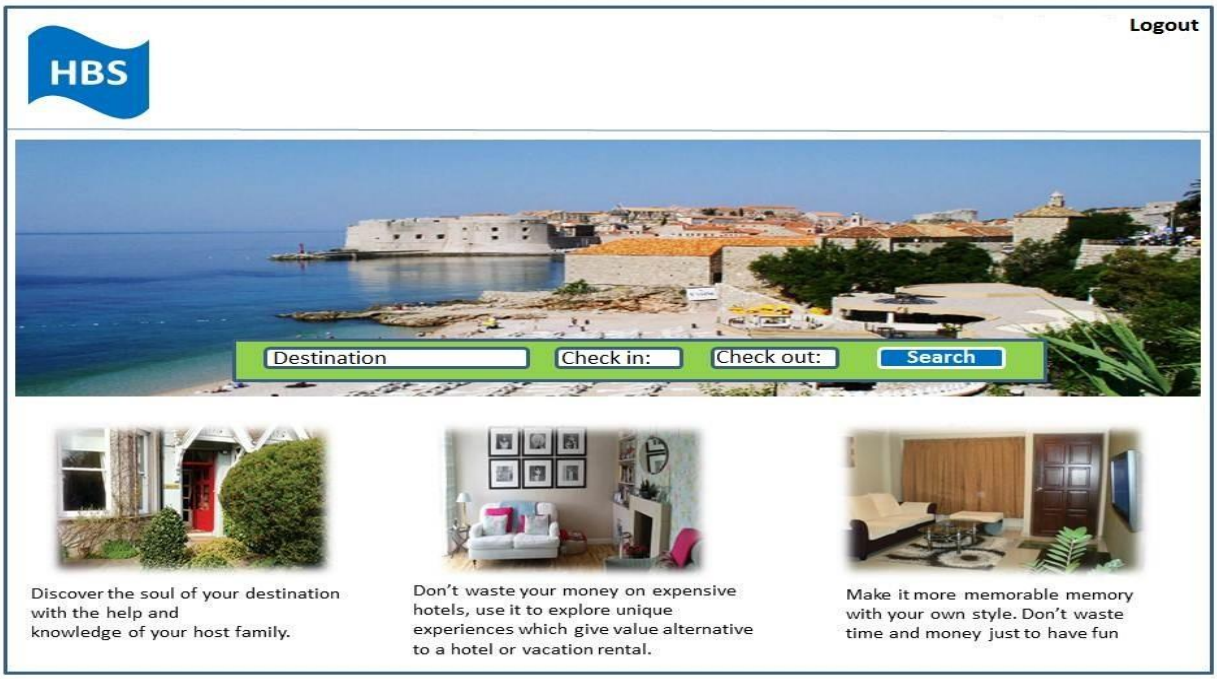


Figure 3.1

c. Graphic User Interface



Screen 1



Screen 2

3.18 Process 15.0: logout process (User , Admin , Owner)

a. Pseudo code

1. Start
2. Select log out
3. Log out from the system
4. End

b. Flow chart

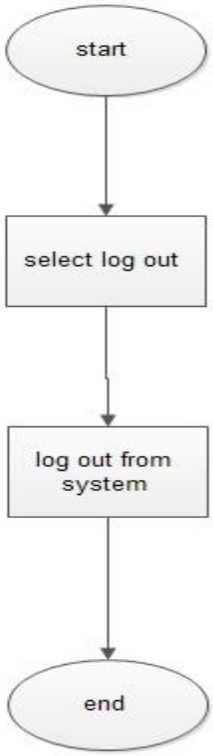
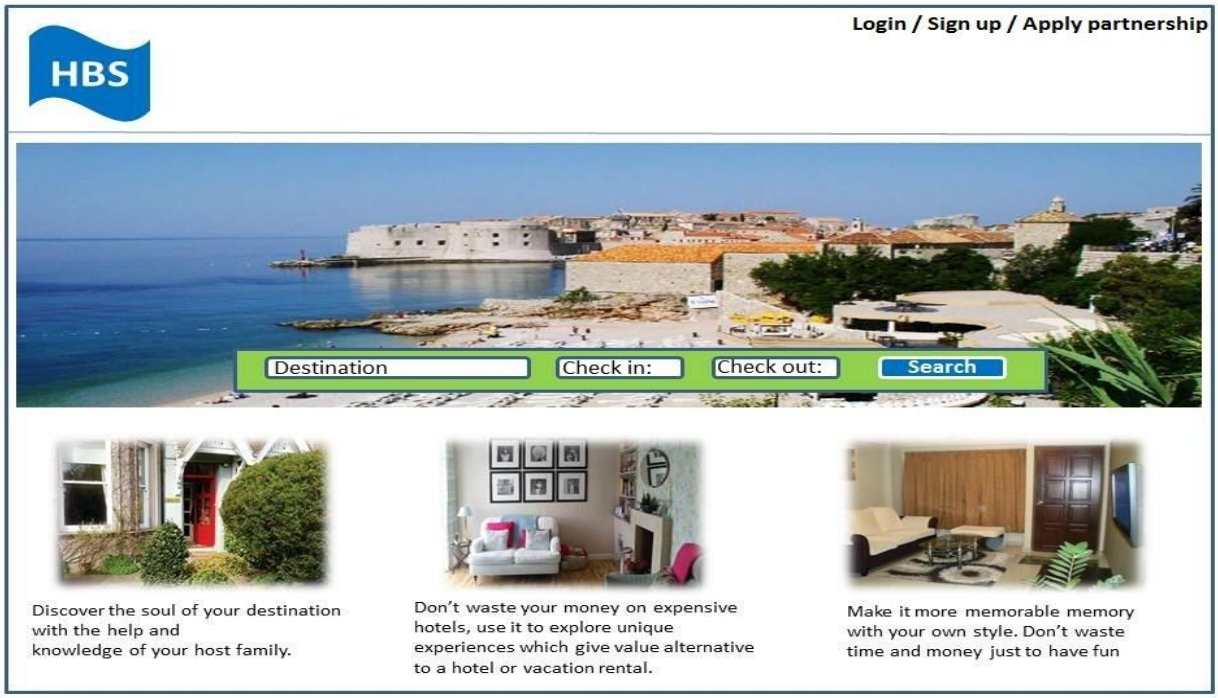


Figure 3.18

c. Graphic User Interface



Screen 1

CHAPTER 4 : DATABASE DESIGN

4.1 ENTITY RELATIONSHIP DIAGRAM

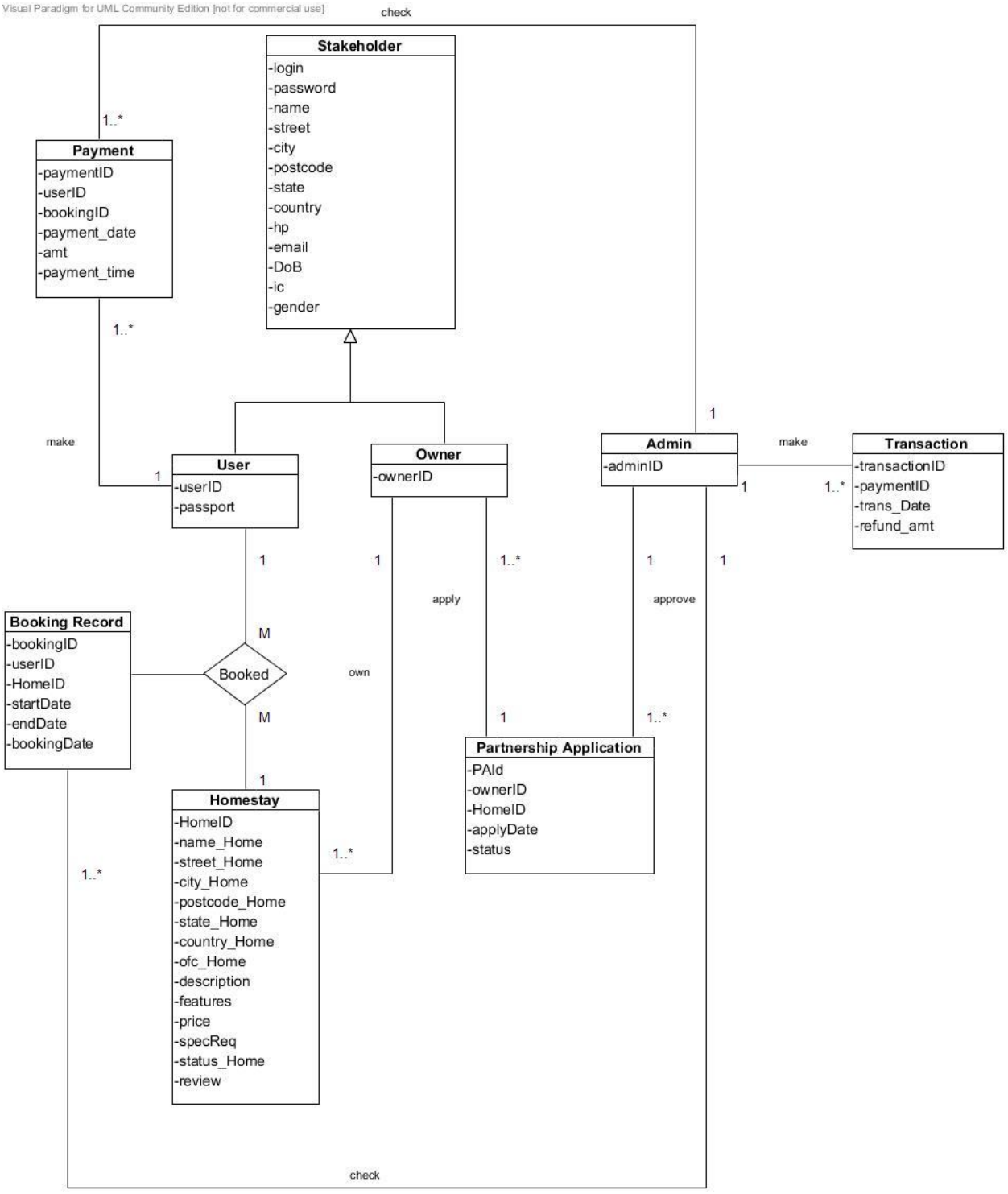


Figure 4.1

Figure 4.1 Entity Relationship Diagram

An entity-relationship diagram (ERD) illustrates system's entities information and entities' relationship. ERD composed of 3 things such as identifying and defining the entities, determine entities' interaction and the cardinality of the relationship.

Interaction of Stakeholder with User and Owner

- Inheritance : Stakeholder is a superclass to subclasses that composed by User and Owner
- All attributes in Stakeholder inherits by all subclasses.
- Only unique attributes that isolated among/between subclasses

Interaction of User and Payment

- One to Many relationship
- A User can make 1 or more than 1 payment
- But, only one payment can be made by a User
- In User : userID (PK)
- In Payment : bookingID (PK)

Interaction of User and Homestay

- Many to Many relationship
- A User may book 1 or more than 1 Homestay
- A Homestay can be booked by 1 or more than 1 User
- Thus, the relationship between entity User and entity Homestay is Many to Many relationship
- Hence, an associative entity is created from these named Booking Record entity and cardinality many to many is removed
- The data in associative entity show which User book in which Homestay.
- In User : userID (PK)
- In Homestay : HomeID (PK)

Booking Record

- Associative entity created from Many to Many relationship
- Contain information relationship between a User and Homestay booked
- Contain bookingID (PK) , userID (FK) , HomeID (FK)

Interaction of Owner and Homestay

- One to Many relationship
- An Owner can own 1 or more than 1 Homestay
- But, a Homestay can only be owned by an Owner
- In Owner : ownerID (PK)
- In Homestay : HomeID (PK)

Interaction of Owner and Partnership Application

- One to Many relationship
- Only 1 Partnership Application can be applied by an Owner
- But, 1 or more than 1 Owner can apply for Partnership Application
- In Owner : ownerID (PK)
- In Partnership Application : PAId (PK) also known as Partnership Agreement Identification Number

Interaction of Admin and Partnership Application

- One to Many relationship
- Only an Admin can approve 1 or more than 1 Partnership Application
- But, only 1 Partnership Application can be approved by an Admin
- In Admin : adminID (PK)
- In Partnership Application : PaId (PK) also known as Partnership Identification

Interaction of Admin and Transaction

- One to Many relationship
- One Admin can make one or many transaction
- But, one transaction can be made only by one admin
- In Admin : adminID (PK)
- In Transaction : transactionID (PK)

Interaction of Admin and Payment

- One to Many relationship
- One Admin can check one or many payment
- But, one payment can be checked by one Admin
- In Admin : adminID (PK)
- In Payment : paymentID (PK)

Interaction of Admin and Booking Record

- One to Many relationship
- One Admin can check one or more than one Booking Record
- But, one Booking Record can be checked only one Admin
- In Admin : adminID(PK)
- In Booking Record : bookingID(PK)

4.2 DATA DICTIONARY

1. **STAKEHOLDER**
2. **USER**
3. **OWNER**
4. **PAYMENT**
5. **HOMESTAY**
6. **BOOKING RECORD**
7. **PARTNERSHIP APPLICATION**
8. **TRANSACTION**

1. **STAKEHOLDER**

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	login	Login allowed user, admin, and owner to access into the system and do permitted action corresponding to their authorization. They can login by using email. e.g aminah32@yahoo.com	varchar	-	50		
2.	password	Password allowed user, admin and owner to access into the system as only correct password will be allowed to access the system and do permitted action corresponding to their validation. Password is changeable by user, admin and owner itself. e.g merah23hat71loi98	varchar	-	30		
3.	name	Name represents the full name of the actor e.g Siti Aminah binti Zakaria	char	-	50		
4.	street	Street is part of address of the actor. e.g 34 Jalan Utama, Blok A, Tingkat 1, Taman Indah	Varchar	-	50		
5.	city	city of actor's address e.g Tapah	Char	-	20		

6.	postcode	postcode of actor's address e.g 35500	Integer	-	5		
7.	state	State represents the state of actor's address. e.g Perak	Char	-	20		
8.	country	Country represent the country name of actor lives in. e.g Malaysia	Char	-	20		
9.	hp	Hp stands for actor's handphone number e.g 0123212345	Int	-	12		
10.	email	Email represent the actor's email e.g 23abds@yahoo.com	varchar	-	50		
11.	DoB	DoB represents date of birthday of the actor e.g 12/04/1991	DateTime	-	10		
12.	ic	Actor's identification card e.g 910304085955	Int	-	12		
13.	gender	Actor's gender	Char(1)	-	1		

2. USER

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	userID	userID represents the user's identification number, which is unique and distinguish from other user, owner and even admin itself. e.g UID01	varchar	PK	5		

2.	passport	Is an optional. Actor can choose either to fill in or leave it empty. It represents actor's passport number.	varchar	-	12		

3. OWNER

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	ownerID	ownerID represents the owner's identification number, which is unique and distinguish from other user or owner. e.g OID01	varchar	PK	5		

4. PAYMENT

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	paymentID	The payment identification number e.g PID01	varchar	PK	5		
2.	bookingID	BookingID represents the booking identification number. It represents the detailed information that lopcated at	varchar	FK	5		

		another place. Also can be said as reference number. e.g BID01					
3.	userID	userID represents the user's identification number, which is unique and distinguish from other user, owner and even admin itself. userID also represents itself in another document so that identification of user when checking process for payment can easily being done. e.g UID01	varchar	FK	5		
4.	payment_date	Current date that show when the payment is made e.g 02/09/2014	varchar	-	10		
5.	amt	amt represents amount of payment that cost for the homestay rental. e.g RM 100.00	varchar	-	7		
6.	time	Time represents the current time when the payment is made.	time	-	5		

5. HOMESTAY

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	HomeID	HomeID represents the homestay identification number. An owner can have many homestay, thus it is important to have unique homeID to represents uniqueness and make ease of referencing action. e.g HID01	varchar	PK	5		
2.	name_Home	name_Home represents the homestay full name (advertisement name) e.g Aminah Homestay Sweet 16	varchar	-	50		
3.	street_Home	Street_Home is part of address of the actor. e.g 34 Jalan Utama, Blok A, Tingkat 1, Taman Indah	Varchar	-	50		
4.	city_Home	city_Home of actor's address e.g Tapah	Char	-	50		
5.	postcode_Home	postcode_Home of	Integer	-	5		

		actor's address e.g 35500					
6.	state_Home	State_Home represents the state of actor's address. e.g Perak	Char	-	20		
7.	country_Home	Country_Home represent the country name of actor lives in. e.g Malaysia	Char	-	20		
8.	ofc_Home	Homestay office's phone number e.g 054347360	Int	-	12		
9.	description	Description represents the description of the homestay, the background or introduction about the uniqueness and facility of the homestay.	Varchar	-	50		
10.	feature	Feature represents the features of the homestay. For example how many rooms in a homestay house, activities that nearer to to the homestay and etc	varchar	-	50		
11.	price	Price represents homestay's rental price (according to Owner). e.g Rm 100 for a	varchar		7		

		night					
12..	specReq	specReq represents the special request that offered to User. The system list out the extra items that available to User if they want it. (in checkbox form) e.g extra bed	checkbox	-	-		
13.	gallery	gallery represents the collection of picture(view) that Owner put as advertisement for his/her homestay.	Jpag	-	1G		
14.	faxNum	faxNum represents fax number of the homestay.	Varchar	-	10		
15.	Status_Home	status_Home represents the availability status of the homestay. It means that, it will show either the homestay is booked (not available) or available.	char	-	20		
16.	review	review represents the comment that left by User that will describe about the effectiveness (user view).	varchar	-	100		

6. BOOKING RECORD

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	bookingID	BookingID represents the booking identification number. It represents the detailed information that lopcated at another place. Also can be said as reference number. e.g BID01	varchar	PK	5		
2.	userID	userID represents the user's identification number, which is unique and distinguish from other user, owner and even admin itself. e.g UID01	varchar	FK	5		
3.	HomeID	homeID represents the homestay identification number. An owner can have many homestay, thus it is important to have unique homeID to represents uniqueness and	varchar	FK	5		

		make ease of referencing action. e.g HID01					
4.	startDate	startDate represents the start date of User rent the homestay. e.g 12/12/2013	DateTime	-	10		
5.	endDate	endDate represents the end date of User stop rent the homestay. e.g 16/12/2013	DateTime	-	10		
6.	bookingDate	bookingDate represents the date of the booking application is made. e.g 08/12/2013	DateTime	-	10		

7. PARTNERSHIP APPLICATION

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	PAId	PAId represents the Partnership Agreement Identificaton Number, which symbolize for each partnership agreement that has been made between Owner and Admin. e.g PAId01	varchar	PK	5		
2.	ownerID	ownerID represents the owner's identification number, which is unique and distinguish from other user or owner. e.g OID01	varchar	FK	5		
3.	HomeID	HomeID represents the homestay identification number. An owner can have many homestay, thus it is important to have unique HomeID to represents uniqueness and make ease of	varchar	FK	5		

		referencing action. e.g HID01					
4.	applyDate	applyDate represents the current date the partnership is apply. e.g 03/11/2012	DateTime	-	10		
5.	status	status represents the status of the application e.g Approved or Disapproved	char	-	11		

8. TRANSACTION

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1.	transactionID	transcID represents the transaction identification number. It represents the detailed information that located at another place. Also can be said as reference number. transactionID only to deal with re-fund money if booking cancellation is made by User followed the terms	varchar	PK	5		

		and condition. e.g TID01					
2.	paymentID	The payment identification number e.g PID01	varchar	FK	5		
3.	transc_Date	transcDate represents the current date of transaction is made. e.g 15/12/2013	varchar	-	10		
4.	refund_amt	refund_amt represents the total amount of refund that being made if User successfully cancel the booking that has been made.	varchar	-	6		

CHAPTER 5 : GLOSSARY

1. PAId : Partnership Aggreement Identification number
2. OID01 : Owner ID 01, 01 represent owner number
3. PID01 :Payment ID 01, 01 represent any number
4. UID01 : User ID 01, 01 represent any number
5. HID01 : Home ID 01, 01 represent any number
6. BID01 : Booking ID 01, 01 represent any number
7. TID01 : Transaction ID 01, 01 represent any number